TFPT — Architecture and the E_8 Compiler

The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map, and the Coxeter–cyclotomic $D_5 \times A_3 \to E_8$ construction

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What this document is about

The architecture layer: how the two axioms $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$ build the Coxeter–cyclotomic compiler — the carrier $C^+ = D_5$, the family geometry $\mathbb{P}^1 \setminus \mu_4 = A_3$, the μ_4 glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, the EM fixed point α^{-1} (with its ablation), and the whole number alphabet $16, 40, 41, 48, 240, 248$ as carrier traces.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$ as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8, 4, 4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading document 1 (Architecture).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([**E**]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

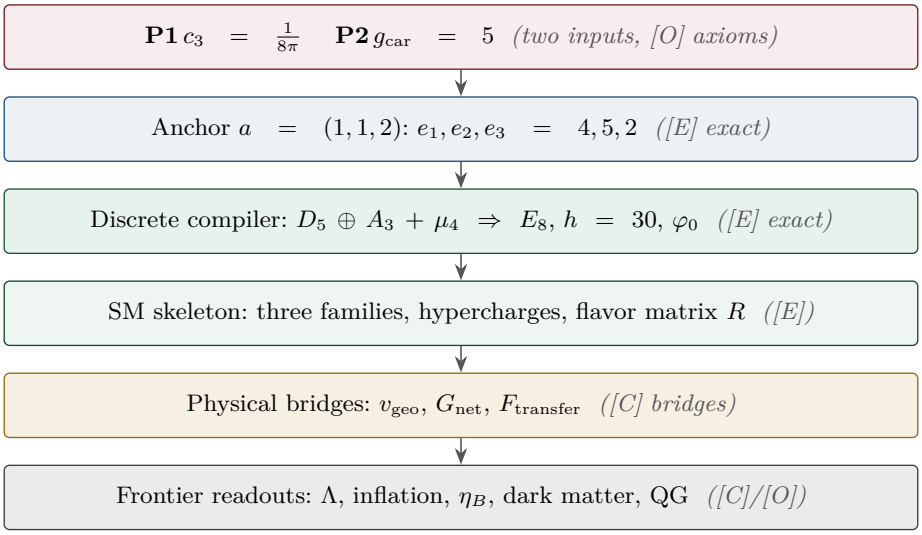
Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD $= 1$ (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ($h_s{=}16/16{=}1{\in}\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM $\mu{=}4/2^2{=}1$; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed'` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test



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Part I
Architecture and the derivation map

Abstract

This document develops the *architecture layer* of TFPT: the two axioms, the derivation map, and the explicit E_8 construction. The thesis is that the bridge layer of TFPT is governed by *three nested grammars*: a **carrier grammar** (from $3+2$ come charges, families, hypercharge, $\Omega_{\rm adm} = 48$, $b_1 = 41/10$), a **seed grammar** (one seed $u = \varphi_0^{\rm ret}$ generates the Cabibbo angle, birefringence, the baryon fraction and the reactor angle), and an **action grammar** (the large scale ratios are exponential actions of $\alpha_\star^{-1}$ with rungs $1/g_{\rm car}$, 1, 2). Two new exact results sharpen the action grammar: the *decuple bridge* $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} [\, v_{\rm geo}/(5\beta_{\rm rad}^2 \bar{M}_{\rm Pl})\,]^{10}$ (the cosmological constant is, in essence, the tenth power of the stripped electroweak scale, with $10 = 2g_{\rm car}$), and the Hubble corollary $H_0/\bar{M}_{\rm Pl} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$. We also give the exact corrected tree form of the Einstein normaliser $\xi = \frac{3}{4}/(1 + 9/128\pi^3)$, the dual-root dictionary that generates all hypercharges and transport cusps from $y_\pm = \{1/2, -1/3\}$, the Λ-anchored metrology closure, the overdetermination of $g_{\rm car} = 5$, and the E_8 carrier atlas in three clearly separated levels. Section 3 adds the strongest structural result: the carrier code is the $D_5 = \mathfrak{so}_{10}$ half-spinor (Aut $= W(D_5)$, order 1920; $R_4 =$ Clebsch graph), the family geometry $\mathbb{P}^1 \setminus \mu_4$ is an A_3, and E_8 is the explicit unimodular glue closure of $D_5 \oplus A_3$ — with the discriminant-form norms $q(D_5) = 5/4$ and $q(A_3) = 3/4$ summing to the E_8 root norm 2 and coinciding with the TFPT constants $\delta_2/\delta_{\rm top}^2$ and $\xi_{\rm tree}$. Status markers used throughout: [E] exact identity, [E] Lie/lattice theorem, [E] formalised, [E] numerical fixed point, [C] physical/conditional, [O] axiom/open. (All markers are explicit; the legacy `[M]`/`[C]` compatibility aliases have been removed.)

1 How to read this document

Notation convention. We write $D_5 \oplus A_3$ for the *root-lattice* direct sum (used in the glue/lattice statements: discriminant forms, μ_4 glue, E_8 as an even unimodular lattice), and $D_5 \times A_3$ for the *group/representation* branching (used for subalgebra embeddings and E_8-rep decompositions). The objects are the same atoms; the symbol marks whether the statement is lattice-theoretic ($\oplus$) or branching/representation-theoretic ($\times$).

Nothing here introduces a new primitive assumption; it reorganises existing outputs and proves the cross-links. The whole bridge layer rests on two inputs,

$$\boxed{c_3 = \frac{1}{8\pi} \ \text{(axiom P1)}, \qquad g_{\text{car}} = 5 \ \text{(axiom P2)}.} \tag{1}$$

Foundational postulates — the two genuine axioms (hardenable, not derivable)

Everything downstream is a theorem *given* two structural inputs that are not reduced further and are honestly flagged as the theory's axioms:

(P1) Boundary kernel.
A single oriented seam Σ carries a primitive, reflection-positive boundary kernel with unit winding $[u_\Sigma] = 1$; its self-linking normalisation fixes $c_3 = 1/(8\pi)$. *Status:* primitive postulate. *Organizing identity (2026-08-06, markers unchanged):* the closed normalization $I \cdot \beta_{\text{angle}} \cdot c_3 = 4 \cdot 2\pi \cdot \frac{1}{8\pi} = 1$ holds exactly with the Jones index $I = 4 = |\mu_4|$ a c_3-free theorem and $\beta_{\text{angle}} = 2\pi$ independently measured — the identity *explains* the value, it does not derive it; the open bridge is the even modular-response normalization (`P1.INDEX.KMS.01` [O], [`verification/v813_p1_index_kms.py`]). *Hardenable by:* the Calderón projector / determinant-character chain (the P1 hardening) and the Lean carrier-rigidity development (`TfptCarrier.*`, boundary-polarization and Calderón interfaces).

(P2) Carrier interface.
This splits into an *axiom* and a *theorem*, which must not be conflated:

- **P2**$_{\text{interface}}$ [O] **(axiom):** the seam is read out by a five-slot carrier $g_{\text{car}} = 5$ (3 colour + 2 weak) — a postulate (the readout interface).
- **P2**$_{\text{algebra}}$ [E] **(theorem):** *given* the carrier, hypercharge ($y_- = -\frac{1}{3}$, $y_+ = \frac{1}{2}$), rigidity and the family count follow from the idempotent/lattice axioms — machine-verified in Lean (`Hypercharge.lean`, `LatticeRigidityGeneral.lean`, `Rigidity.lean`).

So P2 is an *interface axiom* whose *algebraic consequences are a formalised theorem* — "axiom or theorem?" has the precise answer: interface axiom, algebra theorem.

These two are *inputs*: they can be formalised and stress-tested (more Lean, more interface theorems) but not computed away. Every other number in this note — $N_{\text{fam}} = 3$, $\Omega_{\text{adm}} = 48$, $b_1 = 41/10$, $\alpha_\star^{-1}$, the flavor matrix, the E_8 glue, A_s — is a consequence of (P1),(P2).

Anchor-first refinement. The two inputs are themselves the elementary symmetric polynomials of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity of the splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$): $e_1(a) = 4 = |\mu_4|$, $e_2(a) = 5 = g_{\text{car}}$, $e_3(a) = 2 = |\mathbb{Z}_2|$, with $c_3 = 1/(2e_1(a)\pi) = 1/(8\pi)$. The power sums $p_n(a) = 2 + 2^n$ then give $|R(E_8)| = p_1p_2p_3 = 240$, $\dim E_8 = 240 + (p_4 - p_3) = 248$ and the binary ladder $p_{n+1} - p_n = 2^n$. So the input set reduces to $\{a = (1, 1, 2),\ \pi\}$. [`verification/v23_anchor_generator.py`] The whole ladder is one affine map: $p_{n+1} = 2p_n - 2$ identically, with unique fixed point $2 = |\mathbb{Z}_2|$ and the compiler quintet $(4, 6, 10, 18, 34)$ as the T-orbit of 4 — the affine normal form of Theorem 8 ([`verification/v832_anchor_flavor_checksum.py`]).

The parabolic self-code of the anchor ([`verification/v566_parabolic_anchor_selfcode.py`]). The anchor is not only generated — it is *coded twice* by one operator structure that already lives in the compiler. The sheet-diamond direction operators $U = Q\,\text{diag}(1, 0, 0)$ and $V = Q\,\text{diag}(0, 1, 1)$ (v218; $\text{Spec}\, V = \{0, 1, 2\}$) generate the *full* 2|1 parabolic (line-stabiliser) algebra $\mathcal{A}$ (dimension 7, word determinant -81, multiplicatively closed; a perturbed Q breaks it) [E]. The Lagrange projector onto V's dominant eigenvalue gives $H = I + \frac{1}{2}V(V - I)$ with spectrum *exactly* $\{1, 1, 2\} = a$, and $\chi_H(x) = x^3 - 4x^2 + 5x - 2$ carries the double code $(4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|) = (\dim_{\mathbb{R}}(GL_3/P), \dim(\mathcal{A}/J), \dim J)$ — flag, Levi quotient and radical of $\mathcal{A}$ [E]. A uniqueness theorem closes the family: the line-stabiliser structure polynomial factorises as $(x - m)(x^2 - mx + 1)$, and all-positive-integer spectra force $m = 2$

— $a = (1, 1, 2)$ is the *only* anchor this parabolic family can code [E]. Honest typing: this is a *compression* of one operator system, not new independent evidence (no-free-pattern), and it is *not* a $P2$ reduction — the route "$g_{\text{car}} = \dim(\mathcal{A}/J)$ as a derived value" is conditional on the *No Carrier Reconstruction* audit [O]; the axioms' markers are untouched. *Stage A of that audit is since executed* ([`verification/v567_carrier_free_q_reconstruction.py`]): with the v11 carrier budgets *forbidden* (rows $(4, 5, 6)$, columns $(9, 5, 1)$, χ_Q with its 5), the carrier-free inputs — sheet involution, winding normalisation, A_3 exponent spectrum, monodromy budget, seam-line demand — admit exactly 28 parabolic reconstructions, and **all** 28 **code the same anchor** $\chi_{H'} = x^3 - 4x^2 + 5x - 2$: the anchor self-code is carrier-free; the single family invariant $81 = N_{\text{fam}}^4$ then pins Q to a two-element residual family ($q_{32} \in \{2{=}|\mathbb{Z}_2|, 0\}$) on which every classical carrier-free certificate coincides — the one separating certificate is exactly the forbidden column budget. Verdict RECONSTRUCTS-TO-FAMILY; the audit relocates to `GATE.QGEO.01` (derive the geometric inputs themselves) and to the one binary sheet-coupling selector [O] — which is since *located and given a canonical witness* ([`verification/v568_bit_selector_ladder.py`]): the twins generate different integral orders, and the *ladder self-consistency* (the dominant mode reproduces its own eigenvalue as the geometric ladder $(1, 2, 4) = (2^0, 2^1, 2^2)$, the binary AnchorLadder) selects the compiler Q uniquely on all 28 members; the split geometry ($q_{32} = 1$) is excluded by the demands already imposed; *why* the ladder self-consistency should hold physically stays open (the RP-selection slot) [O]. Three follow-ups since executed: the *kinematic RP form* on the twins ([`verification/v572_rp_bit_form.py`]) is spectrally blind (identical inertia for every declared corpus state; never PSD — no kinematic positivity selection exists) but *integrally anchor-visible*: the Smith towers of the anchor-state Gram separate the twins in one slot (18 vs 2, a factor $9 = N_{\text{fam}}^2$) — the bit shows to the RP form only through anchor-coupled states [E]; and *Stage B is narrowed* ([`verification/v574_stageb_winding.py`]): the involution *type* is forced by a rank theorem (a two-dim-positive involution cannot carry the binary trace code), and the winding normalisation $N_{\text{fam}} = 3$ is *forced* by the double self-consistency — the ladder selects one member for every $N \le 3$ (higher N die on the exponent demand), and sat $= N^4$ holds only at $N = 3$ ($81 = 3^4$; $N = 1, 2$ give $32 \neq 1$, $512 \neq 16$): two of the five declared geometric inputs relocate to consistency-forced; `GATE.QGEO.01` is narrowed, not closed [E]/[C]. The involution-*existence* input is since *derived* ([`verification/v590_involution_existence.py`]): the integral V-commuting involutions are exactly the spectral sign group $(\mathbb{Z}/2)^3$ (rigidity, both twins), and the v574 type theorem plus two named demands (seam reversal, zero-mode fixing) single out the explicit S_0 with exact Σ-signature uniquely — existence is unconditional; the two demands' continuum derivation stays the named open [E]/[C]. The *interacting* follow-up ([`verification/v584_ext_source_response.py`]) then separates the twins *directionally*: rank-one Ext-source coupling on the Gibbs pencil has exact rational flip thresholds $J_c(s) = -1/(v^T K(s)^{-1} v)$ whose numerators are exactly the twins' pencil factors; in the crossed phase positive coupling $J_c \le 15/16 < 1$ heals exactly the compiler twin (through either Ext arrow), while the alt twin cannot be flipped by any positive coupling in the open unit box — the healing direction is strictly positive, consistent with the v534 toy RP selection; the response pole sits on the golden polynomial $s^2 - s - 1$ (observed). Honest: a response-level selection, not an OS/RP positivity theorem — the alignment-bit contract stays open [E]/[C]. A *geometric candidate* for the whole `GATE.QGEO` realization has since emerged ([`verification/v597_mu4_cover_model.py`]): the μ_3-cover $y^3 = x^4 - 1$ of $\mathbb{P}^1$ branched at μ_4 hits four compiler fingerprints at once — H_1 is a rank-3 module over the Eisenstein integers; the D_4 rotation closes projectively via the deck ($r^4 = \omega$); the cusp-class group is $(\mathbb{Z}/3)^4$ of order 81 (the saturation index, with the per-mark $\mathbb{Z}/3$ mechanism the self-code program asked for); and the unique invariant hermitian form has Lorentz signature $(1, 2)$ — so the RP-positivity route would become Riemann bilinear positivity if the identification completes. The dictionary since stands at *three* exact entries ([`verification/v598_cover_dictionary.py`]): the winding operator U is a single-cusp Gram, the μ_4 sheet coupling Q_- is the *difference of the opposite cusp Grams* (the diamond diagonal), and the binary AnchorLadder $(1, 2, 4)$ is identity plus the diagonal sum; the grading half (Q_+, hence V and Q) is absent from 7296 scanned canonical combinations — the named suspect for the missing generator is the anti-holomorphic D_4 reflection (the real structure); the gate does not move [E]/[C]. The suspect has since been *constructed* ([`verification/v599_real_structure.py`]: R unique up to scale with $R^2 = +1$; its grading functor closes the three gaps at char level, with the anchor spectrum $(1, 1, 2)$ appearing; the invariant form has $\det J = 8$ and Lorentz signature $(1, 2)$), and the *joint embedding has landed* ([`verification/v600_joint_embedding.py`]): a Γ-even degree-2 pair (U^*, V^*) is $\mathbb{Z}[\omega]$-integral, GL_3-conjugate to the compiler pair (explicit g, exact), restricts to *integer* matrices on the saturated real fixed lattice, and its seven-word $\mathbb{Z}$-order has saturation index $81 = 3^4$ with the exact v566 divisor pattern — the self-code algebra lives on the cover; canonicity stays open and the gate still does not move [E]/[C]. The canonicity question has since been answered ([`verification/v601_equivariant_dual.py`]):

the J-adapted cusp projectors give a presentation that is D_4-*equivariant by construction* (mark choice = winding choice) and realizes the *dual* parabolic exactly (conjugate to the transposed compiler pair), and the v590 Klein four-group lives integrally on the curve [E]/[C]. *The cover program at a glance (v597–v605).* Nine rounds close into one narrative: the μ_3-cover $y^3 = x^4 - 1$ carries the fingerprints (v597), the dictionary (v598), the real structure with R^2=1 and the Lorentz form $\det J = 8$ (v599), the joint pair with saturation index $81 = 3^4$ (v600, Lean-certified), the D_4-equivariant canonical presentation and the integral Klein group (v601), the J-adjoint duality, the verbatim anchor formula $H = 1 + V(V-1)/2$ and exact Weil positivity (v602), the *real* seam line with the geometric seam-reversal demand and the first per-mark $\mathbb{Z}/3$ witness (v603), the anti-isometry $M^\dagger J M = J^T$ with the integer form J_{fix} ($\det 72$), the full four-frame order of index $27 = 3^3$ and the torsion localization (81 at the marks, 27 at infinity) (v604), and the RP translator $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(c)$ with $14c$ integral (v605). The gate does *not* move: the residual is the conformal seam identification (`QGEO.SYM.01`), the zero-mode selection (the zero-mode line is *not* R-real — honest negative), and the geometric identity of c [E]/[C]/[O]. The honest decomposition (spectrum ablation): freeing the exponent spectrum too, the one surviving double-selfconsistent family is exactly $\{1, 2, N\}$ with the compiler shape — the self-consistency forces the *sheet* spectrum $\{1, 2\}$ (the binary code) and the shape, and the A_3 exponent demand contributes exactly *one* datum, $N = 3$ [E]. And the redundancy ablation shrinks the input list further: relative to the self-code demands, the monodromy budget (M) and the seam-line demand (L) are *redundant* (both dropped, the class widens to 1274 members and the self-code still picks the compiler Q uniquely), and the maximal ablation drops (E) too: the winding column $3(1, 1, 1)$ plus integrality plus the self-code already pin the compiler alone. Even the winding *form* is now located: dropped entirely, the self-code leaves a four-column family of which exactly one is uniform — the compiler; the winding demand *is* the uniformity choice (1 of 4). What remains declared: the involution *existence*, the column *uniformity*, and $N_{\text{fam}} = 3$ (independently anchored in the E_8 glue, $240 = 16 \times 5 \times 3$) [E]. On the bit itself, the anchor-Hamiltonian *Gibbs pencil* delivers the first dynamical distinction: the compiler twin's RP kernel degenerates at a finite critical temperature $s^* = (8 - 4\sqrt{19})/15$ (determinant constant 81^2), the alternative twin's never does — and the critical mode is the seam-row, sheet-flow-fixed direction (the A_2-Ext carrier): a rank-drop marker, not yet a positivity selection, but the resonance with the alignment bit now has a shared carrier [E].

The Gaussian code bridge and the quartic half (2026-08-03) — the code section closes two loops

Two exact-arithmetic results tie the binary code layer directly to the E_8 lattice and its complex μ_4 structure. *The Gaussian code bridge* ([`verification/v689_gaussian_code_bridge.py`], 26 checks, exact): the μ_4 complex structure J makes E_8 a rank-4 module over the Gaussian integers $\mathbb{Z}[i]$, and the quotient $E_8(\mathbb{Z}[i])/(1+i)$ *is* $\mathbb{F}_2^4$ — the information space of the Reed–Muller code RM(1, 3) (Smith normal form exact; standard model $|\det| = 256 = 2^8$, SNF $[1^4, 2^4]$, presentation-independent). The 240 roots fall 15×16 over the fifteen nontrivial classes (the zero class provably empty), each class is 4 of the 60 G31 lines ($60 = 15 \cdot 4$), the compiler clock $\sigma = c^4$ acts as the RM(1, 3) *family permutation* (3-cycle plus fixed anchor — the v638 info-bit action) with a residual gauge group of order 18, and the coordinate block is the sum-of-families label $F_1{+}F_2{+}F_3$. Three must-fail controls fire (naive placement, non-πJ placement, plain $\mathbb{Z}[i]^4$) [E]. *The quartic half* ([`verification/v690_quartic_half.py`], 22 checks, no floats): the vanishing of the $W(E_8)$ invariants of degrees $\{2, 14, 18, 30\}$ under holomorphic μ_4 restriction is proven symbolically but *trivial* — $W(J\alpha) = -i\,W(\alpha)$ kills every J-orbit contribution for all $d \not\equiv 0 \bmod 4$ (honest typing); the substance is the surviving half: the restrictions $F_8|, F_{12}|, F_{20}|, F_{24}|$ are G31-invariant (60/60 mirrors), algebraically independent (exact Jacobian), with Molien-unique degrees of product $46080 = |G31|$ — by Chevalley (cited) a system of *basic* invariants, i.e. the holomorphic restriction of the $W(E_8)$ ring generates the full G31 invariant ring; the Jacobian carries the 60 mirror lines point-verified. Controls: a non-root-preserving complex structure breaks the informative half, a root-preserving one keeps it — the statement is about the compiler-compatible conjugacy class [E]. No marker moves; both are structure statements of the code layer, registered as `E8.GAUSSCODE.01` and `E8.QUARTICHALF.01`. *The orbit packet lemma (2026-08-04):* the code bridge's number family closes as one lemma ([`verification/v736_orbit_packet.py`], 26/26, ORBIT-PACKET-EXACT; companion Lean module `OrbitPacket.lean`, 34 theorems, kernel-checked):

σ orbit census (Burnside)	$7 = (15 + 3 + 3)/3$
canonical packet split	$240 = 5 \times 48$ (1 fixed + 4 moved, 12 Gaussian lines each, label-free)
carrier reading	packets $= 1 + 4 = 5 = g_{\text{car}}$, $240 = g_{\text{car}} \times 48$
anchor	leader weights $(0, 1, 1, 2) \to (1, 1, 2)$, pair e_6/e_7
$q = 16$ normal form	$240 = 16 \cdot 15$, $\quad 248 = 240 + 8$

The circularity fence is part of the lemma: P1 ($c_3 = 1/(8\pi)$) and P2 ($g_{\text{car}} = 5$) come back from the code as *exact internal reconstruction* — the packet law reconstructs the axioms, it does *not* derive them; and the $1 + 4$ law is σ-*class*-specific (the 160 free order-3 elements of G31 give Burnside 5, not 7). The sharpening to a canonical intertwiner with the 48 regular five-cycles is *dead* on its preregistered criterion ([`verification/v737_seam48_intertwiner.py`], 24/24 machinery, SEAM48-DEAD: incidence distribution 0:96/1:64/2:64/3:16 instead of $\equiv 1$) — $240 = 48 \cdot 5$ remains a factorization *without* coordinatization **[E]**. *The symplectic code lemma (2026-08-04):* the quotient carries its own incidence geometry ([`verification/v752_projective_hamming_incidence.py`], 36/36, PROJ-HAMMING-EXACT; companion Lean module `ProjectiveHamming.lean`, 32 theorems): $h = H/4$ is $\mathbb{Z}[i]$-valued with hermitian Gram determinant 1 — the *unimodular* hermitian E8 lattice over $\mathbb{Z}[i]$ — and the reduction $\bar{h}$ makes $V = L/(1 + i)L$ a symplectic $\mathbb{F}_2^4$ (doubly-even $\Rightarrow$ alternating; explicit Darboux basis); on the 15 nonzero classes the hyperplane incidence matrix satisfies $BB^{\mathsf{T}} = 4I + 3J$ *exactly*, with design parameters 2/3/4/7/15/16 and eigenvalues $7^1(+2)^9(-2)^5$ (the multiplicities $(1, 9, 5)$ are the symplectic signature), the channel $K = B/7$ obeys $K^2 = (4/49)I + (45/49)\Pi_0$, and the NS/R parity is the point $y_0 = \text{FSUM}$ of the geometry ($\chi_{\text{NSR}} = \bar{h}(\cdot, y_0)$; $112 + 128 = 240$, $248 = 120 + 128$). The lemma also *resolves* the seam-48 kill: the distribution above is derived exactly (16× the 3×5 label table), and the 5×48 coordinatization exists via exactly one $\bar{u}$-invariant spread — the kill is explained (wrong blocks), not revised **[E]**. *The Arf/spinor compiler (2026-08-05):* the code layer now carries its canonical quadratic refinement ([`verification/v774_arf_spinor_compiler.py`], 46/46, ARF-SPINOR-EXACT; companion Lean module `ArfSpinorCompiler.lean`, 23 kernel theorems): of the exactly 16 refinements of $\bar{h}$ the frozen selector (σ-invariance, $q(A)$=1, $q(F_\Sigma)$=0) picks a *unique* q^*, which *is* the parity-lift weight form under $\iota\colon V \cong C_{\text{even}}(5)$ (crosslink $\beta = \bar{h}$ exact in all 256 cells); $\text{Stab}(q^*) \cong S_5$ makes $16 = 1 + \bar{5} + 10$ an *Arf geometry* ($Sp(4, 2) \cong S_6$; K_6 duads; $B = I + A_{KG(6,2)}$ with exact ovoid eigenvectors), and the hypercharge table $X = (-2, -2, -2, 3, 3)$ reproduces the `v310/v14` dictionary with the master-moment chain 120/40/41/240/248; the same σ acts as the family 3-cycle A_3^{Fam} on message bits and as the 3+2 slot split A_3^{PS} on carrier legs — naming discipline, not a physics claim **[E]**. *The fired matter fence (2026-08-05):* the preregistered kill test of the code→matter reading *fired* ([`verification/v775_gaussian_class_d5_purity.py`], 18/18, ROOTCLASS-MIXED): no $D_5 \oplus A_3/SU(5)$ convention makes even one full set of 15 classes pure (≥ 7 classes are adjoint-side for *every* embedding, by the 128-spinor-root counting bound); the code→matter table — including the "empty mailbox = right-handed neutrino" reading — must not enter any main claim **[E]**. *Boundary code uniqueness (2026-08-05):* the non-circular selection chain exists ([`verification/v776_boundary_hamming_uniqueness.py`], 31/31, BOUNDARY-CODE-UNIQUE): among all doubly-even self-dual binary codes of length 8 (direct census 30, no Hamming/E_8 input) the boundary data — four J-paired marks, $\mathbb{Z}_3$ deck pair-coherence, and *one* deck-invariant $\mathbb{Z}_2$ (the product of the four side orientations) — select exactly one code; Construction A of it *is* E_8, and the deck packets output $g_{\text{car}} = 1 + 4 = 5$. The circularity fence above is thereby *relocated and shrunk*, not removed: the residuals named then were the doubly-even self-dual *universe premise* (R1), the semantic deck placement (R2) and the anchor naming (R3); all three moved in round 21 (see the Type-II and metaplectic entries below) — R1 is replaced by three physical axioms, R2 is placed, R3 is named; no P2-removal claim — the sharpened contract form is `SEAM.CODE.TYPEII.01` **[C]**. *The $d = 4$ forcing census (2026-08-05):* the glue pattern itself forces the four marks ([`verification/v781_one_object_clock_census.py`], $21/21 + 23/23$, ONE-OBJECT-PARTIAL+CLOCK-FIBER-DEAD): the TFPT glue conditions (cyclic isotropic glue, evenness, unimodularity $|H|^2 = 4d$) admit an even unimodular *diagonal* glue of $A_{d-1} \oplus D_{d+1}$ iff $d = 4$ (exact census $d = 2..12$; at $d = 4$ exactly the two `v92` Lagrangians with the explicit 240-root E_8 certificate; $d = 9$ dies at isotropy; both relaxation controls fire) — so $g_{\text{car}} = d{+}1 = 5$ and $N_{\text{fam}} = d{-}1 = 3$ are the two functor readouts of *one* four-point boundary object. *Honesty gate:* $c_3 = 1/(2\pi d)|_{d=4} = 1/(8\pi)$ is a corpus-legitimate *rewriting* of the anchor-first refinement ($e_1(a) = 4$), not a new derivation of P1 — P1 stays a primitive postulate. The companion clock census is an honest kill: no $\mathbb{Z}_{12} \times_{\mathbb{Z}_6} \mathbb{Z}_{30}$ realization can act on the 60 Gaussian lines (full line-compatible group order 23040, maximal element order 12) **[E]**. *The transition-bus torsor (2026-08-05):* the $240 = 16 \times 15$ reading is settled at its sharp level

([verification/v782_e8_transition_bus.py], 35/35, TRANSITION-BUS-TORSOR): the roots are 15 coordinate 8-frames; each class fiber is a canonical *torsor* under $v^\perp$ on $L/2L$ (no distinguished origin), the deck acts as translation by v, and the obstruction census is proof-grade — *no* deck-equivariant labeling $R(E_8) = V \times (V \setminus 0)$ exists; the frozen symplectic predicate reproduces all 57600 inner products, and the A_{15} comparison quantifies what a strict transition bus would need (a canonical start state, which E_8 provably lacks) **[E]**. *The two-qubit Clifford identification (2026-08-05):* under one predeclared chart the 60 Gaussian lines *are* the 60 pure two-qubit stabilizer rays, entry-exactly over $\mathbb{Z}[i]$ ([verification/v783_two_qubit_clifford.py], 32/32, CLIFFORD-PARTIAL(H4)): $UG_{31}U^{-1}$ is an index-2 subgroup of the Clifford matrix group, $46080 = \mu_4 \times \mathrm{Pauli}_{16} \times Sp(4,2)$, classes = Pauli contexts and spreads = MUB pentads; the index bit is the Galois $\sqrt{2}$/Hadamard bit (not $J \leftrightarrow -J$, the honest H4 partial; identified in round 21 as the $\tau = i$ metaplectic bit — see below). Carrier mathematics only; the fired matter fence above is unaffected **[E]**. *The curve/code 2-torsion decider (2026-08-05):* the seam-curve realization of the code geometry is honestly *partial* ([verification/v784_curve_code_2torsion.py], 33/33, CURVE-CODE-PARTIAL): $(A[2], e_2) \cong (V, \bar{h})$ as bare symplectic spaces, but the correspondence dies at two exact layers — the deck acts syntheme-side ($[3,3]$-free) while the code acts duad-side, and the curve's canonical theta refinement is even (Arf 0, 1+6+9) while q^* is odd (Arf 1, 1+5+10); joint census 0 with positive control 6 **[E]**. *The canonical outer bridge (2026-08-06):* the two kill layers above are the two faces of one outer twist, and that twist is canonical ([verification/v796_duad_syntheme_bridge.py], 24/24, BRIDGE-CANONICAL): the Sylvester duality β (code odd forms $\to$ curve isotropic spreads) with $\beta\sigma = \tau\beta$ and $\beta(q^*) = S^*$ — the unique $\langle\tau, \rho\rangle$-fixed spread — exists and is unique up to the order-6 code joint stabilizer (census $18 + 18 \to 6 + 6$, one orbit including the τ/τ^2 swap); it resolves the Arf mismatch canonically (q_Θ pulls back constantly to $q^*_{\mathrm{even}} = q^* + \bar{h}(\cdot, A)$ — the anchor shift), transports $\bar{5}/10$ and 3+2 exactly onto the S^*-line geometry, sends the anchor transvection t_A to the curve rotation $\rho = J$ on every census bridge, and types χ_{NSR} as an honest exact two-variant torsor. The inner identification stays dead; structure labels only **[E]**. *Type II from the seam axioms (2026-08-06):* residual R1 of the boundary-code chain is replaced by three physical axioms ([verification/v799_seam_code_typeii.py], 19/19, TYPEII-FORCED): exhaustively over all 308,993 subspaces of $\mathbb{F}_2^8$ ($\dim \le 4$), mutual locality of the Construction-A boundary vertex words $\iff$ self-orthogonal, integer conformal spin $\iff$ doubly even, and holomorphy/index one $\iff$ self-dual — the survivor set is exactly the 30 $\hat{e}_8$ copies, and no boundary-legal non-Type-II code exists; the universe premise reduces to A1 (bose statistics / trivial monodromy), A2 (modular T-alignment), A3 (holomorphic index one), each a standard physical demand; imported lattice-VOA/theta ingredients cited and machine-verified in every instance. No P2 marker move **[E]** + **[C]** (the seam satisfies A1–A3). *The metaplectic S-lift (2026-08-06):* the missing Clifford coset is identified ([verification/v798_seam_clifford_modular_s.py], 27/27, MODULAR-S-PARTIAL): the per-binary-factor Hadamard *is* the metaplectic/Weil lift of modular S at its fixed point $\tau = i$ and lands exactly in $\zeta_8 G_{31}$ ($S^2 = I$, $T^2 =$ Pauli parity, $(TK)^3 = \zeta_8 I$); the total Fourier $H \otimes H$ lies inside G_{31} with metaplectic anomaly $(T_3 K_3)^3 = iI = J$ — *the deck is the anomaly of the total modular lift* (R2 placed), and $\zeta_8 = (1+J)/\sqrt{2}$ is the real half-deck swapping the seam pair $\{L, \zeta_8 L\}$; the strict uniqueness census over all 46080 coset elements is *empty* (no Θ-real metaplectic involution fixes q^* — the Arf defect is unavoidable), one G_{31}-conjugacy class survives retyped, and the σ-orbit Arf defects XOR to the anchor A itself (R3 named, not fixed) **[E]**. *The torsor Fourier modes (2026-08-06):* the origin-free fibers are read at character level ([verification/v800_e8_torsor_fourier.py], 30/30, TORSOR-FOURIER-PARTIAL): the 120 projective Fourier rays are canonical (an origin shift is a global sign), orthogonal, deck/Pauli-diagonal and G_{31}-monomial through $Sp(4,2)$; the carrier block — the 8 R-fibers' 64 rays — factorizes as $(1+5+10) \boxtimes (1+3)$ under $W(D_5) \times W(A_3)$ with σ the family 3-cycle plus fixed anchor — the carrier Pascal at *character* level, provably impossible at point level; the honest cap: the weight and character bases are mutually unbiased (1/8 exactly), so the fired matter fence stands unmoved; the 45 deck-even modes are the 45 context Pauli observables under the certified chart **[E]**. *The jet-code normal form (2026-08-06):* the no-origin torsor becomes a ring-theoretic theorem ([verification/v803_e8_jetcode_affine_nsr.py], 18/18+15/15, JETCODE-EXACT+AFFINE-NSR-EXACT): because 2 ramifies in $\mathbb{Z}[i]$, $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$ (unique iso) and $W = L/2L$ is *free* of rank 4 over the dual numbers — the mod-2 root system is a first-order jet space, the deck is the jet unit $J = 1+\epsilon$, and among all 65536 sections of $0 \to \epsilon W \to W \to V \to 0$ there are *zero* μ_4-equivariant splittings: non-splitness *is* the no-canonical-origin obstruction; $240 = 15 \times 8 \times 2$ (address $\times$ error jet $\times$ orientation). The affine companion: $\mathrm{Stab}(\chi_{\mathrm{NSR}}) = \mathrm{AGL}(3,2)$ of order $1344 = |\mathrm{Aut}(H_8)|$ in the same natural 8-point action, and affine functions on the NS/R torsor reconstruct $\mathrm{RM}(1,3) = H_8$ — a typed internal reconstruction, not a P2 removal **[E]**. *The Petersen carrier machine (2026-08-06):* the K_5 edge geometry is certified ([verification/v808_petersen_sixthroot.py], 19/19 + 16/16, PETERSEN-RADIAL-

EXACT+SIXTHROOT-SPECTRAL-ONLY): edge charges $(-4,+1,+6)$ = the certified $SU(5)$ decuple, the edge graph is Petersen SRG$(10,3,0,1)$, the flavor hexagon *is* its distance-2 shell edge-by-edge, and $\mathrm{spec}(P^6) = \{1, 64/729, 1/729\}$ *is* the deployed transport spectrum exactly — a graph-geometric origin for the $\{1,(2/3)^6,(1/3)^6\}$ triple; the sixth root is spectral-only (the corpus freezes the spectrum, not the basis) **[E]**. *[Dated correction, 2026-08-06 round 23: the sixth-root census had missed the deployed* `v221` *evaluation — T_{v221}, the exact spectral sum of* `v221`*'s transfer $(1-w)^6$, equals B^6* bit-exactly*, so the verdict upgrades to* BITEXACT-*vs-v221 (*`[verification/v814_k5_sixstep_transport.py]`*); the frozen* SPECTRAL-ONLY *stands as that probe's adjudication.] The doily incidence identity (2026-08-06):* the outer bridge sharpens (`[verification/v809_curve_code_doily.py]`, 21/21, DOILY-EXACT): $NN^T = B + 2I$ on the 15×15 duad–syntheme doily, $N/3$ with singular values $\{1, 2/3, 0\}$ — the recovery base rate $2/3$ as the unique nontrivial singular value; every census bridge factors exactly $N_{\mathrm{code}} = P_\Delta^T N_{\mathrm{curve}}^T P_\Gamma$ (a transpose duality, not an isomorphism); the $(2/3)^6$ identification stays typed pending the six-step composition **[E]** *[decided 2026-08-06 round 23: clock-only — see the six-step entry below]. The cut code (2026-08-06):* the Arf split is the cut classification of K_5 (`[verification/v810_cutcode_moment.py]`, 17/17 + 16/16, CUTCODE-EXACT+MOMENT-INCIDENCE-PARTIAL): the cut map is injective-isometric on all 256 cells, $16 = 1 + \bar{5} + 10$ *is* $(0|5) + (1|4) + (2|3)$, Clebsch $40 = 5 + 20 + 15$ over the Arf shells with within-10 = Petersen; the $210 = \binom{10}{4}$ map is decorative per the frozen rule (the equivariant bijection killed exhaustively); Arf/cut grading, not matter semantics **[E]**. *The six-step composition and the T_{10} proposal (2026-08-06):* the $(2/3)^6$ question of the doily entry is decided (`[verification/v814_k5_sixstep_transport.py]`, 23/23, SIXSTEP-CLOCK-ONLY+TRANSPORT10-PROPOSED): the rate arises canonically on *three* spatial routes (doily alternation, hexagon compression, hexagon circulation) but each carries a typed obstruction (no unit mode; the bipartite sign $-2/3$; the wrong space), and sixth-power blindness (C_6 vs $2K_3$) makes per-step data mandatory — the deployed six is the *clock* exponent (`v124`/`v486`, fixed eigenbasis); the canonical proposal $T_{10} = P_{10}^6$ in the Λ^2 basis passes all gates with the equitable `v774` quotient $= (Q_{\mathrm{Pet}}/3)^6$ bit-exact — a proposal, no deploy claim **[E]**. *The $(1{+}i)$ ramification ladder (2026-08-07):* the four documented roles of the ramified Gaussian prime $\pi_2 = 1{+}i$ are *one* machine-checked object with four rungs, each re-certified from scratch (`[verification/v833_gaussian_ramification_ladder.py]`, 33/33, RAMIFICATION-LADDER-EXACT). *Rung 1, norm doubling:* $N(\pi_2) = 2$, $(\pi_2^2) = (2)$; on the rebuilt Gaussian lattice $(1{+}J)^{\mathsf{T}}(1{+}J) = 2I_8$ exactly, $\min L = 4$ by census, so the zero class of $L/(1{+}i)L$ carries *no* root (0/240 directly), $240 = 15\times 16$, each class 4 of the 60 Gaussian lines, elementary divisors $(1^4, 2^4)$, index $16 = N(\pi_2)^4$. *Rung 2:* $L/(1{+}i)L \cong \mathbb{F}_2^4$ with the deck acting *trivially* — the 4-bit address. *Rung 3, the non-split jet:* $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$ is the unique ring iso among all 24 bijections, $W = L/2L$ is free of rank 4, the deck is the jet unit $1{+}\epsilon$, and exactly 0 of the 65536 sections are deck-equivariant (the `v803` obstruction re-proved); the 120 root pairs are exactly the $q = 1$ shell of the plus-type form (136/120, Arf 0). *Rung 4, the metaplectic lift:* $\zeta_8 = (1{+}i)/\sqrt{2}$, $(1{+}J)^2 = 2J$, $(SH)^3 = \zeta_8 I$ per binary factor and $(T_3K_3)^3 = iI$ = the deck (`v798`); the exact $\mathbb{Z}[\zeta_8]$ BFS census gives $|C_2/\mu_8| = 11520 = 16\times 720$ with *zero* Galois-mixed classes, hence $|C_2| = 92160$ and $C_2 = K \sqcup \zeta_8 K$ (the `v783` chart theorem cited), with witness parities $\chi(H\otimes I) = 1$ (the RM–CSS ζ_8 phase bit) and $\chi(H\otimes H) = \chi(CZ) = 0$. Controls: the split ring $\mathbb{F}_2\times\mathbb{F}_2$ admits 0 ring isos (2 *must* ramify) and the split self-extension admits 65536/65536 equivariant sections (non-splitness is contentful). One ladder — the same ramified prime read at (π_2), (π_2^2) and $\pi_2/\sqrt{2}$; a compression of `v689`/`v803`/`v783`/`v798`, no marker moves **[E]**. *The message ladder (2026-08-07):* the budget lines of this paper are *one formula* (`[verification/v844_message_doily_rank.py]`, 23/23, MESSAGE-LADDER-EXACT): $M_n = 15\cdot\Delta p_n = 15\cdot 2^n = (15, 30, 60, 120, 240)$ — the anchor's binary shift (`v832`) times the 15 nonzero classes of $V = L/(1{+}i)L$ (`v833`) — with each rung realized by a *named structure of the same rebuilt object*: $M_0 = 15$ the address space; $M_1 = 30 = h(E_8)$, derived here *from the rebuilt root system* (exactly 8 simple roots, connected simply-laced Cartan matrix with $\det = 1$ and the E_8 degree sequence, exact nonnegative integer simple-root coordinates, the highest root unique at height 29 with marks $\{2,2,3,3,4,4,5,6\}$, $h = 1 + 29 = 30 = |R|/8$); $M_2 = 60$ the Gaussian root lines (μ_4 orbits, 4 per class); $M_3 = 120 = |R^+|$ (the sign pairs); $M_4 = 240 = |R|$ with the census $240 = 15\times 16$ (zero class empty); and the coda $248 = 15\cdot 16 + 8 = M_4 +$ rank against the actual census and the actual HNF rank. The structural doubling $60 \to 120 \to 240$ (each line exactly two sign pairs, each pair two roots) and the halving $60 \to 30 = |\mu_4|$ vs. rank are typed. Controls fire: the wrong anchor $(1,2,2)$ gives $(45, 105, 225, 465)$ — all four targets missed — and the wrong multipliers 14 (the weight-4 codeword count) and 16 (the class size) miss all four targets and break the coda **[E]**. *The Doily–Pascal rank theorem (2026-08-07):* the doily incidence identity above sharpens into a *rank theorem* (`[verification/v844_message_doily_rank.py]`, 16/16, DOILY-PASCAL-RANK-EXACT): under the K_6 duad model of the 16 quadratic refinements (Arf census $6{+}10$, rebuilt by brute force over 2^{16}), the duad–syntheme incidence N of the Cremona–Richmond doily carries $NN^{\mathsf{T}} = B + 2I$ *entrywise* (all 225 cells) — duad collinearity *is* symplectic orthogonality,

and the spectrum shift $7/2/-2 \to 9/4/0$ identifies the ovoid (-2)-eigenspace of `v774` with the kernel: $\mathrm{charpoly}(NN^{\mathsf{T}}) = (x-9)(x-4)^9x^5$ exactly, so $\operatorname{rank} N = 10 = A_\Lambda = \binom{g_{\mathrm{car}}}{2}$ (the decuple sector), $\dim\ker N = 5 = g_{\mathrm{car}}$ (the five-slot carrier; the six ovoid indicators $3\cdot\mathbf{1}_{O_q} - 1$ *span* it), multiplicity $9 = N_{\mathrm{fam}}^2$, top singular value $3 = N_{\mathrm{fam}}$ — *the P2 integers are the singular-value data of the Cremona–Richmond incidence.* The recovery value: $\mathrm{sing}(N/3) = \{1,\ 2/3^{\times 9},\ 0^{\times 5}\}$ exactly, $2/3 = (N_{\mathrm{fam}}-1)/N_{\mathrm{fam}}$; the six-step exponentiation identity gives exactly $(2/3)^6 = 64/729$ as the deployed λ_2, and the typed gap is re-affirmed with no upgrade: $1/729$ is *not* a root of the six-step charpoly — the deployed $(1/3)^6$ stays on the clock side (SIXSTEP-CLOCK-ONLY, cited). Both wrong-pairing controls (circulant, frozen-seed LCG; same row sums 3) fire with rank $13 \neq 10$ **[E]**. *The Arf–vacuum syndrome theorem and the code pencil (2026-08-07):* the flat counts and orbit splits of this code section compress into *one two-bit syndrome table* ([`verification/v852_arf_vacuum_ovoid.py`], 24/24, ARF-VACUUM-SYNDROME-EXACT). The Arf bit is a *position-independent direction bit*: for every affine flat $t{+}U$ of $(\mathbb{F}_2^4, \hbar)$ the sum $a(t{+}U) = \sum_{x\in t+U} q^*(x) \bmod 2$ equals $\hbar(u,v)$ of the underlying plane, for *all* 140 flats; the pair $(s_{\mathrm{vac}}, s_{\mathrm{Arf}})$ = (origin bit, direction bit) splits the 140 flats exactly into 15/20/45/60 — the doily lines, the non-isotropic lines, and the two off-origin plane classes, i.e. the `v834` counts as rows of *one* table. The code side is a *pencil*: $H = [15, 11, 3]$ (weight-3 words = the 35 lines, weight-4 = the 105 off-origin flats, set equalities) $\supset \{D = [15, 10, 4],\ C_{\mathrm{iso}},\ C_{\mathrm{non}} = [15, 10, 3]\} \supset K = [15, 9, 4]$, all pairwise intersections K, all pairwise sums H, syndrome fibers 4×512; the *canonicity lemma*: all 16 refinements differ from q^* by simplex $(= H^\perp)$ words, hence induce *one* Arf functional on H; and puncturing/shortening $\mathrm{RM}(2,4) = [16, 11, 4]$ at the vacuum coordinate gives exactly H/D — *the vacuum bit is the parity bit.* Controls fire (dot form: no refinements, ill-defined direction bit; Arf-0: table invariant but labeling broken, zero set $10 \neq 6$; frozen-LCG column permutation: parameters preserved, line labeling destroyed) **[E]**. *The bent phase code and the two CSS codes (2026-08-07):* the canonical q^* is *bent* ([`verification/v853_bent_css_mangoldt.py`], 22/22, ARF-BENT-CSS-EXACT): its phase vector satisfies $\hat{s}_q = -4\, s_q$ on all 16 characters ($W^2 = 16I$; perfect autocorrelation $16\,\delta_{a,b}$ on all 256 pairs), the zero set $\{0\} \cup \overline{5}$ is a $(16, 6, 2)$ Hadamard difference set (every nonzero shift meets it in exactly 2 points — the arithmetic face of the ovoid), and $A_q = C_{\mathrm{iso}}^\perp = [15, 5, 6]$ carries the 31-word structural census (15 weight-8 = the characters, 10 weight-6 = the Arf-0, 6 weight-10 = the Arf-1 refinements). Two CSS codes follow: $A_q \subset C_{\mathrm{iso}}$ self-orthogonal gives $[[15, 5, 3]]$ (distance words = the doily lines), and $A_{16} = \mathrm{RM}(1,4) + \langle q^*\rangle = [16, 6, 6]$ inside $C_{16} = A_{16}^\perp = [16, 10, 4]$ gives $[[16, 4, 4]]$ whose 60 minimal words are *exactly* the 60 isotropic affine planes — the $s_{\mathrm{Arf}} = 0$ rows of the syndrome table; the *vacuum transformation* $[[15, 5, 3]] \to [[16, 4, 4]]$ is typed (one physical slot added, one logical qubit paid, distance $3 \to 4$ bought: shortening A_{16} at the vacuum slot $= A_q$, puncturing $C_{16} = C_{\mathrm{iso}}$). The bent Gauss sum and Walsh eigenvalue are both -4; the 16 bent translates form a second Gram-$16I$ frame *mutually unbiased* against the 16 characters (all 256 cross products $\pm 4 = \sqrt{16}$) — the unbiased partner of the `v800` torsor Fourier rays, reported with no upgrade. Controls fire (Arf-0 flips the Walsh sign to $+4$ and kills the $(16, 6, 2)$ set; the dot form has no bent seed and its $[16, 6, 6]$ extension dies at dimension $5 \neq 6$) **[E]**. *The ovoid decoder (2026-08-07):* the doily rank theorem above gains its *integral* layer ([`verification/v852_arf_vacuum_ovoid.py`], 23/23, OVOID-DECODER-EXACT): $N^{\mathsf{T}} v_a = 0$ for all six ovoid vectors $v_a = 3\cdot\mathbf{1}_{O_a} - 1$ (each syntheme covers each letter once — kernel facts from perfect matchings; any five of the six independent, $\sum_a v_a = 0$); the Gram is $V^{\mathsf{T}}V = 36I - 6J$ entrywise; the Smith normal form is $\mathrm{SNF}(N) = \mathrm{diag}(1^{10}, 0^5)$ exactly over $\mathbb{Z}$ — rank 10, kernel 5, *no torsion*, with a $\det = \pm 1$ minor exhibited; the Moore–Penrose inverse has the closed form $N^+ = N^{\mathsf{T}}/4 - J/36$ (all four Penrose axioms in exact $\mathbb{Q}$), the carrier projector is $P_5 = I/2 - B/4 + J/12$ with $\operatorname{tr} P_5 = 5 = g_{\mathrm{car}}$ and $NN^+ = I - P_5$, and the closed-form decoder $\hat{m} = (I - P_5)m$ is exact on the predeclared test set (kernel modes invisible, $\operatorname{im} N$ modes fully visible) — the recovery channel now has an explicit integral inverse, not just a spectrum. The kernel *is* the S_6 standard representation ($\operatorname{tr}(P_5\Pi(\pi)) = \mathrm{fix}(\pi) - 1$ on all 11 conjugacy classes), and the mod-2 shadow $v_a \bmod 2 = q_a$ identifies $\ker_{\mathbb{F}_2}(N^{\mathsf{T}}) = A_q = [15, 5, 6]$ — *three rings, one kernel*: the $\mathbb{R}$ ovoid span, the torsion-free $\mathbb{Z}$ kernel, and the $\mathbb{F}_2$ code of the bent layer. The wrong-pairing control fires (torsion appears, the $\mathbb{F}_2$ kernel collapses to 4) **[E]**. *The simplex-Fourier character theorem (2026-08-08):* the census $240 = 15 \times 16$ (zero class empty) *is* a Fourier law ([`verification/v857_simplex_fourier_winding.py`], 26/26, SIMPLEX-FOURIER-EXACT + INTERTWINER-IDENTIFIED): the Walsh transform of the root multiplicity is $\hat{r} = (240, -16^{15})$, i.e. the uniform-nonzero channel $P = (J-I)/15$ with spectrum $\{1, (-1/15)^{15}\}$ and two-step contraction $1/225$ (exact in Fractions). Ramification (the zero class is empty at *every* odd theta level — one-line proof) and equidistribution (one value over the 15 nonzero classes at every level ≤ 16) force the character theorem. Typed honestly: the identification is *spectral* (the packet grading $(52, 60, 128)$ and the code grading 15×16 do not refine each other); controls fire, and the law is invariant under the 24 explicit isometries and the other

invariant placement C^{**} **[E]**. *The G31 clock alphabet (2026-08-08):* the compiler clock alphabet is a *normal form* of G_{31} ([verification/v858_g31_clock_alphabet.py], 19/19, CLOCK-ALPHABET-SELECTIVE): $\mathrm{Deg}(G_{31})/4 = \{2,3,5,6\} = \{|Z_2|, N_{\mathrm{fam}}, g_{\mathrm{car}}, |R^+(A_3)|\}$ with $\gcd = 4 = |\mu_4|$, $\mathrm{lcm} = 120 = |R^+(E_8)|$, $\mathrm{lcm}/\gcd = 30 = h(E_8)$, and the audit identity $(\sum d/4)(\sum(d-1)/4) = 16 \times 15 = 240$ *realized* on the roots (the 60 reflections are the 60 Gaussian lines, 4 per class — the same 15×16 census as the code side); $(8, 12, 20, 24)$ is the *unique* integer quadruple with product 46080 and sum 64; the kill scan over all 607 rank-4 Shephard–Todd groups leaves G_{31} the sole full-battery passer (weak-triple impostors and raw-sum-64 coincidences typed by name; honesty line: degree-only audits cannot see the reducible torus), and the $W(D_5) \times W(A_3)$ order coincidence is re-killed by computed centers, $|Z(G_{31})| = 4 = \langle J \rangle$ vs 1 (the structural refutation cited) **[E]**. *The global Gaussian shell theorem (2026-08-08, round 36):* the $-1/15$ census law above is now a *theorem for all shells* ([verification/v875_shell_global_lean.py], 15/15, SHELL-GLOBAL-THEOREM): on $V = L/(1+i)L$, for every $n \geq 1$, $\Theta_v(n) = (\Theta_L(n) - \Theta_0(n))/15$ for all $v \neq 0$ (the division exact), $\hat{\Theta}_u(n) = 16\,(16 \cdot \mathbf{1}_{2|n}\,\sigma_3(n/2) - \sigma_3(n))$, and at every *odd* n: $\hat{\Theta}_u(n)/\Theta_L(n) = -1/15$ *exactly* — no census cap. The three inputs, each warded: the G_{31} class action is the 60 symplectic transvections generating $\mathrm{Sp}(4,2)$ (order 720), *transitive* on the 15 nonzero classes and identical to the shell-module action; norm doubling by $(1+i)$ ($\Theta_0(2m) = \Theta_L(m)$, $\Theta_0(\mathrm{odd}) = 0$, exact from $2 = -i(1+i)^2$); and $\Theta_L = 240\,\sigma_3$ (classical Eisenstein E_4; the $n \leq 16500$ census retained as the exact ward). A non-transitive control subgroup admits an invariant *non*-uniform census (the law breaks — transitivity is load-bearing), and the even-shell formula differs correctly (7/135 at $n = 2$). The consequence chain is kernel-checked from the named hypothesis bundle in `TfptCarrier/GaussianShells.lean` (7 theorems, exact integer identities, standard axioms, no `sorry`) **[E]**. *The winding decoder of the clock alphabet (2026-08-08, round 36):* the Paper-2 winding quadratic $q_{\mathrm{wind}}(t) = t^2 - g_{\mathrm{car}}t + |Z_2|$ *decodes* the clock alphabet ([verification/v874_feedback_decoder_lean.py], 11/11, WINDING-DECODER-EXACT): $q(\{2,3,5,6\}) = (-|\mu_4|, -|\mu_4|, |Z_2|, h(D_5)) = (-4, -4, 2, 8)$, with $q(2) = q(3)$ *forced* by the palindrome axis $g_{\mathrm{car}}/2$ and $2 + 3 = 5 = g_{\mathrm{car}}$; the decoder polynomial $D(y) = (y+4)^2(y-2)(y-8) = y^4 - 2y^3 - 48y^2 - 32y + 256$ carries the budget coefficients ($|Z_2|$, $\Omega_{\mathrm{adm}} = 48$, $2^{g_{\mathrm{car}}} = 32$, $2^{\mathrm{rank}(E_8)} = 256$); and the Fourier bridge lands: $|\mu_4|\,q(|Z_2|)/|R(E_8)| = 4 \times (-4)/240 = -1/15$ is *exactly* the nonzero-character Walsh eigenvalue of the census above ($-16 = |\mu_4|\,q(|Z_2|) = \hat{r}(u \neq 0)$). Sibling deformation directions collapse the bridge and alternative alphabets change the decoder; honest lines typed: the -32 coefficient alone is not selective (the selective object is the full vector), and the 6-vs-7 tension carries over verbatim. Type: exact *audit* theorem, not a physical functor — an explicit intertwiner would be needed to make it causal **[E]**.

The finite compiler normal form $C_{\mathrm{fin}} = (V, \hbar, q^*, \sigma, \iota)$ (2026-08-07) — one object, the kill built in [E] ([verification/v845_cfin_normal_form.py])

The four-bit symplectic compiler that this paper assembles piecewise — quotient census, Arf refinements, selector, parity lift, S_5 geometry, Pascal counting — is now *one* machine-checked object, read off the Gaussian quotient and not postulated (28/28, NORMAL-FORM-ASSEMBLED, SPEC v2 declared): $V = L/(1+i)L$ (census $240 = 15 \times 16$, zero class empty); $\hbar$ the descended symplectic form with a *found* family/anchor basis (F_1, F_2, F_3, A) of Gram $J - I$, equal to the bit form on all 256 label pairs; σ the descended order-3 deck ($\sigma^3 = \mathrm{id}$, symplectic, exactly three fixed nonzero classes $\{A, F_\Sigma, F_\Sigma{+}A\}$ and four free 3-orbits — the family cycle); q^* the *unique* selector-picked quadratic refinement (exactly 16 refinements by brute force over 2^{16}, Arf census $6 + 10$; σ-invariant: $4 \to q(A){=}1$: $2 \to q(F_\Sigma){=}0$: 1; Arf type 1, σ-preserved); ι the parity lift, bijective onto the even five-slot code $C_{\mathrm{even}}(5)$ with $\beta = \hbar$ on all 256 cells, $q_{\mathrm{wt}} = q^*$, and σ acting on the five carrier slots as the 3+2 split — message register and carrier register intertwined. *Derived from the one object:* $|Sp(4,2)| = 720$ by full census, $\mathrm{Stab}(q^*) = 120$ acting *faithfully* as S_5 on the five, orbits $[1, 5, 10]$; $16 = 1 + 5 + 10$ and $15 = 5 + 10$; the Pascal reading (the ι-weight census $(1, 10, 5) = (\binom{5}{0}, \binom{5}{2}, \binom{5}{4})$); the projective reading (the $\bar{5}$ is an *ovoid* of the doily in $PG(3,2)$: it meets every isotropic line exactly once); and the budget against the actual lattice objects: $240 = 16 \cdot 15$, 60 lines $= 4 \cdot 15$, $30 = 2 \cdot 15 = h(E_8)$ (rung 1 of the message ladder), rank 8, $248 = 16 \cdot 15 + 8$. *The kill built in* (mandatory negative, `v775` re-anchored): the exact counting bound — every $D_5 \oplus A_3 \subset E_8$ has 128 spinor-side roots, so at most $\lfloor 128/16 \rfloor = 8$ all-spinor classes and at least 7 of the 15 classes *always* contain adjoint-side roots; the concrete doubled-model census *saturates* both bounds ($8 + 7$, zero mixed classes — the J-quotient respects the D_8 grading), and the weight-level kill at the canonical split shows each spinor-pure class restricting to 8 distinct half-spinor weight pairs, never one — the 15 mailboxes are *not* 15 matter multiplets; ROOTCLASS-MIXED re-verified, not overturned. *What the normal form does not remove:* all 30 placements are Type II $[8, 4]$ codes (the `v799` echo), and C_{fin} does *not* derive the three seam axioms, the OPE-linearity scope note, or the `v776` selection of one

copy — P2 ($g_{\text{car}} = 5$, entering as the ι slot count) is *narrowed* to the `v799` residual R1′ ([C] class), not eliminated. Controls fire (the non-alternating dot form admits zero refinements; the trivial deck leaves four selector candidates; the wrong lift breaks $\beta = \hbar$). A normal form, not a new claim; no marker moves [E].

The uniqueness of C_{fin} and $\text{Aut}(C_{\text{fin}}) \cong C_6$ (2026-08-07) — one orbit, six automorphisms [E] ([`verification/v849_cfin_unique_cofinal_lean.py`])

The normal form above is not merely assembled — it is *unique* (22/22, CFIN-UNIQUE, exhaustive finite verification over $\mathbb{F}_2^4$). The WLOG ladder is exact: the 28 nondegenerate alternating forms are one $GL(4,2)$ orbit; $Sp(4,2)$ (order 720 by full enumeration) is transitive on the 6 Arf-1 refinements; per refinement there are exactly 20 admissible σ (one orbit under the order-120 stabilizer) and per (q^*, σ) exactly 120 admissible equivariant ι (one orbit). *The uniqueness theorem*: the full admissible tuple census at $\omega = \hbar$ has 14400 members and the $Sp(4,2) \times S_5$ action has *exactly one* orbit (orbit–stabilizer verified: $14400 \times 6 = 86400 = |G|$) — every admissible compiler object is isomorphic to C_{fin}. *The automorphisms*: $|\text{Aut}(C_{\text{fin}})| = 6$ with element orders $[1,2,3,3,6,6]$, abelian and cyclic — $\text{Aut}(C_{\text{fin}}) \cong C_6$, the projection to $Sp(4,2)$ faithful and the S_5 slot permutation *determined* by the symplectic part. The Gaussian-reduction diagram is typed: the parent lattice fixes the carrier triple $(V, \hbar, \sigma)$ up to exactly the 18-element centralizer torsor, the selector pins q^* uniquely, the 120 ι-completions form a single orbit — and the *strict* initial/terminal reading fails honestly (120 completions against 6 automorphisms: unique up to *non*-unique isomorphism, a one-object groupoid, not a terminal object). Controls fire: dropping alternation kills the object (0 refinements); the wrong Arf class kills ι (the $C_{\text{even}}(5)$ weight census $(1, 10, 5)$ *forces* Arf 1); dropping σ kills *canonicity* (the completion census becomes $16 \to 8 \to 4 \neq 1$) while the bare (q, ι) orbit count survives — σ is load-bearing for the selector, typed either way. P2 sharpens accordingly: the interface object is unique up to isomorphism with C_6 automorphisms; what remains of P2 is unchanged — the `v799` residual R1′ (why *this* normal form at all). No marker moves [E].

The four finite anchor theorems (2026-08-09) — the census-free q^*, the exclusive anchor classification, the doily as a multiplication table, and the $g = 5$ exhaustion [E] ([`verification/v880_finite_anchor_theorems.py`])

Four theorems close the finite-anchor layer of this section at identity grade (26/26, FINITE-ANCHOR-CLOSED; the discovery probe embedded byte-exact). **(1) The census-free normal form.** The selector-picked refinement has the closed weight-only form

$$q^*(x) \;=\; \sum_{i<j} x_i x_j + \sum_i x_i \;=\; \binom{|x|+1}{2} \bmod 2,$$

value table $(0, 1, 1, 0, 0)$ on weights 0–4 — and the `v845` selector *forces* the linear part $(1, 1, 1, 1)$: the 2^{16} brute-force search of the normal-form keybox above is replaced by a theorem. **(2) The exclusive anchor classification.** Every line of $PG(3,2)$ is classified by its q^*-weight multiset, and the classification is *exclusive*:

lines	count	weight multiset	anchor
doily (isotropic)	15	$\{2,1,1\}$	$a = (1,1,2)$
secants	10	$\{2,2,1\}$	the *wrong* anchor $(1,2,2)$
external	10	$\{1,1,1\}$	—

All 15 doily lines carry the anchor $a = (1,1,2)$ with per-line characteristic polynomial $(t-2)(t-1)^2 = t^3 - 4t^2 + 5t - 2$ and Vieta data $(e_1, e_2, e_3) = (4, 5, 2)$ — $e_2 = g_{\text{car}}$ and $e_3 = |Z_2|$ read off one line; the 20 non-isotropic lines split *exactly* as 10 secants carrying the wrong anchor $(1,2,2)$ (Vieta $(5, 8, 4)$ — the `v844` control multiset) plus 10 external lines $\{1,1,1\}$. The incidence budget lands as counting: $15 = M_0$ and $30 = M_1 = h(E_8)$ (counting only, no new structure claimed). The anchor–Vieta report closes the loop: $p_2(a) = 6$, $\chi_L - \chi_R = -p_2(a)\,(t^2 - e_2 t + e_3)$, and $M_n = 15\,(1, e_3, e_1, e_1 e_3, e_1^2)$. **(3) The doily is a multiplication table.** The doily *is* the multiplication table $\Lambda^2 E \times \Lambda^2 E \to \Lambda^4 E$ of the five-slot carrier: each doily line has support partition $\{i\} \cup \{j,k\} \cup \{l,m\}$, the wedge $b_{jk} \wedge b_{lm} = \pm f_i$ holds exactly, the assignment is a bijection onto all $5 \times 3 = 15$ such configurations, and the wedge is nonzero *exactly* on the doily lines. **(4) The $g = 5$ exhaustion.** The quadratic one-step exhaustion $\Lambda^2 \wedge \Lambda^2 = \Lambda^{g-1}$ *forces*

$g_{\text{car}} = 5$ uniquely among odd $g \in \{3, \ldots, 11\}$. The honest split is explicit and load-bearing: the *theorem* (exhaustion $\Rightarrow g = 5$) is [E]; the *physical premise* (why the carrier must satisfy the quadratic one-step exhaustion — the Calderón reading) stays [O] — P2 is *narrowed*, not eliminated; no marker moves.

The finite Pfaffian/K_6 signature (2026-08-09, rounds 44–49) — the $Gr(2,5)$ Plücker reading, the canonical sign character, the self-hosting counting theorem, the flavor hexagon as the automorphism orbit, and the vacuum selection [E] ([`verification/v888_finite_pfaffian_signature.py`])

Rounds 44–49 close the finite Pfaffian/K_6 layer of the compiler at identity grade (27/27 + 19/19 + 42/42; verdicts S6PLUCKER-CLOSED, C6FLAVOR-MEASURED + INTERTWINED, K6PFAFFIAN-CLOSED; three discovery probes embedded byte-exact; exact integer / Fraction / sympy arithmetic in every decision — no floats, no RNG, no fits). **(1) The $Gr(2,5)$ reading.** The 15 doily lines *are* the 15 monomials of the five Plücker quadrics of $Gr(2,5)$: the bijection is 15/15 with *perfect* sign match — every quadric reads $(+,-,+)$, the `v880` Hodge table (converse warded: secants/externals produce no monomial). The S_6 duad representation decomposes as $15 = 1 + 5 + 9$ (Specht $S^{(6)} + S^{(5,1)} + S^{(4,2)}$, characters from scratch by Murnaghan–Nakayama) and the normalized doily operator acts on the isotypics by the exact scalars $(1, 0, 2/3)$, with the anchor identities $2/3 = e_3/(e_2 - e_3)$ and $15 = 1 + e_2 + (e_2 - e_3)^2$ at $(e_1, e_2, e_3) = (4, 5, 2)$; and the q^* zero set is a $(16, 6, 2)$ Hadamard difference set with the anchor signature $(v, k, \lambda, v - k) = (2^{e_1}, p_2(a), e_3, \binom{e_2}{2}) = (16, 6, 2, 10)$. **(2) The K_6 motif theorem.** Under the duad labeling the 35 lines of $PG(3,2)$ are *exactly* the 15 perfect matchings of K_6 (the doily) plus the 10 triangles through the vacuum vertex (the secants — the wrong-anchor class) plus the 10 triangles avoiding it (the external lines, 20 triangles exhausting $\binom{6}{3}$), and the edge law is $q^*(e) = 0$ iff e touches the vacuum vertex — q^* *is* vacuum incidence. **(3) The canonical sign character.** The fermionic Pfaffian signs of the symbolic 6×6 expansion (15 monomials, coefficients ± 1) are gauge-matched to the Hodge table via the canonical global character $\chi(i) = (-1)^{i+1}$ on the five vacuum edges — triple-identified as the vacuum-row Laplace sign of $\mathrm{Pf}(A) = \sum_i (-1)^{i+1} a_{0i} \mathrm{Pf}(A_{\hat{0}\hat{i}})$, the Λ^4-vs-Λ^5 volume-form conversion (verified on all 15), and the C_6 crossing character ($\mathrm{Pf}(PAP^{\mathsf{T}}) = \det(P)\,\mathrm{Pf}(A)$ symbolically) — a global gauge, not a line-dependent fudge (the flipped-χ control produces exactly the 3 expected mismatches); each 4×4 minor Pfaffian *equals* the corresponding $Gr(2,5)$ Plücker quadric. **(4) The self-hosting counting theorem.** The pairing count $\binom{g+1}{2} = g!!$ has exactly the solutions $\{1, 5\}$ over odd g ($g = 1$ trivial, stated; growth separation $g!! > \binom{g+1}{2}$ for all odd $g \geq 7$ by symbolic induction), so $g = 5$ is the *unique* nontrivial fixed point, with $(g-2)!! = 3 = N_{\text{fam}}$. The honest split is explicit and load-bearing: the counting *theorem* is [E]; the *physical premise* (the boundary grammar is a self-hosting Wick pair compiler) stays [O] — P2 is narrowed to *one* premise, not eliminated; no marker moves. **(5) The flavor hexagon is the automorphism orbit.** $\mathrm{Aut}(C_{\text{fin}}) \cong C_6$ is rebuilt exactly; the pinned generator g satisfies $g^2 = \sigma$ (the deployed family 3-cycle) *exactly* and $g^3 =$ the W-edge sheet swap; its action on the ten pair-messages has cycle type $1 + 3 + 6$ with orbits equal to the charge shells, the g-orbit of $\{f_1, w_1\}$ traverses the deployed flavor hexagon edge by edge (D_6-equivalent to the transport cycle), and the recovery chain reads $(|Z_2|/N_{\text{fam}})^{|\mathrm{Aut}(C_{\text{fin}})|} = (2/3)^6 = 64/729$ — the frozen deployed transfer eigenvalue as an exact Fraction (typed honestly as a consistent *new reading*: the corpus derivation of the exponent via $p_2 = |R_+(A_3)|$ stands; 0 of 600 symplectic non-automorphisms factor, scrambled transport fails). **(6) The vacuum-selection theorem.** $Sp(4,2) \cong S_6$ acts transitively on the six Arf-1 refinements; $\mathrm{Stab}(q^*)$ has order 120 and acts faithfully *and* surjectively as S_5 on the remaining five, with orbits on the 16 messages exactly the ι-weight classes $1 + 5 + 10 = (\Lambda^0, \Lambda^4, \Lambda^2)$ — the five-slot carrier is the residual symmetry after choosing one of six equivalent vacua. Two side reports: the anchor power-sum clock $p_n = 2 + 2^n$ has mod-30 orbit $4 \to 6 \to 10 \to 18$ of period exactly $4 = |\mu_4|$, typed CLOCK-DISTINGUISHED (the purely period-4 moduli are exactly $\{5, 10, 15, 30\}$, all dividing 30); the operative-Coxeter-30 reading is separately dead (sandbox, both sides). All controls fired (Arf-0 refinements, secant substitution, wrong scalar, flipped sign character, $g = 3/7$ counterexamples). No marker moves [E].

The Wick functor arc (2026-08-09, rounds 50–55) — the six compiler roles in the deployed seam, the proved scalar obstruction, the block resolution, the seam-diagonal demand, and the constructed channel-mixing candidate [E]/[O] ([`verification/v896_wick_block_functor.py`], [`verification/v898_kms_schur_mixing.py`])

Rounds 50–53 take the one remaining P2 premise of the keybox above — the physical Wick-compiler reading, [O] — and close its *structure* level (27/27+33/33+26/26; verdicts WICKFUNCTOR-PARTIAL, WICKMIX-MEASURED (WICKMIX-OBSTRUCTED), WICKBLOCK-MEASURED (BLOCKFUNCTOR-CONSTRUCTED / SEAM-

DIAGONAL); three discovery probes embedded byte-exact; exact integer / rational / symbolic checks in every decision typed PROVED or SYMBOLIC). **(1) The six roles exist.** The compiler roles are present in the deployed seam one-particle space — five D_5 carrier slot pairs plus the A_3 boundary block (dims $2, 2, 2, 2, 2, 6$ from the `v155/v156` split $16 = 10 + 6$); the $\mathfrak{so}(16)$ sector law holds (current-sector dims 12 on vacuum edges / 4 on pair edges = the q^* edge grading, ratio $3 = N_{\text{fam}}$); and the C_6 intertwining is *exact*: the $\mathrm{Aut}(C_{\text{fin}})$ generator g ($g^2 = \sigma$) induces the channel permutation of cycle type (1)(2)(3) fixing channel 0. But the deployed *vacuum kernel* is channel-diagonal, so all 15 duad two-point functions vanish (DUADMAP-DEGENERATE-ON-VACUUM): the structure level of the functor exists, the value level needs a channel-mixing covariance — WICKFUNCTOR-PARTIAL, the honest typing (the candidate is constructed in round 55, item (5) below). **(2) The scalar obstruction is proved.** On the 6-dimensional scalar channel space *every* C_6-covariant antisymmetric covariance vanishes on the transposed duad — the C_6 2-cycle $\{4, 5\}$ forces a duad zero, so the scalar route reaches at most 14/15 nonzero duads, never 15/15: a *theorem of the deployed action*, exhaustively verified over the isotypic decomposition (the multiplicity list $[3, 0, 1, 1, 1, 0]$ re-derived by hand and by exact root-of-unity sums; the frozen expected list carried a transcription error, caught fail-first); no equivariant scalarization rescues the scalar space, and both relaxation witnesses are constructed. **(3) The block resolution.** The full block-valued functor on the 16-dim one-particle space is BLOCKFUNCTOR-CONSTRUCTED — block commutant dimension 33; all 15/15 duad cross-blocks nonzero; all 15/15 block Wick monomials obey the canonical sign law $\chi(i) = (-1)^{i+1}$ (the round-49 gauge of the keybox above); the interleaving character $+1$ *symbolically*; the review's five conditions hold at block level. **(4) The seam-diagonal demand.** The decisive fact: the *deployed* seam covariance (the `v113` family rebuilt exactly) has *zero* cross-blocks everywhere (SEAM-DIAGONAL) — the value level of the reality bridge needs a *physical channel-mixing mechanism* (a boundary/transport-twisted state, not the vacuum); with it, the K_6/Pfaffian grammar of `v888` would be the measured correlation grammar of the seam. **(5) The constructed mixing candidate (round 55, [`verification/v898_kms_schur_mixing.py`], 30/30; verdict kmsmix-measured [kmsmix-found, schurmix-generates, rp-theta-open]).** The demanded state now *exists as a candidate*, by two routes. *The KMS route:* the Majorana-compatible C_6-invariant Hamiltonians $K = ih$ are parametrized exactly (the hermitian O_{16}-commutant has dimension $112 = 62 + 50$; the antisymmetric slice splits as 33 cross — exactly the block space of (3) — plus 17 diagonal, with the two forced zeros of the transposed $\{4, 5\}$ duad), and already the *first* point of the frozen grid, $h = -(A_{16}^{\text{dep}} + \frac{1}{8}A_{\text{int}})$ at $\beta = 1$, passes *all* gates: strict CAR positivity, exact C_6 block covariance, all 15 duad cross-blocks at the frozen floor, the q^*/sheet grading carried exactly (5/5 vacuum + 10/10 carrier), and all 15 block Wick monomials nonzero with the *canonical* Pf_4 sign pattern of (3); robust at $\beta = 2$. *The Schur route (exact, sympy):* eliminating the boundary of the coupled diagonal state $A_{\text{full}} = [[\kappa A_{CC}, tV], [-tV^{\mathsf{T}}, mJ_3]]$ gives $A_{\text{eff}} = \kappa A_{CC} + \frac{t^2 m}{1-m^2} V J_3 V^{\mathsf{T}}$ *exactly*, and $S = V J_3 V^{\mathsf{T}}$ deposits nonzero correlation on *all ten* carrier duads (uniform $3J$ block, canonical Pf_4 signs) from the bare diagonal state — the same Schur/Feshbach elimination that creates the RH port creates the carrier mixing: on both fronts the real correlation lives in the *eliminated boundary*. The frozen 15-signature table of the winner is the registered prediction any physical seam state must reproduce; the OS-positivity under a *spatial* reflection is the named open check (RP-THETA-OPEN — no spatial θ is deployed on the 16-dim space; the Majorana real structure satisfies the `v426`/`v440` dictionary automatically but is particle-hole, not spatial). The honest split: the obstruction theorem, the block construction and the Schur generation identity [E]; the constructed states are *candidates* — whether the *actual* seam realizes the signature pattern is untouched, so the physical Wick-compiler premise (`SEAM.CFIN.WICKFUNCTOR.01`, `SEAM.CFIN.KMSMIX.01`) stays [O], unmoved — the physical realization is the named demand. *Dated addendum (2026-08-10, rounds 57–59):* the open check RP-THETA-OPEN is now *measured* at family level and the answer is an exclusion — strict reflection positivity under both deployable spatial reflections forces $t = 0$ (OS positivity $\iff$ the mixing floor falls: mutually exclusive on this family), the complete C_6-twisted census is exclusionary (0/6), and the parent-dilation route yields only a marginal cone-boundary witness (with the exact floor identity $t^2 \cdot 3m/(1 - m^2) = 1/200$) while the strict collar route is obstructed by an exact linear $2t$ law ([`verification/v903_seam_rp_exclusion.py`]; a measurement on the complete finite candidate set, not a universal no-go); the [O] premise stays unmoved. *Dated addendum (2026-08-10/11, rounds 60–63):* the exclusion is now *anatomized* — strict 2-cycle RP is impossible on the *whole* C_6-covariant class (the seat law reduces exactly to $\mathrm{diag}(a_J, -a_J)$; the only escape is covariance-breaking at exact linear price 2ε, with the floor exchange $\mathrm{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$), while the strict-collar obstruction is a two-seat linear law whose *equilibrium* kernel witnesses carry the full 1/200 mixing at zero entropy production — no non-equilibrium state is needed, and whether the deployed PURE-I wiring is compiler-forced became the registered successor contract ([`verification/v908_seam_equilibrium_wiring.py`], `SEAM.STATE.WIRING.SELECTOR.01`); the

[O] premise stays unmoved. *Dated addendum (2026-08-12, rounds 69–71):* the successor contract is *closed*, and the answer is a *freedom theorem* ([verification/v911_wiring_freedom.py]): the exact Gröbner census of the orbit/edge rules leaves exactly *one* admissible three-dimensional wiring component modulo the rule gauge (the rotation cell); Z/X wirings are excluded gauge-invariantly by orientation propagation; the deployed PURE-I point is an *interior* point of that cell, connected to the pure-J ray by an integer rule-gauge element — PURE-I IS A DEPLOYMENT REPRESENTATIVE, NOT A COMPILER THEOREM. No compiler-side demand distinguishes the θ_S sheet-swap frame within the rule-gauge algebra (five demand classes, all silent on the angle), the strict-collar RP Grams transform *exactly covariantly* under the rule gauge, and the same deployed parent that fails strict collar RP at θ_S *passes it exactly* in the integer-conjugated frame: the rounds-58/59 obstruction is frame-relative (a statement at the fixed θ_S frame, not a wiring no-go), and a canonical wiring selection would require an additionally *derived* physical frame. The [O] premise of the Wick-compiler demand stays unmoved. No marker moves.

The current reduction state of P2 (2026-07-22) — text consolidation, markers unchanged

The axiom P2 stays *declared* (`AX.P2.01`); what has changed is how much of it is now carried by theorems. The honest ledger state: P1 ($c_3 = 1/(8\pi)$) and the *four marks* are [E]-anchored (Gauss–Bonnet $n = 2\chi = 4$, [verification/v216_marks_gauss_bonnet.py]); given the four marks, $g_{\mathrm{car}} = 5$ follows from the partition $4 = 2{+}1{+}1$ plus the Deligne/Birkhoff–Grothendieck/stability typing ([verification/v491_p2_partition_corollary.py], [verification/v499_p2_weights_deligne_bg.py]; residual R1, the module identification, [C]); and the clock's *order*, the $\mathbb{Z}_8$ tower, the deck's *class* (a half-period translation) and the *position* half of the alignment bit are theorems ([verification/v506_seam_clock_rigidity.py], [verification/v507_seam_bit_origin.py], [verification/v510_seam_bit_freedom.py]). What remains as carrier input is *one discrete symmetry-lift bit*: flag transitivity of the four marks (marks *and* their two sides indistinguishable, $V_4 \to D_4$) $\iff$ $\tau = i$ (equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle) — on the free seam circle bare mark-transitivity is automatic for *every* configuration (the pair-exchanging V_4) and no local jet sees the side bit, so the flag form is the sharp discrete statement of the datum and its negation is concretely measurable (odd-doublet split $(2/\pi)\ln\cot(\delta/2) \neq 0$) ([verification/v512_seam_tau_flag.py], `SEAM.TAU.FLAG.01`). The bit has since survived *ten* side-blind derivation attacks on that scoreboard — free RP/Θ existence (the eighth, v521), the exhausted Wick-computable free-plus-twist state class (the ninth, v525) and the first genuinely interacting seam toy (the tenth, v529) — and is now physically *defined* as the *twist-class choice*: facets #14/#15 extend the web to 15 exact equivalences (both directions each), the flip-axis fraction is a gauge-robust order parameter ($O = \frac{1}{2}$ at $m = 4$, 0 elsewhere), and its η holonomy is in principle interferometrically readable (v528, `SEAM.BIT.TWISTCLASS.01`; the bit remains formal input — a definition replaces no derivation, no marker moves); and it now carries the first *positive* selection datum: reflection positivity, imposed on the interacting mark family, keeps exactly *one* member alive — $\delta = \pi/2$ with positive coupling, the carrier of that order parameter ([verification/v534_seam_straddle_cone.py], `SEAM.STRADDLE.CONE.01`; toy-level evidence for a dynamical origin, not a derivation); plus the [C] identifications R1/R2 of v499. In one sentence: **the compiler's dimensionless inputs reduce to four marks (derived from P1-side topology), one discrete symmetry-lift bit — the twist-class choice (flag transitivity $\iff$ $\tau = i$) — and π.** This is a reduction *state*, not an abolition of the axiom: `AX.P2.01` remains the declared interface input, and its dynamical realisation (`QGEO.REALIZE.01`) stays [C]/[O]. *[2026-08-07 addendum:* the finite normal form of the interface is now assembled as *one* object — $C_{\mathrm{fin}} = (V, \hbar, q^*, \sigma, \iota)$, the keybox above ([verification/v845_cfin_normal_form.py]) — with the code-to-matter kill built in as a checked negative; its honesty section types exactly what the assembly does *not* remove: P2's remaining content is the v799 residual R1′ (the three seam axioms, the OPE-linearity scope, the v776 uniqueness selection). The assembly narrows; it does not eliminate. Markers unchanged.]

Seed normal form, the factorial spine, and the degree-2 pattern [E]/[C] ([verification/v106_review_validation.py])

Three review-validated sharpenings of the anchor picture. **(1) Seed normal form [E].** The seed formula is exactly

$$\boxed{\varphi_0^{\mathrm{ret}} = \frac{|\mu_4|}{N_{\mathrm{fam}}}\, c_3 \;+\; \Omega_{\mathrm{adm}}\, c_3^{|\mu_4|}} \qquad \left(\tfrac{4}{3} = \tfrac{|\mu_4|}{N_{\mathrm{fam}}},\;\; 48 = \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+\right),$$

a *linear seam term* plus a *topological fourth-order correction* over all admissible fermion states — a structured anchor readout, not a fitted expression. **(2) Factorial spine [E].** With $X = 6Y$ on one 16-state generation, $\mathrm{Tr}\, X = 0$ and $\mathrm{Tr}\, X^2 = 120 = 5!$ (computed from the explicit charge table); the *same* factorial moment is $|R^+(E_8)| = \sum(E_8 \text{ exponents}) = 120$, doubles to $|R(E_8)| = 240$, and scales to $|W(D_5)| = 2^4 \cdot 5! = 1920$ (the Hawking-power fingerprint) — one moment, four projections (hypercharge, roots, Weyl group, horizon). **(3) The degree-2 inventory.** The compiler repeatedly selects *degree two*: the Pascal closure $2^{g-1} = \sum_{k\le 2}\binom{g}{k}$ truncates at $K{=}2$ with the *unique* solution $g = 5$ (scan 1..40); the glue norms add to the root norm, $\frac{5}{4} + \frac{3}{4} = 2$; $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$ (variational half × two-dimensional-slice Gauss–Bonnet); the pair sector $\binom{5}{2} = 10 = \mathcal{A}_\Lambda$; the hypercharge moment is a *second* moment; the seam is codimension *two*. *Named hypothesis [C] (recorded, not claimed):* **Quadratic Boundary Locality** — the seam supports only bilinear data, so the truncation $K = 2$ is *forced* rather than chosen; if proven, the Pascal *selection* (red-team Target B residual, typed [O]/[C]) upgrades to [E] and the chain locality $\Rightarrow K{=}2 \Rightarrow g{=}5 \Rightarrow 16 \Rightarrow D_5 \Rightarrow E_8$ closes the carrier choice. *Audit:* the even-code transition reading $240 = 16 \times 15$ is exact arithmetic but non-unique ($240 = 16 \times 5 \times 3$ is the established glue factorisation) and stays recorded, not promoted.

The Sheet-Pairing Lemma (`[verification/v109_sheet_pairing.py]`) — the first exact anchor of the hypothesis. Character-exact weight-multiset combinatorics of the D_5 spinor weights $(\pm\frac{1}{2})^5$ gives: *(i)* **no scalar within a sheet** — the zero-weight multiplicity of $S^+ \otimes S^+$ is 0, because the odd slot count $g = 5$ flips the sign parity of $-w$ (a parity theorem of the five-slot code); *(ii)* **the scalar lives across the sheets**:

$$\boxed{S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4} \quad (256 = 1 + 45 + 210),$$

exact as weight multisets, with zero-mode grading $(1, 5, 10)$ = the carrier code; *(iii)* **sheet-diagonal = odd forms**: $(S^+ \otimes S^+) \cup (S^- \otimes S^-) = 2\Lambda^1 + 2\Lambda^3 + \Lambda^5$ exactly — within-sheet pairings are odd-form-valued, never scalar; *(iv)* the scalar-bearing tower **tops at** Λ^{2K} with $K = 2$ — bilinear seam data reach exactly the Pascal-ladder degree (the source-layer transport theorem; document 3). *Consequence [C]:* the seam's single scalar two-point kernel is necessarily a **sheet pairing** (chirality-off-diagonal — the same $\mathbb{Z}_2$ as the glue chiralities and the branch-kernel selection); the remaining QBL step is to show the Calderón kernel supplies exactly this pairing and nothing beyond.

The Calderón-sheet selection theorem (`[verification/v110_calderon_sheet.py]`) — which involutions certify a scalar kernel. On $H = S^+ \oplus S^-$ the certified bilinear data of a Calderón involution ε ($\varepsilon^2 = 1$) are the matrix elements $\langle f, \varepsilon g\rangle$. The lemma's singlet counts per sheet block, $(0, 0, 1, 1)$, give the exact selection statement [E]:

$$\boxed{\text{a scalar two-point datum exists} \iff \varepsilon \text{ is sheet-odd}}$$

— a sheet-odd involution (swapping $S^+ \leftrightarrow S^-$) certifies exactly $1 + 1 = 2 = |\mathbb{Z}_2|$ invariant scalar kernels (one per orientation), a sheet-even involution certifies *none*. The kernel count $2 = |\mathbb{Z}_2|$ matches exactly the *two Lagrangian glues* (swapped by single spinor conjugation): the Calderón scalar channel carries precisely the glue ambiguity, resolved by the same sheet choice. *Anti-overclaim control [E]:* the parity theorem is generic over the odd ladder ($g = 3, 5, 7$: within-sheet count 0, cross count 2^{g-1}) — sheet-oddness forces the *half-spinor relation* $K = (g-1)/2$, **not** the value $g = 5$; the g-selection stays with the independent rank-8/integrality closure. *Consequence:* the QBL residue splits into two named parts [C] (recorded, not claimed): **(a) interface** — "the seam Calderón involution is sheet-odd" (structurally natural: the one-sided collar/double-cover deck is the *same* $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2}(4\pi)^{-1}$, and exchanging cover sides is sheet-odd by construction); **(b) analytic core** — "the two-point kernel certifies only slot-degrees ≤ 2". Proving both closes QBL $\Rightarrow K{=}2 \Leftrightarrow g{=}5$.

The Quadratic-Transport Theorem (`[verification/v111_quadratic_transport.py]`) — the transport half of the analytic core *is* a theorem. Grade seam transport by Clifford degree in the ten field operators on the five-slot Fock space ($S = \Lambda^\bullet(\mathbb{C}^5)$, S^+ the 16-state code; integer Jordan–Wigner model, spanning claims by full-rank certificates mod p, valid over $\mathbb{Q}$). *Parity* [E]: all linear words are sheet-odd, all 45 quadratic words sheet-even — code-preserving transport has *even* Clifford degree (the same $\mathbb{Z}_2$ again). *Minimality* [E]: sheet-even words of degree ≤ 1 are the scalars alone. *Generation and completeness* [E]: the 45 quadratics ($= \mathfrak{so}(10)$, exactly the Λ^2 term of the certified tower $1 + 45 + 210$) generate the whole code from the vacuum *and* span the full operator algebra $\mathrm{End}(S^+)$ ($\dim = 256$) at word length two — every code operation whatsoever is a word in pair transport, and all higher even-degree elements (e.g. the 210 quartics) are redundant. Hence

$$\boxed{\text{the transport degree is exactly 2}}$$

— degree ≤ 1 generates nothing, degree 2 generates everything: a theorem, not a hypothesis. *Anti-overclaim control **[E]**:* the $g = 3$ world behaves identically (15 quadratics generate $\mathrm{End}(S^+) = 16$) — the theorem selects the *degree*, not the rank; the g-selection stays with Pascal closure + rank-8. *Residue after v109–v111* **[C]**: QBL's remaining content is two *interface* statements — (a) the seam Calderón involution is sheet-odd; (b′) the certified *state* inventory is the ≤ 2-slot tower. Nothing about transport remains hypothetical.

The Self-Counting Channel (`[verification/v112_selfcounting_channel.py]`) — the Pascal closure is an *identity* of the certified channel. For odd g, negation $w \mapsto -w$ maps the S^+ weights bijectively into the opposite sheet, so the Cartan-neutral kernels of the certified channel are exactly the pairs $(w, -w)$ — *one per code state*: $\#\{\text{neutral kernels}\} = \dim S^+ = 2^{g-1}$ exactly **[E]**. Grading the *same* neutral set by pair-degree m gives sizes $\binom{g}{m}$, $m = 0..K$ — for $g = 5$ the triple $(1, 5, 10)$ **[E]**. Two countings of one set:

$$2^{g-1} = \sum_{m \leq K} \binom{g}{m} \quad \text{is the channel counting itself}$$

(symbolically verified for all odd $g = 3..13$) — the Pascal *closure* (`v2/v108`) is not an extra requirement but an identity of the certified channel. *Hodge fold* **[E]**: the code's minus-sign grading $\binom{g}{2j}$ equals the folded binomial $\binom{g}{\min(2j,\, g-2j)}$ with $\min(2j, g-2j) \leq K$ — every code sector appears at pair-degree $\leq K$. *Anti-overclaim:* the identity holds for every odd g and carries no g-selection; the selection stays with the rank-8 closure $g + N_{\mathrm{fam}} = 8$ and ladder integrality. *Residue re-type **[C]** (recorded, not claimed):* clause (b′) loses its closure burden — given the sheet-odd kernel, the channel *automatically* contains one neutral kernel per code state at pair-degree $\leq K$; what remains physical is the single input "the seam certifies through one scalar two-point kernel" (for the established free $c = 8$ seam net, Wick extends the one kernel to the entire even tower by determinants). QBL is thereby the structural consistency frame of the carrier choice; the *selection* is carried by rank-8 + integrality.

One kernel is the whole net (`[verification/v113_quasifree_kernel.py]`) — the QBL input merges with the holomorphy premise. The remaining input is now anchored in the established Gate-A objects. *Bookkeeping* **[E]**: $c(D_5)_1 = \frac{45}{9} = 5 = g_{\mathrm{car}}$, $c(A_3)_1 = \frac{15}{5} = 3 = N_{\mathrm{fam}}$, and the carrier net $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1$ is $10 + 6 = 16$ *free Majorana fermions* ($c = \frac{16}{2} = 8$); the whole extension tower carrier $(\mu{=}16) \to SO(16)_1(\mu{=}4) \to (E_8)_1(\mu{=}1)$ carries the *same* sixteen fermions — only the certified glue grows (index $2 \times 2 = 4 = |\mu_4|$). *Wick/Pfaffian* **[E]**: in the exact ten-Majorana model *all* 210 vacuum four-point and all 210 six-point functions equal the Pfaffian of the single two-point kernel — one kernel determines all correlations. *The kernel is a Calderón involution of rank g* **[E]**: $M = \mathbf{1} + iA$ with $A^2 = -\mathbf{1}$, so $P = M/2$ is a projection with $\mathrm{rank}\, P = 5 = g_{\mathrm{car}}$ and $\varepsilon = iA$ an involution; at seam level (sixteen Majoranas) the rank is $8 = \mathrm{rank}\, E_8$:

the central charge is the rank of the one kernel

($c = 5$ carrier block, $c = 8$ seam hull); and the polarization fixes the vacuum *uniquely* (joint annihilator kernel one-dimensional). *Premise merge **[C]** (recorded, not claimed):* "single scalar two-point kernel" = "the seam state is quasi-free" = the defining property of the free-fermion net the holomorphy gate already posits — QBL adds *no* assumption beyond Gate A; when the gate closes, the carrier choice closes **[E]** along free net ⇒ one kernel ⇒ sheet-odd ⇒ self-counting tower ⇒ complete transport ⇒ carrier. Gate typing **[C]**/**[O]** unchanged.

Hardening (P1): the Gauss–Bonnet origin of $c_3 = \frac{1}{8\pi}$

$c_3 = 1/(8\pi)$ is not an arbitrary number: it is the *one-sided Gauss–Bonnet normalisation of the seam.* The seam construction compactifies the oriented two-dimensional normal slice N_Σ to a sphere S^2, so Gauss–Bonnet gives the seam curvature integral

$$\oint_{S^2} K\, dA = 2\pi\, \chi(S^2) = 4\pi \qquad (\chi(S^2) = 2),$$

and the boundary datum is *one-sided* (the Calderón/double-cover construction uses a single collar),

contributing a factor $1/|\mathbb{Z}_2| = \frac{1}{2}$. Hence

$$\boxed{c_3 = \frac{1}{|\mathbb{Z}_2| \oint_{S^2} K\, dA} = \frac{1}{2 \cdot 4\pi} = \frac{1}{8\pi}}$$

equivalently $c_3 = \frac{1}{2}(4\pi)^{-1}$, the one-sided 2D heat-kernel coefficient. This locates both factors precisely:

- the *discrete* $8 = |\mathbb{Z}_2| \cdot 2 \cdot \chi(S^2) = 2 \cdot 2 \cdot 2$ *coincides with* $h(D_5) = \operatorname{rank} E_8 = \varphi(30)$ — the geometric and the bootstrap readings of the "8" agree;
- the *continuous* π is the irreducible Gauss–Bonnet / 2D-Gaussian primitive ($\int e^{-x^2} = \sqrt{\pi}$), the single primitive the self-consistency loop also isolates.

Status: this *hardens* (P1) — it gives c_3 a named geometric origin and reconciles the "8" with E_8 — but does *not* eliminate π; π remains the one continuous primitive. **[C]**

The one-paragraph version. There is a boundary (the seam). It yields $c_3 = 1/(8\pi)$ and a five-slot carrier $g_{\text{car}} = 5$, split 3 colour $+2$ weak. Two base charges $y_- = -1/3$ and $y_+ = 1/2$ generate the whole fermion family; the 16 even-occupation states are one generation, three families give $3 \times 16 = 48$. One seed $u = \varphi_0^{\text{ret}}$ drives four low-energy observables. Finally $\alpha_\star^{-1} \approx 137$ is the exponential engine: the electroweak scale takes one fifth of it ($e^{-\alpha_\star^{-1}/5}$), the cosmological constant takes it twice ($e^{-2\alpha_\star^{-1}}$), and the Hubble scale is the square root of Λ. A small grammar generates many sectors.

2 Headline: the Pascal compiler on five carrier slots

The strongest new statement is that three of the theory's separate structures are the *same* truncated row of Pascal's triangle on the five-slot carrier.

Family = action ladder = transition count (one Pascal row)

The even exterior algebra of E ($\dim E = g_{\text{car}} = 5$), the action ladder, and the E_8 root count are one Pascal row:

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E, \qquad (\dim \Lambda^0, \dim \Lambda^2, \dim \Lambda^4) = (1, 10, 5), \qquad \dim S^+ = 1 + 10 + 5 = 16,$$

$$A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10 = (\dim \Lambda^0 E,\ \dim \Lambda^4 E,\ \dim \Lambda^2 E),$$

$$|R(E_8)| = \dim S^+ (\dim S^+ - 1) = 16 \cdot 15 = 240.$$

Family (exterior degree), cosmology (action charge) and E_8 (root transitions) read the same $\{1, 5, 10\}$. **[E]**

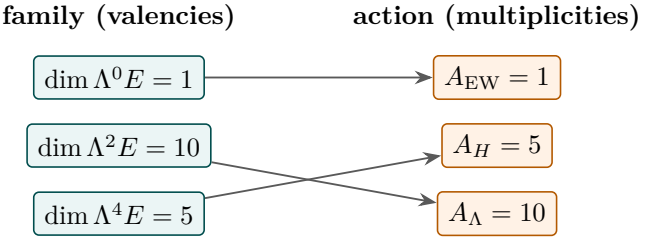


Figure 1: Exterior–Action duality. The Standard-Model family is the valency side of the even Hamming scheme on the five-slot code; the cosmological action ladder is its spectral (multiplicity) side. The Λ (pair) sector $\Lambda^2 E$ has dimension $\binom{5}{2} = 10$, the origin of the decuple exponent.

Natural sector assignment:

$$\boxed{EW \leftrightarrow \Lambda^0 E \text{ (singlet)}, \quad H \leftrightarrow \Lambda^4 E \cong E^\vee \text{ (dim 5)}, \quad \Lambda \leftrightarrow \Lambda^2 E \text{ (dim 10)}.}$$

The cosmological constant lives on the *pair* sector $\Lambda^2 E$, dim $= \binom{5}{2} = 10$; this is why the decuple bridge has exponent 10 — Λ is a product over all $\binom{5}{2}$ carrier pairs (K_5 edges), not over the full 16-state family.

2.1 $g_{\rm car} = 5$ as the Pascal closure

A family is exactly the complete Pascal truncation to degree two:

$$\boxed{\dim S^+ = 2^{g_{\rm car}-1} = \binom{g_{\rm car}}{0} + \binom{g_{\rm car}}{1} + \binom{g_{\rm car}}{2} \iff g_{\rm car} = 5} \qquad (16 = 1+5+10), \quad \text{[E]}, \ (2)$$

the unique positive-integer solution of $f(g) := 2^{g-1} - 1 - \frac{g(g+1)}{2} = 0$. *Proof of uniqueness.* Direct evaluation gives $f(1){=}-1$, $f(2){=}-2$, $f(3){=}-3$, $f(4){=}-3$, $f(5){=}0$, $f(6){=}+10$; so $g{=}5$ is the only root in $\{1,\dots,6\}$. For $g \geq 6$ the forward difference is $f(g{+}1){-}f(g) = 2^{g-1}{-}(g{+}1) \geq 2^5{-}7 = 25 > 0$ and increasing, so f is strictly increasing on $[6,\infty)$ from $f(6){=}10 > 0$; hence $f(g) > 0$ for all $g \geq 6$ and there is *no* further root. Thus $g_{\rm car} = 5$ is the unique solution (no finite-range caveat). [verification/v2_carrier_pascal.py] This is sharper than $2g_{\rm car} = \binom{g_{\rm car}}{2}$: the family code (even states, 2^{g-1}) equals the action code (sum of the first three Pascal entries) only at $g_{\rm car} = 5$.

Carrier uniqueness theorem: $g_{\rm car} = 5$ and the 3+2 split are forced [E]

The carrier rank and its colour/weak split are not chosen — they are the unique solutions of two integer conditions tied to the E_8 closure and the SM gauge algebra:

$$\boxed{g_{\rm car} = 5 \text{ is the unique } g \text{ with } N_{\rm fam} = \tfrac{2^{g-1}-1}{g} \in \mathbb{Z}^+ \text{ and } \operatorname{rank} E_8 = g + N_{\rm fam} = 8} \qquad (3)$$

$$\boxed{(b,s) = (3,2) \text{ is the unique split of } \ b+s = g_{\rm car} = 5, \ \ b^2+s^2 = 13 = |R(A_3)|{+}1} \qquad (4)$$

The integer-family carriers are $g \in \{3,5,7,\dots\}$ (rank $4,8,16$), so rank $E_8 = 8$ picks $g_{\rm car} = 5$ uniquely; and $b^2+s^2 = 13$ with $b{+}s = 5$ forces $(3,2)$, reproducing $\dim \mathfrak{g}_{\rm SM} = (b^2{-}1)+(s^2{-}1)+1 = 12 = |R(A_3)|$. The induced hypercharge is anomaly-free ($\operatorname{Tr} Y = \operatorname{Tr} Y^3 = 0$ over the 16 states). This hardens P2: the *rank* and the *split* are forced; only the seam→carrier interface itself remains a declared input. [verification/v14_carrier_uniqueness.py]

The Pascal grammar: one $\Lambda^\bullet$ mechanism in four places (anti-numerology) [E]. The recurring small integers are *not* coincidences — they are the *same* exterior-algebra (Pascal) mechanism read in four spaces, which is why the binomials of rows 4 and 5 keep reappearing:

space	Pascal reading
carrier (one generation)	$\dim S^+ = \Lambda^{\rm even}(5) = \binom{5}{0}{+}\binom{5}{2}{+}\binom{5}{4} = 1{+}10{+}5 = 16$
scale ladder (cosmology)	$A_{\rm EW} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1:5:10$
flavor u/d readout	$\lVert \operatorname{Pl}(K) \rVert_1 = \binom{4}{0}{+}\binom{4}{1}{+}\binom{4}{2} = 1{+}4{+}6 = 11 = 16{-}g_{\rm car}$
mass volume	$\det M_{\rm SM} \sim (\varphi_0^{\rm ret})^{6+9+10} = (\varphi_0^{\rm ret})^{25} = (\varphi_0^{\rm ret})^{g_{\rm car}^2}$

So the carrier is the even row 5, the scale ladder is the lower row 5, the u/d leg is the cumulative row 4 up to $\deg u^c = 2$, and the mass volume closes on $g_{\rm car}^2$. *One* grammar (Λ^k of the 5-slot carrier and the $4 = |\mu_4|$ family fundamental), not four lucky hits. [verification/v44_carrier_exterior.py] [verification/v45_family_exterior.py] [verification/v46_grand_mass_volume.py]

2.2 One master moment generates 40, 41, 240, δ_2

With $X = 6Y$, the quadratic hypercharge trace on one family is $\mathrm{Tr}_{S^+} X^2 = 120 = 5!$, and it generates the whole integer chain:

$$\sum_{f,j} L_{f,j} = \frac{\mathrm{Tr}_{S^+} X^2}{3} = 40, \qquad 10b_1 = \frac{\mathrm{Tr}_{S^+} X^2}{3} + 1 = 41,$$
$$|R(E_8)| = 2\,\mathrm{Tr}_{S^+} X^2 = 240, \qquad \delta_2 = 4!\ \mathrm{Tr}_{S^+} X^2\, c_3^8. \tag{5}$$

The flavor budget, the $U(1)$ budget, the E_8 count and the second seam defect all hang on the second hypercharge moment 5!. **[E]** *Round-22 sharpening:* the moment itself is one symbolic identity, $\mathrm{Tr}_{S^+} X^2 = 4|X|^2 = 4h(E_8) = 120$ — the Coxeter number carries the whole chain ([`verification/v810_cutcode_moment.py`]) **[E]**.

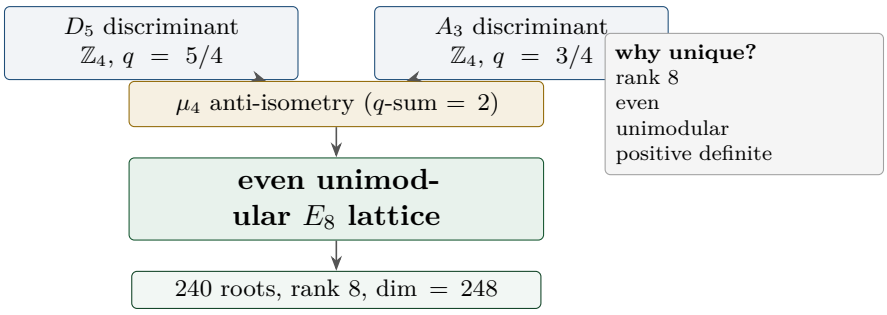
2.3 Compact E_8 compression and the role of ν^c

$$\boxed{\begin{aligned} |R(E_8)| &= \dim S^+ \cdot g_{\mathrm{car}} \cdot N_{\mathrm{fam}} = 16 \cdot 5 \cdot 3 = 240, \\ \dim E_8 &= \dim S^+(\mathcal{A}_H + \mathcal{A}_\Lambda) + (g_{\mathrm{car}} + N_{\mathrm{fam}}) = 240 + 8 = 248 \end{aligned}} \tag{6}$$

with $N_{\mathrm{fam}} = (2^{g_{\mathrm{car}}-1} - 1)/g_{\mathrm{car}} = 15/5 = 3$ and $\mathrm{rank}\, E_8 = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$. **[E]**. The right-handed neutrino ν^c is the Pascal singleton $\Lambda^0 E = (1,1)_0$ that closes $16 = 1 + 5 + 10$; removing it breaks the truncation and shifts α^{-1} by 2.16×10^{-3}.

3 The $D_5 \times A_3$ compiler: E_8 as a closure, not an input

This section upgrades the E_8 atlas (Section 10) from a counting coincidence to an explicit lattice construction. The whole content is consolidated and verified in the standalone companion note *"The Even Carrier Code and the $\mathbb{Z}_{30}$ Coxeter Compiler."* The thesis: the carrier code is the D_5 half-spinor, the family geometry is an A_3, and E_8 is the standard unimodular glue closure of $D_5 \oplus A_3$ — with the four punctures μ_4 as the glue code.



$D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ — the unique even unimodular rank-8 glue [E] exact (lattice).

The compiler in one line:

> five carrier slots $\Rightarrow$ D_5, four family corners μ_4 $\Rightarrow$ A_3, $(D_5 \oplus A_3) + \mu_4$-glue $\Rightarrow$ E_8.

The glue is *one* operation in two categories. The same order-$|\mu_4|$=4 discriminant glue closes $D_5 \oplus A_3$ into E_8 *twice over*: once on *root lattices* (the unimodular pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, v1) and once on *chiral conformal nets* (the simple-current / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$, v92/v281). These are not two coincidences but the two functorial faces of a single commuting square (v234/v277/v260); the boundary realisation on which the strict-TOE residual turns (`SEAM.EQUIV.01`) is exactly the statement that the raw seam realises this square — now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (the lattice/S_3 stack pins it at every computable level, Lean `FORM.SEAM.MMST.01`; only the cited

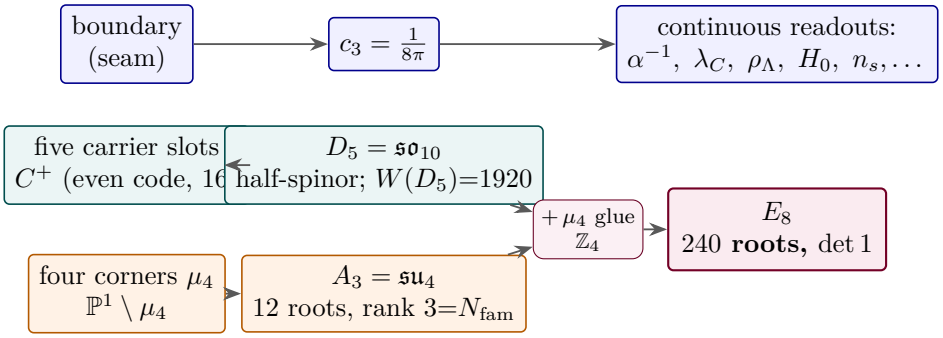
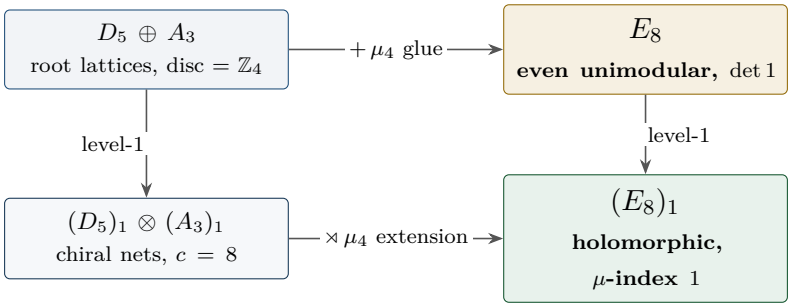


Figure 2: The explanation hierarchy. The boundary fixes the seam seed c_3; the five-slot even carrier code is the D_5 half-spinor; the four punctures μ_4 give A_3; their common $\mathbb{Z}_4$ discriminant (μ_4) glues $D_5 \oplus A_3$ into the unimodular E_8. E_8 is the *closure*, not the input; c_3 dresses the continuous observables.

MMST/OS continuum-existence theorems stay [O], v367/v368/v376–v379/v336; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]). The extension leg of that citation package is itself re-founded at net level ([`verification/v469_seam_crossedproduct_route.py`]): the 128-spinor step is the *local* $\mathbb{Z}_2$ simple-current crossed product ($h_s(SO(16)_1) = 16/16 = 1 \in \mathbb{Z}$, the Longo–Rehren locality criterion; KLM $\mu = 4/2^2 = 1$), on 1995–2001 peer-reviewed subfactor theory, with the lattice-VOA preprint route demoted to an independent second witness, and the realisation input reduced to invariant level (R1′). Premise (i) of that package (c_-=8) now also carries a *ground-state* witness ([`verification/v489_seam_modular_commutator.py`]): the Kim–Shehab–Kim modular commutator $J(A,B,C) = (\pi/3)c_-$, evaluated on the exact BdG ground state of the collar model, gives $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$ for the 16-layer carrier (per layer 0.500000 at L=48; trivial and orientation-flip controls pass) — independent of both the Chern route (v367) and the KLM count (v378). `SEAM.EQUIV.01` stays [O]. *Follow-up (2026-08-03, the G_{net} witnesses):* the index-4 statement now carries two *exact* arithmetic/index witnesses. The NS/R grading *is* the parity character of $E_8(\mathbb{Z}[i])/(1{+}i) = \mathbb{F}_2^4$ — $(1{+}i)L$ lies in the even sublattice (proved on a $\mathbb{Z}$-basis), all 15×16 root classes are parity-pure (7 NS + 8 R), so the Ramond projection is $(1{+}i)$-*adic* at the one ramified edge of the $\mathbb{Z}[i]$-E8 Hecke tower ([`verification/v722_phys_ramified_ns_r.py`], 9/9, RAMIFIED-SEES-NSR); and on the CAR ladder the μ_4-average expectation has optimal Pimsner–Popa constant exactly 1/4 and Watatani index exactly 4 via an explicit quasi-basis with *forced* weights, with μ_4 *derived* from the clock and the Ramond sector healed state-preservingly by the Klein zero-mode dressing ([`verification/v726_phys_car_pp_index.py`], 57/57, T2-SLICE-GO). The remaining half is the *identification* of the ladder Q-system with the constructed $\mathbb{C}[\mathbb{Z}_4]$ object (`GNET.RAMIFIED.01`, kills registered); `GATE.METRIC.08/.10` typing unchanged. *Follow-up (2026-08-05, the continuum strand):* the finite-level precursors of the GNS limit are measured *coherent* ([`verification/v779_gnet_gns_arf.py`], 27/27 + 27/27): the Haar scaling tower intertwines exactly, the index-4 expectation squares commute exactly (the finite half of the Q-system identification), the state converges geometrically, and the exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1}\cos(\pi c/2)$ types the $\mathbb{C}[\mathbb{Z}_4]$ obstruction at exactly $2^{-\ell}$ (explaining the $\ell = 1$ three-sector anomaly); the honest negative: the $16 = 1{+}5{+}10$ counting echo with the Arf geometry is exact but *non-geometric*, and the two fours (index 4 vs channel 4/49) stay in their registers. The round-21 martingale probe measures the limit-existence hypotheses of the continuum strand as met ([`verification/v802_gnet_martingale.py`], 15/15, MARTINGALE-HYPOTHESES-MET; the analytic remainder stays typed). Typing unchanged.



One operation, two categories: the order-$|\mu_4|$=4 glue is the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and the simple-current / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$; the square commutes. [E] exact.

3.1 Carrier $= D_5 = \mathfrak{so}_{10}$ (proved)

The 16 even-occupation states $C^+ = \{x \in \mathbb{F}_2^{g_{\rm car}} : |x| \text{ even}\}$ are exactly the weights of the D_5 half-spinor $\frac{1}{2}(\pm 1, \ldots, \pm 1)$ (even number of minus signs). The flip-weight-4 relation is the **Clebsch graph** SRG$(16, 5, 0, 2)$ and the flip-weight-2 relation is its complement SRG$(16, 10, 6, 6)$; the *affine* automorphism group of the scheme (translations $\rtimes$ coordinate permutations) is the D_5 Weyl group,

$$\boxed{\mathrm{Aut}_{\rm aff}(C^+) = C^+ \rtimes S_5, \quad |\,\mathrm{Aut}_{\rm aff}(C^+)| = 2^{g_{\rm car}-1} g_{\rm car}! = 1920 = |W(D_5)| = r\,|R(E_8)|} \quad \text{[E]}, \tag{7}$$

(the linear code automorphism group alone is only S_5; the order-1920 group is the affine/scheme one) and the Clebsch edge count is the flavor budget $|E(R_4)| = \frac{16\cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$. *Round-22 grading (2026-08-06):* the Arf split $16 = 1 + \bar{5} + 10$ *is* the cut classification $(0|5) + (1|4) + (2|3)$ of K_5 (the cut map injective-isometric on all 256 cells), and the Clebsch edges grade as $40 = 5 + 20 + 15$ over the Arf shells with the within-10 block equal to the Petersen graph — an Arf/cut geometry, exact but carrying no matter semantics ([verification/v810_cutcode_moment.py]) [E].

3.2 Family $= A_3 = \mathfrak{su}_4$ from $\mathbb{P}^1 \setminus \mu_4$ (proved, with the metric settled)

The carrier geometry fixes $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, $H_1 = \mathbb{Z}^3$, rigid D_4 generated by $z \mapsto iz$, $z \mapsto 1/z$. The twelve puncture-difference cycles $\gamma_i - \gamma_j$ carry the residue pairing $\mathrm{Res}(\omega_i, \gamma_j) = \delta_{ij}$ and form the A_3 root system:

$$\boxed{\#\{\gamma_i - \gamma_j\} = 12 = |R(A_3)| = \dim \mathfrak{g}_{\rm SM}, \quad \text{rank} = 3 = N_{\rm fam}, \quad \text{Gram} = \text{Cartan}(A_3),\ \det = 4} \quad \text{[E]}. \tag{8}$$

The corner rotation $z \mapsto iz$ acts on $H^1 = \mathbb{C}^3$ as the A_3 Coxeter element (order $4 = h(A_3) = |\mu_4|$, spectrum $\{i, -1, -i\}$, $\chi = \Phi_4\Phi_2$), an isometry of the Cartan form — the exact analogue of the E_8 Coxeter element reading Φ_{30}.

The μ_4 clock is the order-4 character of the E_8 cycle, and $(2, 3, 5)$ is the icosahedron ([verification/v223_coxeter_totative_clock.py], [verification/v219_icosahedral_mckay.py]). The two "Coxeter" clocks are one object read at two scales. The E_8 exponents are the totatives $(\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$, and the multiplicative order-4 element 7 (7^2=19, 7^4=1 mod 30) generates a literal μ_4 clock $\langle 7\rangle = \{1, 7, 13, 19\}$ that permutes the four conjugate Coxeter 2-planes — the same μ_4 as the seam corner rotation. And the period $30 = 2{\cdot}3{\cdot}5$ itself is not a free factorisation: by the McKay correspondence E_8 is the exceptional *top* of the tower of finite $SU(2)$ subgroups ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$), so its branch data is the *icosahedron* and the three atoms $(|\mathbb{Z}_2|, N_{\rm fam}, g_{\rm car}) = (2, 3, 5)$ are its rotation-axis orders. The binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has nine irreducible-representation degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ equal to the affine-E_8 Kac marks, with $\sum d_i = 30 = h(E_8)$ and $\sum d_i^2 = 120$. This is a *backward certificate* of the closed E_8 [E] (it explains the $(2, 3, 5)$ choice [C]; it does *not* replace the P2 input).

The metric, settled (former [C]). The transcendental geometric (Green) metric $G_{ij} = -\log|p_i - p_j|$ on μ_4 is only D_4-symmetric, with a regularisation-independent anisotropy $-\log 2$ on the root

space, so it is *not* proportional to the A_3 Cartan form. But it does not need to be: the E_8-relevant A_3 is the canonical *residue* pairing (integral, S_4-symmetric, = Cartan(A_3)), and the geometric metric is a D_4-equivariant deformation preserving the root system, Weyl group and Coxeter element. So the lattice structure is exact [E]; the analytic metric is a characterised deformation, not an open gap.

3.3 The μ_4 glue and the $5/4 + 3/4 = 2$ discriminant theorem (proved)

The two factors have equal discriminant groups and E_8 is unimodular, so the glue index is exactly $|\mu_4|$:

$$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4, \qquad [E_8 : D_5 \oplus A_3] = \sqrt{\frac{\det D_5 \det A_3}{\det E_8}} = \sqrt{\frac{4\cdot 4}{1}} = 4 = |\mu_4|. \tag{9}$$

The discriminant-form norms sum to the E_8 root norm, and — strikingly — they are pre-existing TFPT constants:

$$\boxed{\begin{aligned} &q(D_5) = \tfrac{5}{4} = \frac{\delta_2}{\delta_{\mathrm{top}}^2}, \qquad q(A_3) = \tfrac{3}{4} = \xi_{\mathrm{tree}}, \\ &q(D_5) + q(A_3) = 2 = |E_8 \text{ root}|^2 \end{aligned}} \qquad \text{[E] [verification/v1_e8_glue.py]} \tag{10}$$

This is literally the E_8 spinor-root split $\frac{1}{2}(\pm)^8 = \frac{1}{2}(\pm)^5 \oplus \frac{1}{2}(\pm)^3$, $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$.

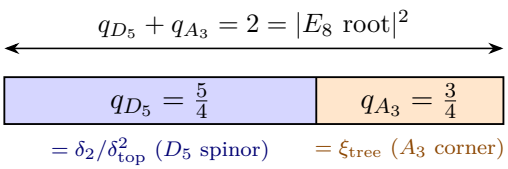


Figure 3: The discriminant-norm glue. The E_8 spinor root $\frac{1}{2}(\pm)^8$ (norm 2) splits across $D_5 \times A_3$ as $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4}$; the two halves are the pre-existing TFPT constants $\delta_2/\delta_{\mathrm{top}}^2 = \frac{5}{4}$ and $\xi_{\mathrm{tree}} = \frac{3}{4}$.

The standard branching is realised explicitly:

$$\mathfrak{e}_8 = (45,1)\oplus(1,15)\oplus(10,6)\oplus(16,4)\oplus(\overline{16},\overline{4}), \qquad 248 = 45{+}15{+}60{+}64{+}64, \quad 240 = 40{+}12{+}60{+}64{+}64, \tag{11}$$

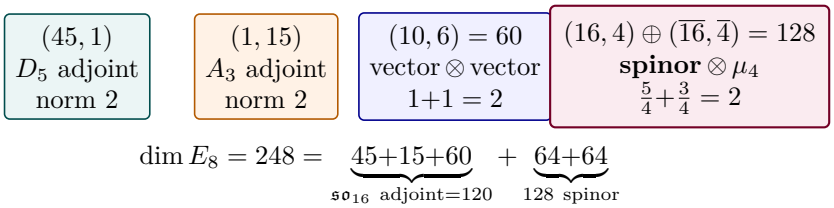


Figure 4: The $E_8 \supset D_5 \times A_3$ branching as norm-2 blocks. The adjoint half is the hypercharge moment $\mathrm{Tr}_{S^+} X^2 = 120 = \dim \mathfrak{so}_{16}$; the spinor half is two sheets over the four corners μ_4.

and assembling $D_5 \oplus D_3 \subset D_8$ with the even spinor coset produces all 240 roots (norm 2, Cartan det = 1, every root an integer combination of one simple-root basis): *the* 240*-metrisation is no longer blind — E_8 is built.* [E]

3.4 Glue-norm arithmetic, and three readings of 41, 60, 120

The two glue norms generate the carrier integers through sum, difference and product. With $\dim S^+ = 16$:

$$\boxed{\begin{aligned} &16(q_D{+}q_A) = 32 = 2^{g_{\mathrm{car}}} = h(D_5)h(A_3), \qquad 16(q_D{-}q_A) = 8 = r(E_8) = h(D_5), \\ &16\, q_D q_A = 15 = \dim S^+ {-} 1 \end{aligned}} \quad \text{[E]}, \tag{12}$$

so the glue already contains the full carrier exhaustion $2^{g_{\rm car}}$, the E_8 rank, and the non-trivial transition count $\mathcal{A}_H+\mathcal{A}_\Lambda = 15$; and $|\mu_4| = (q_D-1)^{-1} = (1-q_A)^{-1} = 4$ is the inverse displacement of the two norms from the unit norm.

Spine Quotient Lemma: the recurring factors are adjacent quotients of $(2,3,4,5)$ [E] ([verification/v74_compiler_micro_lemmas.py])

The integers of the spine are $(|\mathbb{Z}_2|, N_{\rm fam}, |\mu_4|, g_{\rm car}) = (2,3,4,5)$. Every "mysterious" $O(1)$ factor in TFPT is an *adjacent quotient* of this chain, not a separate special number:

$$\tfrac{|\mathbb{Z}_2|}{N_{\rm fam}} = \tfrac{2}{3}, \quad \tfrac{N_{\rm fam}}{|\mu_4|} = \tfrac{3}{4} = q(A_3), \quad \tfrac{|\mu_4|}{g_{\rm car}} = \tfrac{4}{5}, \quad \tfrac{g_{\rm car}}{|\mu_4|} = \tfrac{5}{4} = q(D_5), \quad \tfrac{|\mu_4|}{N_{\rm fam}} = \tfrac{4}{3}.$$

These are exactly the factors that recur across the theory: $\lambda_2 = (\frac{2}{3})^6$ (the transport gap = Page recovery rate, and the Koide target $Q_\star = \frac{2}{3}$); $\varepsilon = \frac{3}{4}\varphi_0 = q(A_3)\varphi_0$ (the solar misalignment); $\varphi_0 = \frac{4}{3}c_3 + 48c_3^4$ with leading $\frac{4}{3}c_3 = \frac{1}{6\pi}$; and $q(D_5)+q(A_3) = \frac{5}{4}+\frac{3}{4} = 2$ (the E_8 root norm). So $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3})$ are five projections of one discrete chain — not scattered constants. The spine is moreover *one* finite object: as a tetrahedron on $T = \{e_3(a), p_0(a), e_1(a), e_2(a)\}$ its edge and face products and its volume $120 = |R^+(E_8)| = 5!$ reproduce the central integer grammar, with the honest negative control that the graph reading $240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$ breaks at K_6 ([verification/v91_spine_tetrahedron.py]).

The carrier adjoint $\dim D_5 = |R(D_5)| + g_{\rm car} = 45$ then gives a second reading of the $U(1)$ integer,

$$\boxed{10b_1 = \dim D_5 - |\mu_4| = 45 - 4 = 41 = \underbrace{40}_{\sum L} + \underbrace{(g_{\rm car} - |\mu_4|)}_{=N_\Phi=1}} \quad [\mathbf{E}], \tag{13}$$

i.e. the single Higgs doublet is the carrier rank minus the glue corners, $5-4=1$.

A fourth, metric reading of 41 *([verification/v222_cm_norm_duality.py]).* On the square seam (j=1728, complex multiplication by the Gaussian integers $\mathbb{Z}[i]$) the $U(1)$ index is the Gaussian *norm* of the carrier-glue vector,

$$10b_1 = N_{\mathbb{Z}[i]}(g_{\rm car}+i|\mu_4|) = g_{\rm car}^2 + |\mu_4|^2 = 5^2 + 4^2 = 41,$$

and the hexagonal partner (j=0, Eisenstein $\mathbb{Z}[\omega]$) reads the same $(3,2)$ carrier split as the scalaron exponent $N_{\mathbb{Z}[\omega]}(3+2\omega) = 3^2-3\cdot2+2^2 = 7$. So 41 (EM) and 7 (scalaron) are the two exceptional CM norms of the one carrier, square vs. hexagonal — not two unrelated $O(1)$ numbers [E]. The start defect 60 and the hypercharge moment 120 are "bilingual":

$$\begin{aligned} D_{\rm start} &= \dim D_5 + \dim A_3 = 45 + 15 = 60 = \mathcal{A}_\Lambda \cdot |R^+(A_3)| = 10 \cdot 6, \\ \mathrm{Tr}_{S^+} X^2 &= 120 = \underbrace{60}_{\text{adjoint}} + \underbrace{60}_{\text{mixed (10,6)}}, \end{aligned} \tag{14}$$

and $\dim E_8 = 248 = \mathrm{Tr}_{S^+} X^2 + 2\dim S^+|\mu_4| = 120 + 2\cdot16\cdot4$. Finally the hypercharge norm and the EW exponent are A_3-rational,

$$\boxed{\gamma_{\rm car} = \frac{\mathcal{A}_\Lambda}{|R(A_3)|} = \frac{10}{12} = \frac{5}{6}, \qquad I_1^{\rm EW} = \frac{|E(K_5)|}{|R(A_3)|} + \frac{1}{|\mu_4||R(A_3)|} = \frac{10}{12} + \frac{1}{48} = \frac{41}{48}} \quad [\mathbf{E}]. \tag{15}$$

Transport reduction. The flavor matrix L has its mod-6 content fixed by a single combinatorial datum: $\{0,1,3\}$ is the *unique* (up to D_6) 3-subset of $C_6 = |R^+(A_3)|$ with all distinct cyclic distances $\{1,2,3\}$, and the three sectors are its D_6 orbit ($u \equiv \{0,1,3\}$, $d \equiv 2-\{0,1,3\}$, $e \equiv 2+\{0,1,3\}$). The total is fixed by the carrier, $\sum L = 40 = |R(D_5)|$, with winding unit $6 = |R^+(A_3)|$. Given the residue *set* of a sector, the full ordered packet is reproduced by two universal rules — the lightest generation carries the single $+6$ winding, and the entries are sorted in decreasing length (monotone hierarchy). The *only* remaining datum is then, per sector, which residue is the first-generation anchor that receives the winding:

$$\boxed{\text{anchors } (a_u, a_d, a_e) = (1,1,2) \;\Rightarrow\; L_u = (7,3,0),\; L_d = (7,5,2),\; L_e = (8,5,3)} \quad [\mathbf{E}] \text{ (given anchors)}. \tag{16}$$

So the open step of a full "Clebsch walk" theorem is sharply isolated: it is exactly these three per-sector anchors. They are *not* free: the three transport cusps are exactly the D_5 dual-root combinations, and each sector is tied to its cusp by the right-handed hypercharge,

$$\boxed{\{1, \tfrac{2}{3}, \tfrac{1}{3}\} = \{2y_+,\ -2y_-,\ 2(y_+ + y_-)\}, \qquad \text{lepton} \leftrightarrow 2y_+,\ \text{up} \leftrightarrow -2y_-,\ \text{down} \leftrightarrow 2(y_+ + y_-)} \quad \textbf{[E]}. \tag{17}$$

The canonical-transport-kernel theorem (document 3, Part II) derives the residue packets — and hence the anchors — as the *word-lengths of the family-connection holonomy* $\mathrm{Hol}_{\mathcal{G}_{D_4}}(\nabla^\star_F)$ along the D_4-invariant geodesic spine of $\mathbb{P}^1 \setminus \mu_4$ with these cusps as boundary data. The anchors are therefore a genuine $D_5 \times A_3$ readout (cusps from the D_5 carrier, spine holonomy from the A_3 four-punctured sphere), not an extra parameter. A compact candidate closed form is the nearest-integer cusp map

$$\boxed{a_f = \mathrm{round}(2\,\mathrm{cusp}_f): \quad (a_u, a_d, a_e) = \mathrm{round}(\tfrac{4}{3}, \tfrac{2}{3}, 2) = (1, 1, 2)} \quad \textbf{[C]}, \tag{18}$$

which reproduces the matrix exactly. The remaining open step is now precise and purely geometric: an *independent* closed-form evaluation of the D_4-spine holonomy word-lengths from the lattice data alone (the same $\mathbb{P}^1 \setminus \mu_4$ whose residue pairing is Cartan(A_3) and whose corner rotation is the A_3 Coxeter element), rather than from the family-connection theorem (document 3).

Riemann–Hilbert check, and what it does (and does not) fix. Setting up the rigid $\mathbb{Z}_4$-symmetric rank-3 $SU(3)$ local system explicitly — four puncture monodromies $M_k = R^k C R^{-k}$ with the order-3 cusp class $C \sim \mathrm{diag}(1, \omega, \omega^2)$ ($\omega = e^{2\pi i/3}$, the three cusps as eigenvalue-exponents) and the rotation R ($R^4 = \mathbb{1}$) — the closure $M_0 M_1 M_2 M_3 = \mathbb{1}$ has a solution to machine precision, and the unique R has eigenvalues $\{i, -1, -i\}$: the A_3 Coxeter element. This *independently* confirms the rigidity of the family connection and the corner-rotation = Coxeter identity. However, the flat monodromy *alone* does not fix the integer word-lengths $\{0, 1, 3\}$: those are *parabolic degrees* (the cusp exponents as parabolic weights, together with the $\mathbb{Z}_2$ sheet parity), i.e. Hodge/metric data on the spine, not topological monodromy. *Status update (the computation has since been carried out):* the exact residue normal form $A_0^\star$ with spec $= \{0, \frac{1}{3}, \frac{2}{3}\}$ is pinned algebraically ([`verification/v115_anchor_residue.py`], [`verification/v116_resonance_uniqueness.py`]), the unitarised holonomy is the exact 24-element $W(A_3) = S_4$ ([`verification/v117_monodromy_weyl_a3.py`]), and the parabolic weights are the μ_4-character degrees of $H^1(\mathbb{P}^1 \setminus \mu_4)$ — one spectrum in four readouts ([`verification/v137_qplus_cohomology.py`]); the deck itself is *derived* — the integer model selects the sheet-twisted class, no discrete freedom left ([`verification/v141_deck_selection.py`]), and the full Möbius D_4 of the curve matches the integer model parity by parity ([`verification/v146_moebius_d4.py`]). What remains of the old residual is the much smaller selector establishment R4′ (research contracts): derive the *one* lift map of the torsion normal — the pairing values are atom identities ([`verification/v142_frame_integrality.py`]/[`verification/v145_pairing_atoms.py`]). **[E]**/**[C]**

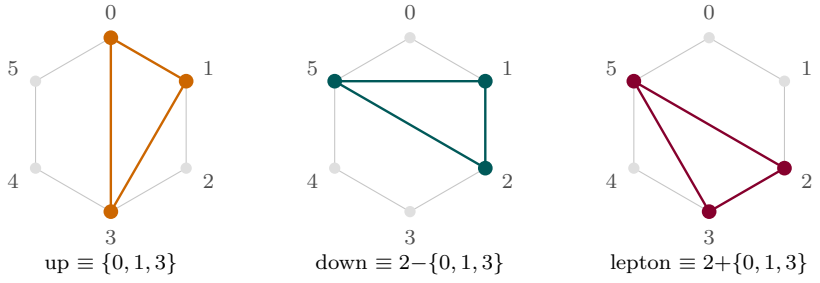


Figure 5: Flavor transport on the hexagon $C_6 = Z_6 = |R^+(A_3)|$. The unique distinct-distance triple $\{0, 1, 3\}$ (cyclic distances $\{1, 2, 3\}$) and its D_6 orbit give the three sectors' residues mod 6; the total $\sum L = 40 = |R(D_5)|$ and the first-generation +6 winding come from the carrier.

3.5 What the compiler outputs for the Standard Model and our reality

Every entry below is an *output* of $(c_3, g_{\rm car})$ through the $D_5 \times A_3$ compiler; none is an independent input. "Reads" names the lattice object the observable comes from.

SM / cosmology observable	Compiler output	Reads	St.
Number of generations	$N_{\rm fam} = 3 = \operatorname{rank} A_3 = \dim H^1(\mathbb{P}^1 \setminus \mu_4)$	A_3	[E]
Gauge algebra dimension	$\dim \mathfrak{g}_{\rm SM} = 12 = \lvert R(A_3)\rvert = \gcd(6r, 2h)$	A_3	[E]
One generation (16, incl. ν^c)	$S^+ = D_5$ half-spinor $= \Lambda^{0,2,4}E$	D_5	[E]
Hypercharge spectrum	all Y from $y_\pm = \{\frac{1}{2}, -\frac{1}{3}\}$; $\operatorname{Tr} X^2 = 120$	D_5 root data	[E]
Abelian index b_1	$10b_1 = 1 + \lvert\mu_4\rvert\lvert E(K_5)\rvert = 1 + 4 \cdot 10 = 41$	$\mu_4 \times K_5$	[E]
Fermion family occupancy	$\Omega_{\rm adm} = N_{\rm fam} \dim S^+ = (N_{\rm fam}+N_\Phi)\lvert R(A_3)\rvert = 48$	A_3	[E]
Fine structure $\alpha_\star^{-1} = 137.0359992168$	root of $F_{U(1)}(\alpha) = 0$ (explicit, unique)	EM closure	[E]
Cabibbo angle	$\lambda_C = \sqrt{u(1-u)}$, $u = \frac{4}{3}c_3 + 48c_3^4$	seed	[E]
CKM CP phase	$\delta_{\rm CKM} = \frac{2\pi}{\lvert R^+(A_3)\rvert} + N_{\rm fam} u(1-u) = \frac{\pi}{3} + 3\lambda_C^2$	A_3^+, seed	[E]/[C]
PMNS angles/phase	$\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2}$, $\sin^2\theta_{13} = e^{-5/6}u$, $\delta^\nu = \frac{4\pi}{3}$	A_3^+, seed	[E]/[C]
Charged-lepton hierarchy	$\det M_\ell \propto u^{h/N_{\rm fam}} = u^{10}$; $(5,3,2)$	decuple	[E]/[C]
Mass-hierarchy transport	$\sum L = 40 = \lvert R(D_5)\rvert$; entries on $C_6 = \lvert R^+(A_3)\rvert$	$D_5 \times A_3$	[E]
Strong CP	$\theta_{\rm eff} = 0$ (determinant erasure, given RP; document 3, Part III)	—	[E]/[C]
Weak CP	holonomy: $\delta_{\rm CKM} - \frac{\pi}{3} = N_{\rm fam}\, u(1-u)$	A_3^+	[C]
Gravity tree normaliser	$\xi_{\rm tree} = \frac{\dim \mathfrak{g}_{\rm SM}}{\dim S^+} = \frac{3}{4} = q(A_3)$	A_3 glue	[E]
Cosmological constant	$\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} [v_{\rm geo}/(5\beta_{\rm rad}^2 \bar{M}_{\rm Pl})]^{10}$	decuple	[E]
Hubble anchor	$H_0\sqrt{\Omega_\Lambda} = \bar{M}_{\rm Pl} e^{-\alpha_\star^{-1}}/(2\pi)$	action	[E]
Baryon density	$\omega_b = \lambda_C/\mathcal{A}_\Lambda = \lambda_C/10$	decuple	[E]/[C]
Inflation (scalaron)	$M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^{\Omega_{\rm adm} - 10b_1} = c_3^7$	deficit $r-1$	[E]
Inflation amplitude	$A_s = N_\star^2 c_3^7/24\pi^2 \approx 2.2 \times 10^{-9}$	deficit	[C]/[O]

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

Reading. The discrete sector — generations, gauge dimension, family content, hypercharge, b_1, occupancy — is now *group-theoretic output* of $D_5 \times A_3$. The continuous sector — α^{-1}, λ_C, the mixings, ρ_Λ, H_0, inflation — is the seam seed c_3 and the action exponentials read on the *same* carrier graphs (K_5 pairs, the C_6 hexagon $= \lvert R^+(A_3)\rvert$, the decuple $\mathcal{A}_\Lambda = \binom{g_{\rm car}}{2} = 10$). What used to be a list of separate hits is, structurally, one machine: D_5 carries the family and the flavor budget, A_3 carries the generations and the gauge dimension, μ_4 glues them (and reappears as the 41 chain, the discriminant norms, and the CP phase step $2\pi/6$), and c_3 dresses everything.

4 Grammar I — the carrier grammar $(3+2)$

Machine checks for this section: [`verification/v2_carrier_pascal.py`]

From $g_{\rm car} = 5$ and the split $(b, s) = (3, 2)$ the carrier data follow:

$$\begin{aligned} \gamma_{\rm car} &= \frac{g_{\rm car}}{g_{\rm car}+1} = \frac{5}{6}, \quad N_{\rm fam} = \frac{g_{\rm car}+1}{2} = 3, \quad \dim S^+ = 2^{g_{\rm car}-1} = 16, \\ \Omega_{\rm adm} &= N_{\rm fam}\, 2^{g_{\rm car}-1} = 48, \qquad b_1 = \frac{g_{\rm car}\, 2^{g_{\rm car}-2}+1}{10} = \frac{41}{10}. \end{aligned} \tag{19}$$

4.1 Dual-root dictionary (everything from two charges)

Two carrier roots generate the family and the transport cusps [E]. With $y_+ = \frac{1}{2}$ and $y_- = -\frac{1}{3}$ (the determinant-normalised carrier eigenvalues), the full one-family hypercharge spectrum and the flavor-transport cusps are sums of the two roots:

$$\begin{aligned} Y(Q_L) &= y_+ + y_- = \tfrac{1}{6}, \quad Y(u^c) = 2y_- = -\tfrac{2}{3}, \quad Y(d^c) = -y_- = \tfrac{1}{3}, \\ Y(L_L) &= -y_+ = -\tfrac{1}{2}, \qquad Y(e^c) = 2y_+ = 1, \qquad Y(\nu^c) = 0, \end{aligned}$$

$$\{1, \tfrac{2}{3}, \tfrac{1}{3}\} = \{\, 2y_+,\ -2y_-,\ 2(y_+ + y_-)\,\}, \qquad \gamma_{\text{car}} = \operatorname{Tr}_E Y^2 = 3y_-^2 + 2y_+^2 = \tfrac{5}{6},$$

$$\varphi_{\text{base}} = \frac{1 - \gamma_{\text{car}}}{\pi} = \frac{1}{6\pi}.$$

All residuals are exactly zero. The Standard-Model charge table is not assumed; it is built by addition from $\{1/2, -1/3\}$.

The $X = 6Y$ quadratic and the hypercharge Pascal-sign duality [E]

Clearing denominators with $X = 6Y$ turns the two base charges into the integer roots $x_+ = 3$, $x_- = -2$ of

$$\boxed{x^2 - x - 6 = 0}: \qquad x_+ + x_- = 1 = N_\Phi, \quad x_+ - x_- = 5 = g_{\text{car}},$$

$$-x_+ x_- = 6 = |R^+(A_3)|, \qquad \sqrt{\Delta} = 5,$$

so the Higgs singleton, the carrier rank and the A_3 hexagon all fall out of one quadratic. Counting the signs of X over a generation ($X = 1^{\times 6}, 2^{\times 3}, 6^{\times 1}, (-4)^{\times 3}, (-3)^{\times 2}, 0^{\times 1}$) gives the *Pascal triple* again:

$$\boxed{(\#X{=}0,\ \#X{<}0,\ \#X{>}0) = (1, 5, 10) = \tbinom{5}{0}, \tbinom{5}{1}, \tbinom{5}{2}},$$

$$10{-}5{=}g_{\text{car}}, \quad 5{-}1{=}|\mu_4|, \quad 10{-}1{=}\operatorname{tr} R,$$

with $\nu^c \leftrightarrow \binom{5}{0}$ the neutral singleton — the right-handed neutrino is the Pascal zero with a job. The moment ladder adds $\operatorname{Tr} X^2 = 120 = 5!$ and $\operatorname{Tr} X^4 / \operatorname{Tr} X^2 = 2280/120 = 19$ (an E_8 exponent; audit). **[E]**

4.2 Carrier arithmetic (proved identities)

The discrete coefficients are not isolated numbers; they are exact carrier traces:

$$10\, b_1 = 41 = g_{\text{car}}^2 + (g_{\text{car}} - 1)^2 = \Omega_{\text{adm}}\, \gamma_{\text{car}} + 1 = 1 + h_{E_8} \tfrac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}} = 1 + 30 \cdot \tfrac{16}{12} = 40 + 1, \quad \textbf{[E]}$$

(20)

$$I_1^{\text{EW}} = \frac{41}{48} = \operatorname{Var}_{S^+}(Y) \cdot b_1 = \frac{5}{24} \cdot \frac{41}{10} = \gamma_{\text{car}} + \frac{1}{\Omega_{\text{adm}}}, \quad \textbf{[E]}$$

(21)

$$\operatorname{Tr}_{S^+}(X^2) = 120 = 5! \quad (X = 6Y), \qquad \delta_2 = \tfrac{5}{4}\delta_{\text{top}}^2 = 2880\, c_3^8 = 4! \cdot 5!\, c_3^8 = 4!\, \operatorname{Tr}_{S^+}(X^2)\, c_3^8, \quad \textbf{[E]}$$

(22)

$$\frac{5}{4} = \underbrace{\tfrac{3}{2} \cdot \tfrac{5}{6}}_{B\gamma} = \underbrace{\tfrac{D_{\text{start}}}{\Omega_{\text{adm}}}}_{60/48} = \underbrace{\tfrac{g_{\text{car}}}{g_{\text{car}} - 1}}_{5/4} = \frac{\delta_2}{\delta_{\text{top}}^2}. \quad \textbf{[E]}$$

(23)

So: b_1 is a Pythagorean carrier identity ($5^2 + 4^2$); equivalently the anchor-product form $10b_1 = p_1 p_3 + (g_{\text{car}} - p_1) = 4 \cdot 10 + (5 - 4) = 41$ reads "all four glue corners times all ten carrier pairs, plus the one leftover Higgs slot"; the electroweak exponent $41/48$ is the hypercharge variance times the $U(1)$ budget; the second seam defect $2880\, c_3^8$ is a product of factorial traces $4! \cdot 5!$; and the universal compression $5/4 = g_{\text{car}}/(g_{\text{car}} - 1)$ recurs whenever the five-slot carrier projects onto four effective directions. As a minor curiosity, the SU(5)-tree weak mixing angle rewrites as

$\sin^2\theta_W^{\rm tree} = \frac{3}{8} = N_{\rm fam}/(g_{\rm car}+N_{\rm fam}) = N_{\rm fam}/\operatorname{rank}(E_8)$ (the value is the standard one; the rewrite is a coincidence to note, not a claim). [C]

5 Grammar II — the seed grammar (one decoder u)

Machine checks for this section: [verification/v106_review_validation.py]

The seed is, exactly, a pure function of c_3:

$$\boxed{u = \varphi_0^{\rm ret} = \frac{4}{3}\,c_3 + \Omega_{\rm adm}\,c_3^4 = \underbrace{\frac{1}{6\pi}}_{\varphi_{\rm base}} + \underbrace{\frac{3}{256\pi^4}}_{\delta_{\rm top}} = 0.0531719521768} \qquad \text{[E]}. \tag{24}$$

Its two coefficients are themselves carrier data: $\frac{4}{3} = \dim S^+/\dim \mathfrak{g}_{\rm SM} = 16/12 = 8/(g_{\rm car}+1)$ (so $\varphi_{\rm base} = 1/((g_{\rm car}+1)\pi)$) and $48 = \Omega_{\rm adm}$. The seed is thus a pure function of c_3 and the carrier counts.

One seed, four readouts (and their inverses) [E].

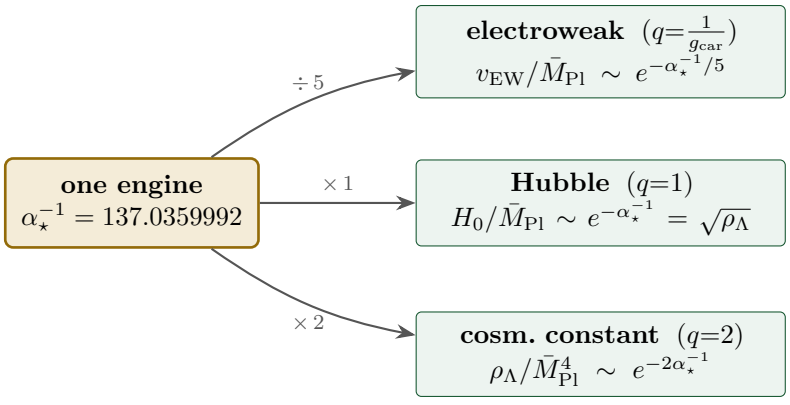
$$\lambda_C^2 = u(1-u), \quad \beta_{\rm rad} = \frac{u}{4\pi}, \quad \Omega_b = \Big(1-\frac{1}{4\pi}\Big)u, \quad \sin^2\theta_{13} = e^{-5/6}\,u,$$

$$u = \frac{1-\sqrt{1-4\lambda_C^2}}{2} = 4\pi\,\beta_{\rm rad} = \frac{\Omega_b}{1-1/(4\pi)} = e^{5/6}\sin^2\theta_{13}$$
$$= \sum_{n\ge 0} C_n\,\lambda_C^{2n+2} = \lambda_C^2 + \lambda_C^4 + 2\lambda_C^6 + 5\lambda_C^8 + \cdots$$

One knob u turns four dials simultaneously: Cabibbo mixing (the Bernoulli width $\sqrt{u(1-u)}$), cosmic birefringence, the baryon fraction, and the reactor angle. Turning one forces the others; there is no free per-observable tuning. The exponent $5/6$ is the carrier trace $\gamma_{\rm car} = \operatorname{Tr}_E Y^2$, *not* the Euler–Mascheroni constant $0.5772\ldots$ (audit [O]).

6 Grammar III — the action grammar (exponentials of $\alpha_\star^{-1}$)

The relation in plain words. The fine-structure constant is not just "a number near 137": it is the *one exponential engine* behind three wildly separated scales. Take the same $\alpha_\star^{-1} = 137.036$ and divide the carrier into it: $\frac{1}{5}$ of it sets the *electroweak* scale, $1\times$ it sets the *Hubble* scale, $2\times$ it sets the *cosmological constant*. The three action charges are the Pascal row $1:5:10$ of the five-slot carrier, and the Hubble rung is just the square root of the vacuum energy. So one constant fixes the electroweak hierarchy, dark energy and the expansion rate at once.



action charges $1 : g_{\rm car} : 2g_{\rm car} = 1:5:10 = \binom{5}{0} : \binom{5}{1} : \binom{5}{2}$ (Pascal row of K_5)

Read every dimensionless ratio through its *action* and ask for the action charge:

$$\boxed{x = R_x\, e^{-A_x}, \qquad A_x = q_x\, \alpha_\star^{-1} + \Delta_x, \qquad q_x \in \left\{\tfrac{1}{g_{\rm car}},\, 1,\, 2\right\}.} \tag{25}$$

The robust ladder and its residues are

sector	action A_x	charge q_x	residue R_x	value of $-\ln x$
electroweak $v_{\rm geo}/\bar{M}_{\rm Pl}$	$(\alpha_\star^{-1}+\delta_{\rm ph})/g_{\rm car}$	$1/g_{\rm car}$	$g_{\rm car}\beta_{\rm rad}^2$	36.85
Hubble $H_0/\bar{M}_{\rm Pl}$	$\alpha_\star^{-1}$	1	$1/(2\pi\sqrt{\Omega_\Lambda})$	138.68
cosm. const. $\rho_\Lambda/\bar{M}_{\rm Pl}^4$	$2\alpha_\star^{-1}$	2	$3/(4\pi^2)$	276.65

$$A_{\rm EW} = \frac{\alpha_\star^{-1}+\delta_{\rm ph}}{g_{\rm car}} = 27.5259, \qquad \mathcal{A}_\Lambda = 2\alpha_\star^{-1} = 274.072, \qquad \boxed{A_{\rm EW}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda = 1{:}5{:}10}. \tag{26}$$

The leading ratio is exact: $\mathcal{A}_\Lambda/(\alpha_\star^{-1}/g_{\rm car}) = 2g_{\rm car} = 10$ (with the transport correction $\mathcal{A}_\Lambda/A_{\rm EW} = 9.957$). The claim is that the *action charges*, not the raw logarithms, form the ladder.

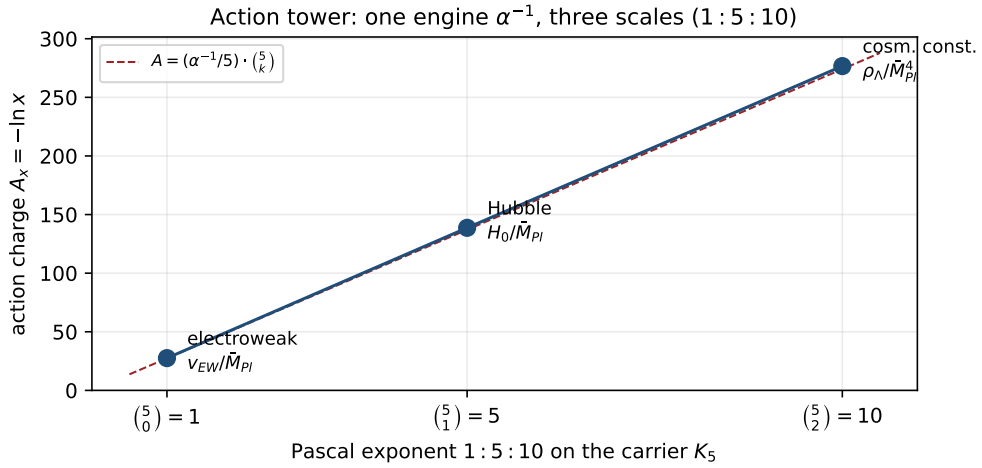


Figure 6: Action tower (data plot, `verification/make_figures.py`). The three action charges $A_x = -\ln x$ for the electroweak, Hubble and cosmological-constant ratios fall on the line $A = (\alpha_\star^{-1}/5)\binom{5}{k}$: one engine $\alpha_\star^{-1}$ generates all three scales with the Pascal exponents $1:5:10$.

6.1 The EM closure: $\alpha_\star^{-1}$ as an explicit fixed point (made explicit here)

The engine $\alpha_\star^{-1}$ is not an input — it is the unique root of an explicit closure equation (the exact electromagnetic closure theorem). With $c_3 = 1/(8\pi)$, $\varphi_{\rm base} = 1/(6\pi)$, $q(\alpha) = \delta_{\rm top}e^{-2\alpha}$ and the seam response $\varphi_{\rm seam}(\alpha) = \varphi_{\rm base} + q(\alpha)(1-q(\alpha))^{-5/4}$,

$$\boxed{F_{U(1)}(\alpha) = \alpha^3 - 2c_3^3\alpha^2 - \tfrac{4}{5}c_3^6\Big(\textstyle\sum_{f,j} L_{f,j} + N_\Phi\Big)\log\frac{1}{\varphi_{\rm seam}(\alpha)} = 0} \qquad \text{[E]/[C]}, \tag{27}$$

where the coefficient is $\frac{4}{5}(\sum L + N_\Phi)c_3^6 = \frac{4}{5}\cdot 41\, c_3^6 = 8b_1c_3^6$, so the fine-structure constant reads the *flavor transport budget* $\sum L = 40$ and the $U(1)$ index $b_1 = 41/10$.

Lemma 1 (Existence and uniqueness of the EM fixed point)**.** $F_{U(1)}$ has *exactly one* positive root, and it is simple.

Proof. Write $F_{U(1)}(\alpha) = h(\alpha) - C\,g(\alpha)$ with $h(\alpha) = \alpha^3 - 2c_3^3\alpha^2$, $C = \frac{4}{5}c_3^6 M > 0$ ($M = 41$), and $g(\alpha) = -\log\varphi_{\rm seam}(\alpha)$. Since $q(\alpha) = \delta_{\rm top}e^{-2\alpha} \in (0,1)$ is strictly decreasing, $\varphi_{\rm seam} = \varphi_{\rm base} + q(1-$

$q)^{-5/4}$ is C^1 and decreasing, so g is C^1, bounded, and slowly increasing with $g'(\alpha) = O(q)$ tiny; numerically $g \in [2.9342, 2.9343]$ on the admissible window. The cubic h has its interior critical point at $\alpha_c = \frac{4}{3}c_3^3 = 8.399 \times 10^{-5}$, with $h(0) = 0$ and $h < 0$ on $(0, \alpha_c)$. *Monotonicity (corrected endpoint).* Since $F'_{U(1)}(\alpha) = 3\alpha^2 - 4c_3^3\alpha - C\,g'(\alpha)$ and at α_c the first two terms *cancel* ($3\alpha_c = 4c_3^3$), one has $F'_{U(1)}(\alpha_c) = -C\,g'(\alpha_c) \approx -5.89 \times 10^{-10} < 0$: the derivative is *not* yet positive at α_c. Its first zero lies slightly above, at $\alpha_0 = 8.6264 \times 10^{-5}$ (where $3\alpha_0^2 - 4c_3^3\alpha_0 = C\,g'(\alpha_0)$), with $F'_{U(1)} < 0$ on (α_c, α_0) and $F'_{U(1)} > 0$ on (α_0, ∞). Now $F_{U(1)}(0) = -C\,g(0) < 0$, and $F_{U(1)}$ *decreases* on $(0, \alpha_0]$ (both h and $-Cg$ decrease on $(0, \alpha_c]$, then $F'_{U(1)} < 0$ on $[\alpha_c, \alpha_0]$), so $F_{U(1)}(\alpha_0) \approx -3.82 \times 10^{-7} < 0$; thereafter $F_{U(1)}$ is strictly increasing to $+\infty$. Hence $F_{U(1)} < 0$ on $(0, \alpha_0]$ and strictly increasing on $[\alpha_0, \infty)$, so it crosses zero *exactly once*. The crossing has $F'_{U(1)}(\alpha_\star) = 1.58 \times 10^{-4} > 0$, so the root is simple. A sign scan on $(0, 0.05)$ confirms a single change. *Rigorous version:* an **interval-arithmetic** enclosure (`mpmath.iv`) brackets the root in [0.00729735256220970, 0.00729735256221000] with $F_{U(1)} < 0$ on the lower and > 0 on the upper endpoint (definite signs), and a 200-cell interval partition of $(\alpha_c, 0.05)$ contains exactly *one* indeterminate (root-bearing) band — a verified uniqueness, not merely a sampled one. [verification/v3_em_alpha.py] □

Solving:

$$\boxed{\alpha_\star = 0.00729735256220985\ldots, \quad \alpha_\star^{-1} = 137.0359992168407\ldots} \quad \textbf{[E]} \tag{28}$$

[verification/v3_em_alpha.py] versus CODATA-2022 $\alpha_\star^{-1} = 137.035999177(21)$ — a relative deviation 2.9×10^{-10} ($\approx 1.9\sigma$ on the tiny experimental error; the older 2018 value 137.035999084 gave 9.7×10^{-10}, so the agreement *improved* in relative terms). The exponent $-5/4$ in φ_{seam} is the carrier discriminant norm $q(D_5)$, and $\delta_{\text{top}} = 48c_3^4$, so the closure is built only from c_3, the carrier, and the flavor budget. This discharges the former open audit point: the EM closure is a stated, existence-/uniqueness-proven, numerically reproduced equation, not a fit.

Why *this* function: the boundary $U(1)$ Ward reading (Theorem C) [E]/[C]. $F_{U(1)}$ is not a chosen ansatz but the reduced Ward identity of the boundary $U(1)$ partition function $Z_\partial = Z_{\text{Maxwell}}\, Z_{\text{Calderón}}\, Z_{\text{transport}}$, with three terms whose coefficients are compiler atoms, not fits:

$$\underbrace{\alpha^3}_{\text{Maxwell moment}} \quad \underbrace{-\,2c_3^3\alpha^2}_{\text{Calderón sheet } (2=|\mathbb{Z}_2|)} \quad \underbrace{-\,8b_1c_3^6 \log\tfrac{1}{\varphi_{\text{seam}}}}_{C_6 \text{ transport det.}} = 0, \qquad 8b_1 = \tfrac{4}{5}M = \tfrac{4}{5}(\textstyle\sum L + N_\Phi) = \tfrac{164}{5}.$$

The Calderón factor $2 = |\mathbb{Z}_2|$ (sheet), the transport power c_3^6 is the C_6 hexagon, and the seam exponent $-\frac{5}{4} = -q(D_5)$ is the carrier discriminant norm. **Typing:** the three-term decomposition and the coefficient identities ($8b_1 = \frac{4}{5}M$, $-\frac{5}{4} = q(D_5)$) are exact **[E]**; the *physical origin* as a boundary Ward closure (the lemmas $\partial_\tau \log Z_{\text{Maxwell}} = \alpha^3$, $\partial_\tau \log Z_{\text{Calderón}} = -2c_3^3\alpha^2$, $\partial_\tau \log \det_\zeta(1 - \mathcal{T}_{C_6,U(1)}) = -8b_1c_3^6 \log\frac{1}{\varphi_{\text{seam}}}$) is a conjectured **[C]** interpretation, not yet machine-proven. [verification/v48_em_ward.py]

The determinant-line (Quillen) form. ([verification/v341_alpha_quillen.py]) The three terms assemble into a single *stationarity* statement for the boundary $U(1)$ determinant line,

$$\underbrace{[\alpha^3 - 2c_3^3\alpha^2]}_{\delta \log\det \Delta_{U(1)}} = \underbrace{[8b_1c_3^6]}_{\text{ctr.}}\ \underbrace{[\log\tfrac{1}{\varphi_{\text{seam}}}]}_{\text{seam resp.}},$$

so $F_{U(1)} = 0$ is $\delta \log\det \Delta_{U(1)} + \text{counterterm} + \text{seam response} = 0$. Every factor is a *named* object, not a knob: the counterterm coefficient is $8b_1c_3^6 = (\text{rank}\, E_8)\, b_1\, c_3^6$ (index × heat-kernel), $c_3 = 1/(8\pi)$ is the Gauss–Bonnet boundary coefficient, the exponent $-\frac{5}{4} = -q(D_5)$ is one carrier discriminant norm and the seam base $\varphi_{\text{base}} = 1/(6\pi) = \frac{4}{3}c_3$ carries the *other*, $c_3/\varphi_{\text{base}} = \frac{3}{4} = q(A_3)$ — both glue forms appear, and the split is verified at the root. **Typing (and: no new gate).** The determinant-line assembly and all coefficient identities are exact **[E]**; $\alpha_\star^{-1}$ itself stays closed **[E]** (unique root + ablation). This adds *no* open item — it *sharpens* the pre-existing **[O]**/physical-origin obligation `EM.WARD.01` (the "why *this* function" residual, already conjectural) by naming

every coefficient. The one still-open part *is* that residual, now *named* as its own tracked target `ALPHA.QUILLEN.EXACT.01` ([`verification/v382_alpha_quillen_exact.py`]): the from-first-principles AQFT/ζ proof that $\delta_\tau(\log\det_\zeta \Delta_{U(1)} + 8b_1c_3^6\log\varphi_{\text{seam}}) = 0$ — i.e. that $F_{U(1)}$ is the *exact* Quillen functional of the seam $U(1)$, the variational principle *derived*, not assembled. It is a *face* of `SEAM.EQUIV.01` (v336) — the seam $U(1)$ determinant line lives on the same holomorphic net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed — hence no second independent gate, just the sharpest remaining EM-Ward obligation, now citable on its own. *Honest progress towards it ([`verification/v391_alpha_quillen_progress.py`]), without closing it:* a solvable model $\det_\zeta(-\partial_\tau^2+m^2)$ on a circle $= (2\sinh(\beta m/2))^2$ has the closed m-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \ldots$, a heat-kernel/Laurent series — so a ζ-determinant's variation is *structured* heat-kernel data, and the counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object. This *reduces* `ALPHA.QUILLEN.EXACT.01` to the named sub-target "the seam $U(1)$ Seeley–DeWitt coefficient $= 8b_1c_3^6$", but the target itself stays [O] (an external math proof, v384); the value $\alpha_\star^{-1}$ stays [E]. *A second honest step ([`verification/v433_alpha_quillen_heatkernel.py`]) bridges this reduction to the heat-kernel origin below, without closing it:* v391's circle reached only the low orders a_0, a_1, whereas a solvable *4D* model — the flat $\Delta = -\partial^2+m^2$ on T^4, $\mathrm{Tr}\,e^{-t\Delta} = V/(4\pi)^2\,t^{-2}e^{-tm^2}$ — has $a_4 = Vm^4/(2(4\pi)^2) \neq 0$, *reaching* the t^0 (conformal-anomaly) order the circle could not, the very order in which the quadratic term is grounded below (v342, Gilkey a_4). So a ζ-determinant's variation is heat-kernel data *at the a_4 order*, and the measure itself is c_3-built, $(4\pi)^{-2} = (2c_3)^2 = 4c_3^2$; the *actual* seam $a_4 = 8b_1c_3^6$ and the cubic α^3 Chern moment stay [O] (v384), the seam F-normalisation [C] (retyped by the fifth step below, v470, as the embedding index k_Y=5/3).

The heat-kernel origin of the terms ([`verification/v342_em_ward_heatkernel.py`]). The determinant-line form is more than bookkeeping: its terms are textbook heat-kernel coefficients. [E] $c_3 = 1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, $c_3 = 1/(|\mathbb{Z}_2|\oint K)$, the same χ=2 as the marks, v216). [E] the α^2 (Calderón) term *is* the Gilkey gauge-curvature coefficient: the $U(1)$ Laplacian's Seeley–DeWitt a_4 contains $30\,\Omega_{ij}\Omega^{ij}/360 = \frac{1}{12}\Omega^2$ (Gilkey, *Invariance Theory*, Thm 4.8.16), and with $\Omega = i\alpha F$ this forces an $-(\alpha^2/12)F^2$ contribution to $\log\det\Delta_{U(1)}$ — so the Calderón $-2c_3^3\alpha^2$ is the gauge-curvature (Quillen-curvature) piece, not an ansatz. [E] the c_3-orders $\{0,3,6\}$ are an arithmetic ladder (step 3 = the three channels; $6 = |R^+(A_3)|$, the C_6 hexagon), and the π-power test confirms it: $2c_3^3 = 1/(256\pi^3)$ carries π^3, i.e. *three* boundary c_3-insertions, not a single bulk coefficient. **Typing:** the boundary structure and the gauge-curvature origin of the quadratic term are heat-kernel-forced [E]; the exact coefficient value is [C] (the seam F-normalisation — retyped as the embedding index k_Y=5/3 by the fifth step below, v470); the cubic α^3 Maxwell moment (a topological/Chern level) and the full "this *is* the seam $U(1)$ ζ-determinant" claim stay [O] — exactly the `EM.WARD.01` residual, unchanged. So the quadratic/curvature half of the EM-Ward origin is now derived; the cubic-moment half remains the honest open part. *A third honest step ([`verification/v434_alpha_quillen_betafunction.py`]) settles those residuals and shows they are not three independent problems.* The counterterm's *matter* factor is heat-kernel-forced too: by the $\beta = a_4$ theorem the one-loop $U(1)_Y$ β *is* the a_4 coefficient, and the carrier hypercharge content gives $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) $\Rightarrow b_1 = 41/10$ (GUT, v159/v246), so $8b_1c_3^6 = b_1\,(\mathrm{rank}\,E_8)\,c_3^6$ is an a_4 content coefficient times a boundary geometry — the same *kind* of object as the Calderón a_4. The exact value then turns only on the seam F-normalisation, so the "actual seam a_4" residual *is* the F-normalisation [C] (one obligation, not two); and the cubic factors $a^3 - 2c_3^3a^2 = a^2(a-2c_3^3)$, exhibiting a^3 as the first moment of the Chern density a^2 — a level whose from-first-principles origin stays [O]. The EM-Ward residual is thus *one* [C] + *one* [O], not three. *A fourth honest step ([`verification/v435_alpha_quillen_chernlevel.py`]) sharpens that single remaining [O] — residual (2), the cubic α^3 Chern level — without closing it.* A π-power / metric-independence test partitions the closure: the three coefficients carry π-powers exactly $\{0,3,6\}$ ($k_0 = 1$; $-2c_3^3 = -1/(256\pi^3)$; $-8b_1c_3^6 = -41/(327680\pi^6)$), so the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-*independent*) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. So the test confirms, now from a test rather than a claim, that a^3 is a level and not a Seeley–DeWitt

curvature integral (the other two terms are heat-kernel boundary data). [E] *Conditional* on the Chern–Simons/η reading of the Maxwell moment (the `EM1` interpretation, v48), large-gauge invariance quantises the seam $U(1)$ level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. [C] What stays [O] is the `EM1` lemma itself — the from-first-principles Chern–Simons-boundary proof that $\partial_\tau \log Z_{\text{Maxwell}} = a^3$ — external math (type A, v384). *A fifth honest step (`[verification/v470_alpha_inflow_level.py]`) upgrades both leftovers, still without closing the target.* [C] *The level is computed, not minimal:* under the same `EM1` reading, the a^3 coefficient $k_0 = 1$ *equals* the FHS bulk Chern invariant $|C| = 1$ of the very p+ip collar that realises S3 of `SEAM.EQUIV.01` (v367/v461; control $C(M{=}3) = 0$) — "why exactly 1" is answered by a computation plus the cited inflow identification (TKNN / Avron–Seiler–Simon quantisation; Callan–Harvey inflow; the APS/Witten η=CS reading of $\delta \log\det$; Quillen/Bismut–Freed determinant-line integrality), *replacing* the minimality assumption of the fourth step. [E] *The F-normalisation is the embedding index:* $k_Y = \text{tr}_{\bar{5}}(Y^2)/\,\text{tr}(T_3^2) = (5/6)/(1/2) = 5/3$ (Ginsparg 1987), so $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$ exactly — the "GUT 3/5" *is* the inverse embedding index, and once the seam net is $(E_8)_1$ at level 1 the $U(1)_Y$ normalisation $\langle J(z)J(w)\rangle = k/(z{-}w)^2$ is *fixed* by it (level-1 current-algebra rigidity, no continuous dial): the seam F-normalisation [C] has *zero independent content* — a face of `SEAM.EQUIV.01`, per the v382 typing. [C] *One phase, two responses:* the collar phase carries $c_- = 8$ (gravitational, feeding `SEAM.EQUIV.01`/S3, v456) *and* $C = 1$ ($U(1)$, feeding `EM1`) — so discharging the one realisation input feeds both named targets at once. *A sixth honest step (`[verification/v472_quillen_detline_moduli.py]`) exhibits that bridge lemma at the finite level.* `ALPHA.QUILLEN.EXACT.01` names the level as the curvature of the *determinant line over the $U(1)$ moduli* (Quillen/Bismut–Freed curvature, Dai–Freed section) — and the fifth step computed the Chern invariant only over the Bloch Brillouin zone, the translation-invariant shortcut. The sixth step computes the Quillen-shaped object itself: the same collar Hamiltonian in real space with twisted boundary conditions — the twist torus (θ_x, θ_y) *is* the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied frame over that torus integrates to $2\pi \cdot 1$ exactly (10^{-9}), size-independently ($L = 4, 6$), with clean controls (trivial collar $\to 0$, orientation flip $\to -1$), the twist-moduli integer equal to the Bloch-BZ integer for all three phases (Niu–Thouless–Wu), and the Fermi gap open over the whole torus (the Dai–Freed section has no zero). So "$\delta \log\det$(seam model) = the inflow response" *holds verbatim at the finite/model level*, where $\log\det$ of the occupied frame replaces $\log\det_\zeta$. [E] (computation) / [C] (the EM1 reading). What stays [O] is the *continuum* identification "$\delta \log\det_\zeta$(seam) = the inflow response" on the abstract seam (the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays [O], and $\alpha_\star^{-1}$ stays [E] regardless. *A seventh step unifies this target with the φ_0^{ret}-puncture target (`[verification/v484_seam_contact_unit.py]`).* The $\{0, 3, 6\}$ c_3-ladder above and the puncture's per-mark weight c_3 (v483) are *one* rule — the seam contact unit — and it is not a new unknown: "c_3 per insertion" *is* the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$ (v239), i.e. one bare boundary propagator ($\frac{1}{2\pi} = 4c_3$) orbit-averaged over the four marks. [E] derived on the seam circle for the finite cycle sector (the bare Green function takes integer multiples of $c_3 \ln 2$ at the μ_4 separations; the log-det contact expansion is exactly graded in c_3 per insertion), with the grammar already suite-wide (the Λ prefactor $\frac{3}{4\pi^2} = 48c_3^2$, v60, carries the same $\Omega_{\text{adm}} = 48$ at two insertions). [C]/[O] Consequence: `ALPHA.QUILLEN.EXACT.01` and `HYP.PHI0.PUNCTURE.01` *merge* into one remaining analytic step — the diagonal zeta-renormalisation at the marks plus the state-multiplicity matching ($41 = 10b_1$ here, $48 = \Omega_{\text{adm}}$ there). *An eighth step settles that remaining step at the computable level (`[verification/v485_contact_diagonal_closed.py]`).* [E] The renormalised coincident-point Green function on a circle of circumference ℓ is $G_{\text{reg}}(0;\ell) = \frac{1}{\pi}\ln(\ell/2\pi)$ exactly — it *vanishes* precisely at $\ell = 2\pi = 1/(4c_3)$, the KMS seam circumference (v239): the seam scale is the unique zero-self-energy scheme point, and the log is the pure *scale* channel (matching the fact that $F_{U(1)}$ carries exactly *one* log term while the φ_0^{ret} split is log-free). [E] With the diagonal zero the mark contact calculus *resums in closed form* (the BFK/gluing route, v151): $\det(I - uC) = (1 - 4u)(1 + 2u)^2$ with $u = \varepsilon c_3 \ln 2$ — all orders, the linear term absent because $\text{Tr}\, C = 0$. [E] And the two multiplicities are *one* state set under two response weights: the

same $48 = 3 \times 16$ admissible states counted flat give the φ_0^{ret} puncture count $48 = \Omega_{\mathrm{adm}}$, and Y^2/loop/GUT-weighted give the alpha budget $41 = 10b_1 = 40 + 1$ ($\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg, v470). [C]/[O] Net: every finite/computable piece of the merged target is proven; the single remaining [O] is the identification of the abstract seam ζ-det with this glued mark determinant — *exactly* the v382 typing (a face of `SEAM.EQUIV.01`), now substantiated computationally rather than asserted. *A ninth step executes both named faces of the target at the computable level (`[verification/v974_alpha_faces_computed.py]`), typing what remains.* [E] *The diagonal-zeta face is executed:* the BFK Mayer–Vietoris determinant formula (v151) now *runs* on the seam circle cut at the four μ_4 marks — gluing constant exactly 2^{-N} (symbolic, $N = 1, \ldots, 4$, general interval lengths), massless reassembly $\det'_\zeta(\Delta_{S^1}) = \ell^2$, jump matrix = the *inverse* mark Green matrix $R(m) = (G_m|_{\mathrm{marks}})^{-1}$; at the KMS scale $\ell = 2\pi = 1/(4c_3)$ this gives $R_0 = 16c_3\,\mathrm{circ}(2, -1, 0, -1)$, $\det' R_0 = 2^{16}c_3^3$ and glued product $(2\pi)^2$ — the v485 diagonal/scale channel reproduced *exactly* via the KMS unit. Against the v484 mark matrix ($-4c_3 \ln 2 \cdot \mathrm{circ}(0, 1, 2, 1)$) there is *no* global proportionality — the two kernels sit in *swapped* μ_4 channels (gauge zero mode vs. sign rep) — but the correspondence is channel-wise exact (same μ_4 Fourier eigenbasis, |spec| multisets $\{0, 2, 2, 4\}$, doublet ratio $4/\ln 2$ with c_3 cancelling, $\mathrm{Tr}\, B^k = 4^k + 2 \cdot 2^k$ mirroring $\mathrm{Tr}\, C^k = 4^k + 2(-2)^k$), so this face's residual *sharpens* to the channel-swap question. [E] *The EM1/topological face carries a computed pump witness:* on the v367 collar, pumped charge = edge spectral flow = Streda count = Chern = 1 per flux quantum (hybrid-Wannier winding residual 0, Streda $\Delta N = C\, n_\phi$ exact integers, the conjugation mutant flips all four readings simultaneously, M=3 controls all zero) — so the cited TKNN/Callan–Harvey step "boundary level = bulk response" of the fifth step now has a computed lattice witness; [C] the identification of the complex-copy $U(1)$ with the physical Majorana collar stays where it was, with the 16-copy bookkeeping (c_-=8, k_Y=5/3, $(1/k_Y)\frac{41}{6} = \frac{41}{10} = b_1$) re-derived exactly. [E] *The finite Quillen identity is two-sided:* over the $U(1)$-twist torus the FHS/Berry curvature integral of the determinant line *and* the metric-side integral (Gaussian overlap norm, Poincaré–Lelong divisor evaluation) both equal +1, the difference an exact form (3×10^{-16}), transition-function winding = Chern. [C] Typed conditional: *if* the seam state is quasi-free (measured, not forced in the strong form — `SEAM.STATE.RPMIXING.01`), *then* Quillen's identity applies literally to the seam determinant line. [E] The computed witnesses are carried in the ledger as `ALPHA.QUILLEN.FACES.01`; the remaining open content of `ALPHA.QUILLEN.EXACT.01` is thereby typed as exactly {rigidity premise} + {channel-swap question} + {continuum Bismut–Freed identification} — the target stays [O], and $\alpha_\star^{-1}$ stays [E] regardless (re-anchored: Quillen split residual 10^{-58} at 50 digits, $8b_1 = \frac{164}{5}$ exact). [E] *Channel-swap update (2026-08-28):* the swap question is resolved at the *finite* level by μ_4-character grading — the deck swap $r \to r{+}2$ maps the jump |spectrum| $[0, 2, 4, 2]$ onto the mark pattern $|[4, -2, 0, -2]|$ exactly (the odd swap fails), the graded $\det'$ difference is c_3-free = $\log(\ln 2/4)$, and the channelwise unit ratio is uniform $4/\ln 2$ (`[verification/v985_quillen_channel_swap.py]`); the open residual contracts to {rigidity premise} + {continuum Bismut–Freed identification}, and the target stays [O].

The full ablation — which inputs are load-bearing and which only set the last digit — is in the Ablation appendix below.

Inversion test (this round). Running the closure backwards — inserting the *measured* α and solving for the integer budget $M = \sum L + N_\Phi$ — returns $M = 41.0000001\ldots$, i.e. the external measurement reconstructs the internal discrete budget $41 = 40 + 1$ to 10^{-7}. This is stronger than "we hit α": the equation reads the discrete 41 out of nature. [E]

6.2 The decuple bridge (new, exact headline)

Eliminating $\alpha_\star^{-1} = g_{\mathrm{car}}\, A_{\mathrm{EW}} - \delta_{\mathrm{ph}}$ between the electroweak and the cosmological-constant lines gives a direct relation between the two scales:

Decuple bridge: Λ is the tenth power of the stripped EW scale [E]

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{2\delta_{\rm ph}} \left[\frac{v_{\rm geo}}{5\,\beta_{\rm rad}^2\,\bar{M}_{\rm Pl}}\right]^{10}} \tag{29}$$

where the exponent $10 = 2g_{\rm car}$ and $5 = g_{\rm car}$. Verified to a relative residual $\sim 10^{-159}$.

Proof. From $v_{\rm geo}/\bar{M}_{\rm Pl} = g_{\rm car}\beta_{\rm rad}^2 e^{-A_{\rm EW}}$, $e^{-A_{\rm EW}} = (v_{\rm geo}/\bar{M}_{\rm Pl})/(g_{\rm car}\beta_{\rm rad}^2)$. From $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2}e^{-2\alpha_\star^{-1}}$ and $\alpha_\star^{-1} = g_{\rm car}A_{\rm EW} - \delta_{\rm ph}$, $e^{-2\alpha_\star^{-1}} = e^{2\delta_{\rm ph}}e^{-2g_{\rm car}A_{\rm EW}} = e^{2\delta_{\rm ph}}(e^{-A_{\rm EW}})^{2g_{\rm car}}$. Substituting and using $2g_{\rm car} = 10$, $g_{\rm car} = 5$ gives the boxed form. □

In words: the electroweak scale is not directly the Λ scale, but once the electroweak ratio is stripped of its small $\beta_{\rm rad}$ prefactor, Λ is its tenth power, and the "ten" is $2g_{\rm car}$, not a fitted exponent. This is the direct algebraic compression of the action ladder.

6.3 The full power tower: x^1, x^5, x^{10} on the carrier graph K_5

The decuple bridge is one rung of a tower. Writing $x := e^{-A_{\rm EW}} = v_{\rm geo}/(5\beta_{\rm rad}^2\bar{M}_{\rm Pl})$ for the stripped electroweak unit, all three cosmological scales are powers of the *same* x with binomial exponents (verified exactly):

$$\boxed{\frac{v_{\rm geo}}{5\beta_{\rm rad}^2\bar{M}_{\rm Pl}} = x^{\binom{5}{0}} = x, \qquad \frac{H_0}{\bar{M}_{\rm Pl}} = x^{\binom{5}{1}}R_{\rm H} = x^5 R_{\rm H}, \qquad \frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = x^{\binom{5}{2}}R_\Lambda = x^{10}R_\Lambda} \tag{30}$$

with residues $R_{\rm H} = e^{\delta_{\rm ph}}/(2\pi\sqrt{\Omega_\Lambda})$ and $R_\Lambda = \frac{3}{4\pi^2}e^{2\delta_{\rm ph}}$. [E]. The exponents are the complete-graph counts of the five-slot carrier K_5:

$$H \leftrightarrow \text{vertices of } K_5 \ (\,\# = \tbinom{5}{1} = 5\,), \qquad \Lambda \leftrightarrow \text{edges of } K_5 \ (\,\# = \tbinom{5}{2} = 10\,).$$

So the cosmological constant is the product of one stripped-electroweak unit over every carrier *pair* (K_5 edge product), and the Hubble scale the product over every carrier *slot* (K_5 vertex product). This is the cleanest single-variable statement of the action ladder.

6.4 Hubble is the square root of Λ (corollary, exact)

On the comparison layer $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = 3\Omega_\Lambda(H_0/\bar{M}_{\rm Pl})^2$; equating to the TFPT line gives

$$\boxed{\frac{H_0}{\bar{M}_{\rm Pl}} = \frac{e^{-\alpha_\star^{-1}}}{2\pi\sqrt{\Omega_\Lambda}}}, \qquad \mathcal{A}_H = -\ln\frac{H_0}{\bar{M}_{\rm Pl}} = \alpha_\star^{-1} + \ln\!\big(2\pi\sqrt{\Omega_\Lambda}\big) = 138.68. \tag{31}$$

So the Hubble rung is *not* an independent third hit: it is the square root of the vacuum energy plus a geometric cosmology factor. The genuine TFPT content is the single $e^{-\alpha_\star^{-1}}$. (The numerical coincidence $\mathcal{A}_H - \alpha_\star^{-1} \approx \pi^2/6$ is just $\ln(2\pi\sqrt{\Omega_\Lambda})$ and is Ω_Λ-dependent; not a claim. [O])

6.5 The ~ 123 orders of the cosmological constant

$$\log_{10}\frac{M_{\rm Pl}^4}{\Lambda_{\rm IR}} = \underbrace{\frac{2\alpha_\star^{-1}}{\ln 10}}_{119.028} + \underbrace{\log_{10}\frac{1}{\delta_{\rm top}}}_{3.920} = 122.948 \approx 123. \tag{32}$$

The "119" is the action $2\alpha_\star^{-1}$; the "≈ 4" is the seam defect $\delta_{\rm top} = \Omega_{\rm adm}c_3^4$.

High-precision cosmological constant: a $\sim 10^{-13}$ prediction [E]

Because $\alpha_\star^{-1}$ is a *derived* number (the EM fixed point, known to 13 digits) and the residue $3/(4\pi^2)$ is exact, the dimensionless vacuum energy is predicted essentially exactly:

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{-2\alpha_\star^{-1}} = 7.12533 \times 10^{-121}} \qquad \textbf{[E]}\ [\texttt{verification/v7_gravity_cosmo.py}], \tag{33}$$

i.e. a dark-energy scale $\rho_\Lambda^{1/4} = \left(\frac{3}{4\pi^2}\right)^{1/4} \bar{M}_{\rm Pl}\, e^{-\alpha_\star^{-1}/2} = 2.23747$ meV (reduced Planck mass $\bar{M}_{\rm Pl} = 2.435323 \times 10^{18}$ GeV). The *dimensionless* ratio carries no observational input — its uncertainty is that of $\alpha_\star^{-1}$, $\sim 10^{-13}$ relative; the absolute meV value inherits only the Planck-mass uncertainty ($\sim 2 \times 10^{-5}$). Both are *far* sharper than the measured vacuum energy (Planck $\rho_\Lambda^{1/4} \simeq 2.24$ meV, known to $\sim 1\%$ and entangled with the H_0 tension). So TFPT pins the cosmological constant several orders of magnitude more precisely than it is currently measured — a genuine, falsifiable forward prediction: any future sub-percent ρ_Λ must land on 2.2375 meV.

6.6 The ladder is binomial; $2g_{\rm car} = \binom{g_{\rm car}}{2}$ singles out $g_{\rm car} = 5$

In units of the electroweak action $\alpha_\star^{-1}/g_{\rm car}$, the three rungs are the first three binomial coefficients of the five-slot carrier:

$$\frac{A_{\rm EW}}{\alpha_\star^{-1}/g_{\rm car}} : \frac{\mathcal{A}_H}{\alpha_\star^{-1}/g_{\rm car}} : \frac{\mathcal{A}_\Lambda}{\alpha_\star^{-1}/g_{\rm car}} = 1 : g_{\rm car} : 2g_{\rm car} = \binom{g_{\rm car}}{0} : \binom{g_{\rm car}}{1} : \binom{g_{\rm car}}{2} = 1 : 5 : 10. \tag{34}$$

The first two are automatic ($\binom{g_{\rm car}}{0} = 1$, $\binom{g_{\rm car}}{1} = g_{\rm car}$). The non-trivial content is the cosmological-constant charge:

$$\boxed{2g_{\rm car} = \binom{g_{\rm car}}{2} \quad \Longleftrightarrow \quad g_{\rm car} = 5} \qquad \textbf{[E]}\ [\texttt{verification/v2_carrier_pascal.py}], \tag{35}$$

since $\binom{g_{\rm car}}{2} = \frac{g_{\rm car}(g_{\rm car}-1)}{2} = 2g_{\rm car} \iff g_{\rm car} = 5$. So the "double overlap" charge $2g_{\rm car}$ equals the number of carrier-slot *pairs* $\binom{g_{\rm car}}{2}$ *only* at $g_{\rm car} = 5$: the cosmological constant is suppressed by one action unit per carrier pair, and this combinatorial reading is exclusive to the physical rank. It is an independent overdetermination of $g_{\rm car} = 5$ from the action grammar (cf. the landscape section).

6.7 Discriminant action theorem: the ladder reads the hypercharge discriminant

The $g_{\rm car}$ in the action ladder is not merely a rank: it is the square root of the discriminant of the carrier (hypercharge) polynomial. For a general split $b + s = g$ the carrier polynomial is $bs\,Y^2 + (s-b)Y - 1 = 0$ with discriminant

$$\Delta_Y = (s-b)^2 + 4bs = (b+s)^2 = g_{\rm car}^2, \qquad \text{so for } 6Y^2 - Y - 1 = 0:\ \Delta_Y = 25,\ \sqrt{\Delta_Y} = 5 = g_{\rm car}. \tag{36}$$

Hence the action ladder is, exactly,

$$\boxed{A_{\rm EW} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \qquad \mathcal{A}_H = \sqrt{\Delta_Y}\, A_{\rm EW}, \qquad \mathcal{A}_\Lambda = \binom{\sqrt{\Delta_Y}}{2} A_{\rm EW}} \qquad \textbf{[E]}. \tag{37}$$

The electroweak hierarchy therefore reads the *discriminant of the hypercharge polynomial*, tying the Standard-Model hypercharge derivation directly to the EW-hierarchy and Λ problems.

6.8 A two-dimensional action lattice

All scales sit on the integer lattice spanned by two generators — the gauge unit $\alpha_\star^{-1}/g_{\rm car}$ and the flavor unit $L_0 = \ln(1/u)$ — as $S_x = a_x(\alpha_\star^{-1}/g_{\rm car}) + b_x L_0 + (\text{residue})$:

scale	a_x	b_x	axis
$v_{\rm geo}/\bar{M}_{\rm Pl}$	$1 = \binom{5}{0}$	0	gauge
$H_0/\bar{M}_{\rm Pl}$	$5 = \binom{5}{1}$	0	gauge
$\rho_\Lambda/\bar{M}_{\rm Pl}^4$	$10 = \binom{5}{2}$	0	gauge
$m_e/\bar{M}_{\rm Pl}$	1	5	flavor
$m_\mu/\bar{M}_{\rm Pl}$	1	3	flavor
$m_\tau/\bar{M}_{\rm Pl}$	1	2	flavor

The cosmological/gauge sector runs along $b = 0$ with binomial charges; the charged-fermion sector runs along $a = 1$ with the integer flavor ladder $n_f = (5, 3, 2)$. This is the compact "periodic table" of TFPT scales: two integer labels per scale. [E] (charges) / [C] (residues).
A third (seam/gravity) generator completes the lattice: $L_c = \ln(1/c_3) = \ln(8\pi)$. The scalaron lives purely on it, $M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2} = e^{-\frac{7}{2}L_c}$, and the seam defect is $\delta_{\rm top} = e^{-(4L_c - \ln 48)}$. So the three log-generators are $\{\alpha_\star^{-1}$ (hierarchy), $L_0 = \ln(1/u)$ (flavor), $L_c = \ln(1/c_3)$ (seam/gravity)$\}$. [E]

7 The Einstein normaliser as a corrected tree value

Machine checks for this section: [verification/v1_e8_glue.py]

The gravitational normaliser $\xi = c_3/u$ has a transparent exact form. Writing $u = \frac{1}{6\pi}\left[1 + \frac{9}{128\pi^3}\right]$,

$$\boxed{\xi = \frac{c_3}{u} = \frac{3}{4}\cdot\frac{1}{1+\dfrac{9}{128\pi^3}} = 0.748303} \qquad \text{[E]}. \tag{38}$$

So $\xi_{\rm tree} = \frac{3}{4} = \dim\mathfrak{g}_{\rm SM}/\dim S^+ = 12/16$ is the exact leading value (the inverse of the $\frac{4}{3} = \dim S^+/\dim\mathfrak{g}_{\rm SM}$ family/gauge compression that sits in the seed $u_{\rm base} = \frac{4}{3}c_3$, so $\xi_{\rm tree}\, u_{\rm base} = c_3$), and the deviation is precisely the small topological seed correction $9/(128\pi^3)$, i.e. 0.226%. Gravity does not pick up a mysterious factor: first comes 3/4, then the seam defect shifts it minimally.

8 Λ-anchored metrology closure, and two anchors

Machine checks for this section: [verification/v60_lambda_metrology_branch.py]

The absolute SI scale is a calibration, not a compiler output. The dimensionless vacuum quotient lets one anchor it to the observed geometric curvature without circularity.

Theorem 1 (Λ-anchored metrology closure)**.** With $\lambda_\Lambda := \rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2}e^{-2\alpha_\star^{-1}}$ and the observed geometric curvature $\Lambda_{\rm geom}^{\rm obs} > 0$,

$$\bar{M}_{\rm Pl}^2 = \frac{\Lambda_{\rm geom}^{\rm obs}}{\lambda_\Lambda}, \qquad G_N^{\rm TFPT} = \frac{c^3}{8\pi\hbar}\frac{\lambda_\Lambda}{\Lambda_{\rm geom}^{\rm obs}} = \frac{3c^3}{32\pi^3\hbar\,\Lambda_{\rm geom}^{\rm obs}}\,e^{-2\alpha_\star^{-1}}.$$

Proof. $\bar{M}_{\rm Pl}^2 = 1/(8\pi G_N)$ and $\Lambda_{\rm geom} = \rho_\Lambda/\bar{M}_{\rm Pl}^2$; the branch sets $\rho_\Lambda = \lambda_\Lambda\bar{M}_{\rm Pl}^4$, so $\Lambda_{\rm geom} = \lambda_\Lambda\bar{M}_{\rm Pl}^2$, giving $\bar{M}_{\rm Pl}^2$ and then $G_N = \lambda_\Lambda/(8\pi\Lambda_{\rm geom}^{\rm obs})$; restoring $c^3/\hbar$ closes it. □

Non-circularity ([O], essential). The anchor must be the curvature $\Lambda_{\rm geom}^{\rm obs}$ (units m^{-2}), *not* the SI vacuum energy density $\rho_\Lambda^{\rm obs} = \Lambda_{\rm geom}c^4/(8\pi G_N)$, which already contains G_N. The curvature follows from $\Lambda_{\rm geom}^{\rm obs} = 3\Omega_\Lambda H_0^2/c^2$.

Two independent anchors (crosscheck with teeth). The metrology can be pinned in two ways that should agree:

1. *Electroweak anchor*: the dimensionless $G_N v_{\rm phys}^2 = 4.06717 \times 10^{-34}$ fixes G_N once $v_{\rm phys}$ is calibrated non-gravitationally.
2. *Λ-curvature anchor* (this note): $G_N = \frac{3c^3}{32\pi^3\hbar\Lambda_{\rm geom}^{\rm obs}}e^{-2\alpha_\star^{-1}}$.

With the example anchor $H_0 = 67.36\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.6847$ ($\Lambda^{\rm obs}_{\rm geom} = 1.08914 \times 10^{-52}\,\mathrm{m^{-2}}$) the Λ anchor gives

$$G_N^{\rm TFPT} = 6.65071 \times 10^{-11}\,\mathrm{m^3 kg^{-1} s^{-2}}, \qquad G_N^{\rm TFPT}/G_N^{\rm CODATA} - 1 = -3.5 \times 10^{-3}. \tag{39}$$

Closure algebra [E]; SI anchor and numerical match [C] (the cosmological anchor is model dependent; the Hubble tension applies). EW crosscheck: the Λ-pinned $\bar{M}_{\rm Pl} = 2.43964 \times 10^{18}\,\mathrm{GeV}$ gives $v_{\rm geo} = 242.6\,\mathrm{GeV}$, needing $Z_{\rm EW} = (246.22/242.61)^2 = 1.030$.

Prefactor branch ([O]). This note uses $\Lambda_{\rm IR}/M_{\rm Pl}^4 = \delta_{\rm top} e^{-2\alpha_\star^{-1}}$. A competing prefactor $2c_3 e^{-2\alpha_\star^{-1}}$ would differ by $2c_3/\delta_{\rm top} = 661.5$ and misscale G_N by the same factor; the branch must be versioned.

Mandatory Λ-layer typing ([O]). Three distinct Λ objects must never be conflated:

$$\underbrace{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}}_{\text{vacuum density quotient (dimensionless)}} \neq \underbrace{\frac{\Lambda_{\rm IR}}{M_{\rm Pl}^4} = \delta_{\rm top} e^{-2\alpha_\star^{-1}}}_{\text{seam-determinant / IR scale (dimensionless)}} \neq \underbrace{\Lambda^{\rm obs}_{\rm geom} = \frac{3\Omega_\Lambda H_0^2}{c^2}}_{\text{observed curvature } [\mathrm{m^{-2}}]} .$$

Only $\Lambda^{\rm obs}_{\rm geom}$ carries a dimension and is the non-circular anchor. Keeping the three typed protects the decuple bridge and the metrology closure from a spurious prefactor objection.

Measurement-near (physical) decuple bridge. Since $v_{\rm phys} = v_{\rm geo}\sqrt{Z_{\rm EW}}$ (charged-current residue, not a fit), the decuple bridge in the observable vev reads

$$\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} \left[\frac{v_{\rm phys}}{5\beta_{\rm rad}^2 \sqrt{Z_{\rm EW}}\, \bar{M}_{\rm Pl}} \right]^{10}. \tag{40}$$

This is the same identity, written in the low-energy charged-current scale rather than the geometric source scale; both forms should appear side by side. [E]/[C]

9 The landscape: $g_{\rm car} = 5$ is overdetermined

Machine checks for this section: [verification/v6_bootstrap.py] [verification/v14_carrier_uniqueness.py]

Varying the carrier rank and rebuilding every constant from the same machinery (c_3 fixed), the EM-closure value $\alpha^{-1}(g_{\rm car})$ is monotone and meets the observed value only at $g_{\rm car} = 5$:

$g_{\rm car}$	$N_{\rm fam}$	$\dim S^+$	$\Omega_{\rm adm}$	$10b_1$	$\alpha^{-1}(g_{\rm car})$	verdict
3	2	4	8	7	258.15	wrong α, wrong family count
4	2.5	8	20	17	187.31	non-integer families
5	**3**	**16**	**48**	**41**	**137.036**	**SM, 3 families, observed α**
6	3.5	32	112	97	101.30	non-integer families
7	4	64	256	225	75.61	wrong α

Several distinct requirements converge on $g_{\rm car} = 5$: integer families ($g_{\rm car}$ odd); three families $(g_{\rm car}+1)/2 = 3$; the observed α^{-1} on the monotone curve; the SM structure $(b, s) = (3, 2)$, $\dim S^+ = 16$; and $\Omega_{\rm adm} = 48$, $10b_1 = 41$. A clean algebraic shadow: $\frac{g(g+1)}{2} = 2^{g-1} - 1$ has $g = 5$ as its non-trivial solution ([C]).

10 The E_8 carrier atlas — stated unambiguously

Machine checks for this section: [verification/v5_e8_cascade.py]

In the series E_8 is a *downstream scale grammar*, not a primitive cause: the boundary kernel fixes α, the carrier rank, families and admissibility without E_8 upstream. Three levels.

10.1 Level 1 — original E_8 data of the series ([E])

$$|R(E_8)| = 248 - 8 = 240 = 5 \cdot 48 = 5\,\Omega_{\rm adm} = \mathrm{lcm}(\Omega_{\rm adm}, D_{\rm start}) = \mathrm{lcm}(48, 60), \tag{41}$$

$$D_{\rm start} = g_{\rm car} \cdot \dim \mathfrak{g}_{\rm SM} = 5 \cdot 12 = 60, \qquad \gcd(\Omega_{\rm adm}, D_{\rm start}) = 12 = \dim \mathfrak{g}_{\rm SM}, \tag{42}$$

$$\frac{|R(E_8)|}{D_{\rm start}} = \frac{240}{60} = 4 = \frac{\Omega_{\rm adm}}{\dim \mathfrak{g}_{\rm SM}}, \qquad 2880 = D_{\rm start}\,\Omega_{\rm adm} = |R(E_8)| \cdot \dim \mathfrak{g}_{\rm SM}, \tag{43}$$

$$\kappa_{E_8} = \frac{\gamma_{\rm car}}{\ln(248/60)} = 0.58723, \qquad X(n, r, I_1) = \frac{\bar{M}_{\rm Pl}}{8} \sin^2\theta_{13} \Big(\frac{60 - 2n}{60}\Big)^{\kappa_{E_8}} e^{-(8-r)/64} e^{-12\pi I_1}. \tag{44}$$

Coxeter/gauge typing of the lcm–gcd pair. Writing $D_{\rm start} = 60 = 2h_{E_8}$ and $\Omega_{\rm adm} = 48 = 6\,\mathrm{rank}(E_8)$, the relations become $\gcd(2h_{E_8}, 6r_{E_8}) = \dim \mathfrak{g}_{\rm SM} = 12$ and $\mathrm{lcm}(2h_{E_8}, 6r_{E_8}) = |R(E_8)| = 240$: the local SM algebra is the gcd of the Coxeter double-period and the $\mathbb{Z}_6$ rank-occupancy, and the E_8 root count is their lcm. Likewise the EW exponent reads $I_1^{\rm EW} = \frac{41}{48} = \frac{h_{E_8}}{36} + \frac{1}{6r_{E_8}} = \frac{5}{6} + \frac{1}{48}$ (Coxeter principal part plus a rank singlet). [E] (arithmetic) / [C] (typing).

10.2 Level 2 — new exact counting identities ([E])

$$|R(E_8)| = 240 = 16 \cdot 15 = \dim S^+\,(\dim S^+ - 1), \tag{45}$$

$$\dim E_8 = 248 = 8 \cdot 31 = (g_{\rm car} + N_{\rm fam})(2^{g_{\rm car}} - 1), \tag{46}$$

$$|R(E_8)| = 8\,(31 - 1) = \mathrm{rank}(E_8)\,(2^{g_{\rm car}} - 2), \qquad \mathrm{rank}(E_8) = 8 = g_{\rm car} + N_{\rm fam} = 5 + 3, \tag{47}$$

$$10\,b_1 = 41 = \frac{|R(E_8)|}{6} + 1, \qquad I_1^{\rm EW} = \frac{41}{48} = \gamma_{\rm car} + \frac{1}{\Omega_{\rm adm}}, \tag{48}$$

with $31 = 2^{g_{\rm car}} - 1$ the number of non-empty occupation words of the $[5, 4, 2]$ carrier code.

10.3 Level 2.5 — the even-flip transition atlas (a falsifiable program)

Identify S^+ with the even-weight binary code on five bits. A transition $x \to y \neq x$ has flip $F = x \oplus y$, even and nonempty, so $|F| \in \{2, 4\}$; per state there are $\binom{5}{2} + \binom{5}{4} = 10 + 5 = 15$ flips, giving

$$240 = 16 \cdot (10 + 5) = \underbrace{160}_{|F|=2} + \underbrace{80}_{|F|=4}, \tag{49}$$

and the 120 undirected flips match the 120 E_8 root *pairs* ([E], counting). This makes the atlas conjecture *testable*: a genuine root map must send each flip to a norm-2 vector in $\mathbb{R}^8$ with inner products in $\{-2, -1, 0, 1, 2\}$ reproducing the E_8 profile (per root: 1 self, 56 at $+1$, 126 orthogonal, 56 at -1, 1 anti).

Honest obstruction ([O]). The flip atlas splits $160 + 80$ by flip size; the standard E_8 (D_8) split is $112 + 128$. Since $160 + 80 \neq 112 + 128$, no flip-size-preserving map is an E_8 root system; a metrization, if it exists, must mix flip sizes. Until a norm-2, E_8-profile embedding is exhibited, E_8 stays a *counting* atlas, not a derived root origin.

Sharper two-stage audit (Coxeter before Weyl). A full root-system isomorphism is hard; a much earlier, cheaper test is the Coxeter spectrum. *Stage A:* find a natural order-30 operator $C_{\rm TFPT}$ on the even-flip atlas (or its 8-dimensional primitive-character space) with characteristic polynomial $\chi_{C_{\rm TFPT}}(t) = \Phi_{30}(t)$. *Stage B (only if A holds):* the norm-2, E_8-profile, Weyl-closure metrization. Stage A can fail or carry far sooner than B, so it is the right first falsifier.

Stage A is closed on the primitive-character space ([`verification/v531_coxeter_tensor_stage_a.py`]). The explicit integer operator

$$T_{30} = \mathrm{Comp}(\Phi_5) \otimes \mathrm{Comp}(\Phi_6)$$

(the 4×4 carrier clock of v418 tensor the 2×2 family/CP hexagon) has *exact* order 30 ($T_{30}^{30} = I$, $T_{30}^{k} \neq I$ for $1 \leq k < 30$) and $\chi_{T_{30}} = \Phi_{30} = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1$ exactly: by CRT ($\gcd(5,6) = 1$) its eigenvalues are the products (primitive 5th)×(primitive 6th) = exactly the eight primitive 30th roots — the carrier clock times the family clock *is* the Coxeter clock, as one integer operator. Its traces are the Ramanujan sums, $\mathrm{tr}(T_{30}^{k}) = c_{30}(k) = c_5(k)\,c_6(k)$ for all k, with the signed divisor table $(-1, 1, 2, 4, -2, -4, -8, 8)$ at $k = (1, 2, 3, 5, 6, 10, 15, 30)$ and the atom readout $(|\operatorname{tr} T^3|, |\operatorname{tr} T^5|, |\operatorname{tr} T^{15}|) = (2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \operatorname{rank} E_8)$ — eight integer traces type the clock *without* diagonalisation, a cheap kill test for any flip-operator candidate (honest scope: Φ_{15} passes the unsigned triple since $\Phi_{30}(t) = \Phi_{15}(-t)$; the *signed* table is the discriminator, and $\Phi_{16}, \Phi_{20}, \Phi_{24}$ die on the triple already). [E] *What stays open ([O]):* a canonical intertwiner from the concrete even-flip atlas onto this primitive space (`COX.FLIP.INTERTWINER`), and Stage B.

Audit only — *non-load-bearing* speculation: the $\dim 496$ **heterotic count [O].** *This box feeds no derivation; it is an arithmetic coincidence kept for completeness only, under the no-free-pattern rule.* The even half S^+ and odd half S^- of the exterior algebra each have dimension $2^{g_{\mathrm{car}}-1} = 16$ and each carry $16 \cdot 15 = 240$ transitions. Together the full carrier ($2^{g_{\mathrm{car}}} = 32$ states) gives

$$\dim S^+(2^{g_{\mathrm{car}}} - 1) = 16 \cdot 31 = 496 = \dim(E_8 \times E_8) = \dim SO(32) = \binom{32}{2} \qquad \text{(count)},$$

i.e. the two halves read as two E_8 root atlases, all $\binom{32}{2} = 496$ pairs as $SO(32)$ — the two anomaly-free heterotic gauge dimensions. The *dimension* equality is exact; that the carrier algebra *is* either gauge group is unproven and not used anywhere. [O] (audit-only). The same fingerprint recurs at checksum level: the dual products of the E_8 invariant degrees sum to $744 = 3 \cdot 248$ (the j-function constant), and the identical construction for D_{16} gives $1488 = 3 \cdot 496 = 2 \cdot 744$ ([`verification/v532_e8_degree_modular_checksum.py`]) — the checksum measures the common heterotic rank-16/Coxeter-30 structure, *not* E_8 uniqueness. [E] (audit-only).

10.4 Level 2.7 — the carrier Coxeter theorem (the strongest new bridge)

Use the universal root-system identity $|R| = \mathrm{rank} \cdot h$ (Coxeter number h). With the carrier values $\mathrm{rank}(E_8) = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$ and $|R(E_8)| = \dim S^+(\dim S^+ - 1) = 240$,

$$\boxed{h_{E_8} = \frac{|R(E_8)|}{\mathrm{rank}(E_8)} = \frac{240}{8} = 30 = N_{\mathrm{fam}} \binom{g_{\mathrm{car}}}{2} = N_{\mathrm{fam}}\, A_\Lambda = 2 g_{\mathrm{car}} N_{\mathrm{fam}} = 2^{g_{\mathrm{car}}} - 2} \qquad \text{[E]}. \quad (50)$$

The Coxeter number of E_8 is exactly three families times the cosmological pair action: each family contributes one Λ-decuple ($A_\Lambda = \binom{5}{2} = 10$) to the Coxeter period. Equivalently the action ladder is a Coxeter ladder,

$$A_H = \frac{h_{E_8}}{2N_{\mathrm{fam}}} = 5, \qquad A_\Lambda = \frac{h_{E_8}}{N_{\mathrm{fam}}} = 10, \quad (51)$$

so Hubble reads half a Coxeter period per family and Λ a full Coxeter period per family. [C]

Rank–Coxeter coupling. Equivalently $\dim S^+ = 2\,\mathrm{rank}(E_8) = 16$ and $h_{E_8} = 2(\dim S^+ - 1) = 2 \cdot 15 = 30$, so $\dim E_8 = (g_{\mathrm{car}} + N_{\mathrm{fam}})(h + 1) = 8 \cdot 31$. The two counting forms $16 \cdot 15 = 8 \cdot 30 = 240$ coincide because $\dim S^+ = 2r$. Dropping ν^c sends $\dim S^+ \to 15$, which breaks *both* $\dim S^+ = 2r$ and $\dim S^+(\dim S^+ - 1) = 240$: ν^c is the state that closes the rank–Coxeter coupling, not an optional singlet. [E]/[C]

10.5 Level 2.8 — positive-root moment: $|R^+(E_8)| = \mathrm{Tr}_{S^+} X^2 = 5!$

The number of positive E_8 roots equals the master hypercharge moment:

$$\boxed{|R^+(E_8)| = 120 = \mathrm{Tr}_{S^+} X^2 = 5!} \qquad \text{[E]}, \quad (52)$$

so the second hypercharge moment of one family *is* the positive-root half-system. The whole integer chain then reads off $|R^+|$:

$$\sum_{f,j} L_{f,j} = \frac{|R^+(E_8)|}{3} = 40, \qquad 10b_1 = 1 + \frac{|R^+(E_8)|}{3} = 41, \qquad \delta_2 = 4!\,|R^+(E_8)|\,c_3^8. \tag{53}$$

The $\mathbb{Z}_6$ slice is then $|R(E_8)|/6 = \frac{|R^+|}{3} = 40 = \Omega_{\rm adm}\gamma$: one $\mathbb{Z}_6$ phase slice of the E_8 root atlas is exactly the full flavor transport budget. [E] (arithmetic) / [C] (interpretation).

10.6 Level 2.9 — the 2·3·5 Coxeter cyclotomic compiler (meta-theorem)

The three hard carrier atoms — weak rank 2, color rank / family count 3, carrier rank $g_{\rm car} = 5$ — generate the E_8 data *cyclotomically*, without taking E_8 as input. Set

$$\boxed{h := 2 \cdot 3 \cdot g_{\rm car} = 30, \qquad r := \varphi(h) = \varphi(30) = 8 = g_{\rm car} + N_{\rm fam},} \tag{54}$$

where h is the Coxeter number and $r = \varphi(2{\cdot}3{\cdot}5) = 8$ the rank. Then the standard $|R| = rh$, $\dim = r(h{+}1)$ give

$$|R(E_8)| = rh = 8 \cdot 30 = 240, \qquad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \qquad h+1 = 2^{g_{\rm car}} - 1 = 31. \tag{55}$$

Moreover the E_8 *exponents* are exactly the primitive residues modulo $h = 30$:

$$\mathrm{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad \sum_{m \in (\mathbb{Z}/30)^\times} m = 120 = \mathrm{Tr}_{S^+}\, X^2 = |R^+(E_8)|, \tag{56}$$

so the second hypercharge moment equals the sum of the E_8 exponents and the positive-root count. The $\mathbb{Z}_6 \times \mathbb{Z}_5 \simeq \mathbb{Z}_{30}$ tact has $\varphi(30) = 8$ primitive characters, whose phases $e^{2\pi i m/30}$ are the Coxeter eigenphases of E_8; its characteristic polynomial is the 30th cyclotomic polynomial

$$\chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1. \tag{57}$$

Reading: E_8 here is not primarily a root count but the *Coxeter cyclotomic readout of the* $2, 3, 5$ *carrier*. [E] (arithmetic) / [C] (interpretation).

10.7 Level 3 — conjectural geometric readings ([C])

Conjecture 1 (Carrier transition atlas)**.** $R(E_8) \cong \{(a, b) \in S^+ \times S^+ : a \neq b\}$ as a structure-bearing transition set, since $16 \cdot 15 = 240$. *Open:* a structure-preserving map (roots, lengths, Weyl data), not only the count.

Conjecture 2 (Least common refinement / rank-8 over the exhausted carrier)**.** $240 = \mathrm{lcm}(\Omega_{\rm adm}, D_{\rm start})$ reads E_8 as the least common refinement of the occupancy lattice and the $\mathbb{Z}_6$ phase lattice; $248 = (g_{\rm car} + N_{\rm fam})(2^{g_{\rm car}} - 1)$ reads it as a rank-8 structure over the 31 non-empty carrier words.

Two routes to E_8 — keep their status separate. Levels 1–2 are exact arithmetic. There are now *two* distinct routes to E_8, and they must not be conflated:

- **Glue route (closed, [E]):** $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is built explicitly in Section 3 ($\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index 4, 240 roots of norm 2, simple-root Gram det 1). This closes the former "blind 240-metrisation".
- **Direct flip-atlas route (open, [O], optional bonus):** the even-flip transition set $240 = 160 + 80$ is *not* a flip-size-preserving E_8 map, since $160 + 80 \neq 112 + 128$. A direct structure-preserving metrisation of the flip graph remains open — a nice-to-have, not load-bearing, because the glue route already delivers E_8.

So E_8 is closed via the glue; only the direct flip atlas is open.

11 $\mathbb{Z}_6$ monodromy, flavor and CP readouts

Machine checks for this section: [verification/v88_cp_phase_audit.py]

A single $\mathbb{Z}_6$ motif recurs in the quotient $G_{\rm phys} = (SU(3)\times SU(2)\times U(1)_Y)/\mathbb{Z}_6$, the transport cusps $\{1,(2/3)^6,(1/3)^6\}$, $\delta_{\rm ph} = z_\star^{1/6}$, and the CP phases

$$\delta_{\rm CKM}^{(0)} = \frac{\pi}{3}, \quad \delta_{\rm CKM} = \frac{\pi}{3} + 3\lambda_C^2 = 1.19823\,\text{rad}, \quad \delta_{\rm CP}^{\rm PMNS} = \frac{4\pi}{3} = 240^\circ. \tag{58}$$

From the documented inputs $s_{12} = \lambda_C$, $s_{23} = u/(1+\lambda_C)$, $s_{13} = \lambda_C^3/3$ one gets the new derived readouts $|V_{td}| = 0.00908$, $|V_{ts}| = 0.04264$, $J_{\rm CKM} = 3.33\times10^{-5}$. On the neutrino side, with $\sin^2\theta_{13} = e^{-5/6}u = 0.02311$, $\sin^2\theta_{23} = 0.4557$, $\delta_{\rm CP} = 4\pi/3$ and the solar candidate $\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2} = 0.30675$, a near-maximal $|J_{\rm CP}^\nu| \approx 0.030$ (sharp readout for DUNE/Hyper-Kamiokande). [C]

11.1 CP as triple seed variance; a closed observable quadrilateral

The Cabibbo angle is the Bernoulli width of the seed. The *exact* CKM phase is the family transport holonomy (document 3), $\delta_{\rm CKM} = \arg\operatorname{Tr}\Pi_\triangle(\nabla_F^\star)$; on the small-area branch this has the compact readout form (with an $O(\lambda_C^4)$ remainder)

$$\lambda_C = \sqrt{u(1-u)}, \qquad \boxed{\delta_{\rm CKM} - \frac{\pi}{3} = N_{\rm fam}\,u(1-u) = 3\lambda_C^2} \qquad \text{[E]/[C] (readout; exact = holonomy).} \tag{59}$$

The coefficient 3 is the family number $N_{\rm fam}$: weak CP is the $\mathbb{Z}_6$ base angle $\pi/3$ plus $N_{\rm fam}$ times the seed (Bernoulli) variance. The $\mathbb{Z}_6$ base angle now has an A_3 root: since $\mathbb{Z}_6 = |R^+(A_3)| = 6$ (the positive-root hexagon of the family algebra), the CP phases are integer steps of the A_3 holonomy quantum $2\pi/|R^+(A_3)|$,

$$\boxed{\delta_{\rm CKM} = \frac{2\pi}{|R^+(A_3)|} + N_{\rm fam}\,u(1-u), \qquad \delta_{\rm CP}^{\rm PMNS} = 4\cdot\frac{2\pi}{|R^+(A_3)|} = \frac{4\pi}{3}} \qquad \text{[E]/[C].} \tag{60}$$

So weak CP = (A_3 positive-root phase step) + (family count × seed variance), while strong CP is the *determinant* sector, erased. The solar angle reads as a carrier root plus a half-seed, $\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2} = -y_- - \frac{u}{2}$ (since $-y_- = \frac{1}{3}$), so PMNS = carrier root + seed + A_3 holonomy phase. [C]Strong CP is killed by the determinant selector ($\arg\det M_u = \arg\det M_d = 0$, $\bar\theta = 0$), while weak CP survives as the seed-deformed $\mathbb{Z}_6$ holonomy: CP is not forbidden, only *determinant* CP is. [C] — the strong-CP closure lives on the admissible QFT branch and is *conditional* on the analytic hypotheses of the QFT closure (document 3, Part III: reflection positivity, temperedness, clustering, mass gap, OS reconstruction); its status is therefore weaker than the discrete lattice identities, even though the semantics "determinant CP forbidden, holonomy CP allowed" is robust. Since $u = \Omega_b + \beta_{\rm rad}$ exactly, eliminating the seed closes the four UV-shadow observables into one cycle:

$$\lambda_C^2 = (\Omega_b+\beta_{\rm rad})(1-\Omega_b-\beta_{\rm rad}), \quad \sin^2\theta_{13} = e^{-5/6}(\Omega_b+\beta_{\rm rad}), \quad \Omega_b = \Big(1-\tfrac{1}{4\pi}\Big)e^{5/6}\sin^2\theta_{13}. \tag{61}$$

One measured member of $\{\lambda_C, \theta_{13}, \Omega_b, \beta_{\rm rad}\}$ fixes the other three. [E] (the four rows remain operationally distinct readout classes).

12 Mass hierarchy as a hexagon with one extra turn

Machine checks for this section: [verification/v18_quark_yukawa.py] [verification/v20_lepton_c_derivation.py]

The transport length packets are

$$L_u = (7,3,0), \qquad L_d = (7,5,2), \qquad L_e = (8,5,3), \qquad \sum_{f,j} L_{f,j} = 40 = \Omega_{\rm adm}\,\gamma_{\rm car} = 10b_1 - 1. \tag{62}$$

Modulo 6 they lie on related triples of the C_6 hexagon; the first generation carries one extra full turn $(u, d : 1+6;\ e : 2+6)$, costing $\lambda_Y^6 = (u(1-u))^3 \approx 1.28 \times 10^{-4}$ and explaining why the first generation is so light — "one turn more", not a free fit. [C]*Flagged proximity ([O], not exact):* the full-turn cost is close to the topological seam defect, $\lambda_C^6/\delta_{\text{top}} = 1.061$ ($\sim 6\%$), hinting that one flavor winding $\approx$ one seam-defect unit $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$ — a structural smell, not a theorem. Writing the length matrix $L = R + W$ with $R = L \bmod 6$ the residue shadow and W the full-winding part,

$$L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}}_{R} + 6 \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{W}, \qquad \sum R = 22,\ \sum W = 18 = 3 \cdot 6,\ \sum L = 40,$$

and the column sums are $(22, 13, 5)$ with $S_1 = 22 = \sum R$ and $S_2 + S_3 = 18 = \sum W$. So the first generation is light because it stores the *entire* residue budget plus the three full C_6 windings; generations 2 and 3 are the winding complement. [E] (arithmetic).

12.1 Lepton flavor decuple: the charged-lepton determinant carries u^{10}

The source masses on the v_{geo} branch are $(\hat{m}_e, \hat{m}_\mu, \hat{m}_\tau) = \frac{v_{\text{geo}}\pi}{\sqrt{2}}(\frac{16}{7}u^5, \frac{4}{3}u^3, \frac{7}{6}u^2)$. Their product is a pure decuple in the seed:

$$\boxed{\frac{\hat{m}_e \hat{m}_\mu \hat{m}_\tau}{(v_{\text{geo}}\pi/\sqrt{2})^3} = \left(\tfrac{16}{7} \cdot \tfrac{4}{3} \cdot \tfrac{7}{6}\right) u^{5+3+2} = \frac{32}{9}\, u^{10}} \qquad \text{[E]}, \tag{63}$$

with $5+3+2 = g_{\text{car}} + b + s = 10 = \binom{5}{2}$. So the charged-lepton Yukawa powers $(n_e, n_\mu, n_\tau) = (5, 3, 2)$ *are* the three carrier numbers (g_{car}, b, s), and the lepton determinant carries the same decuple (u^{10}) that the cosmological constant carries in the gauge sector $(A_\Lambda = 10)$: Λ and the charged-lepton determinant are two realisations of the same degree-$\binom{5}{2}$ sector.

Sum–product reading. The lepton powers $(5, 3, 2)$ encode *both* decuple numbers at once: their sum is the Λ action, their product is the E_8 Coxeter number,

$$5 + 3 + 2 = 10 = A_\Lambda, \qquad 5 \cdot 3 \cdot 2 = 30 = h_{E_8} = 2 \cdot 3 \cdot 5, \tag{64}$$

and $(5, 3, 2)$ is the prime factorisation of $h_{E_8} = 30$. So $\det M_\ell$, ρ_Λ and the E_8 Coxeter period read the same carrier core additively $(= 10)$ and multiplicatively $(= 30)$. [E]/[C]

12.2 Why the top Yukawa is ≈ 1: the $L = 0$ anchor (Froggatt–Nielsen ladder)

The transport length L_f is a Froggatt–Nielsen-type charge with expansion parameter λ_C:

$$\boxed{y_f \simeq \lambda_C^{L_f} \times \Lambda_f \ (\Lambda_f = \mathcal{O}(1)), \qquad y_t \simeq 1 \text{ because } L_t = 0} \qquad \text{[E]/[C]}. \tag{65}$$

The top sits at $L_t = 0$ (the *unwound* carrier direction), so $y_t = \Lambda_t \simeq 1$ ($\sqrt{2}\, m_t/v = 0.991$); every other charged fermion is suppressed by integer powers of $\lambda_C = 0.2244$. Numerically (with $\mathcal{O}(1)$ sector prefactors $\Lambda_f \in [0.43, 1.07]$):

f	t	b	τ	c	μ	s	u, d	e
L_f	0	2	3	3	5	5	7	8
$y_f/\lambda_C^{L_f}$	0.99	0.48	0.90	0.65	1.07	0.94	$\sim$ 0.4–0.9	0.46

So the entire charged-fermion hierarchy is one λ_C-ladder anchored by the top at $L = 0$; the lightness of the first generation $(L = 7, 8)$ and the heaviness of the top $(L = 0)$ are the two ends of the same integer winding spectrum. This is the standard Froggatt–Nielsen mechanism with λ_C as the flavon ratio and L_f as the FN charge — a clean bridge between TFPT transport and an established flavor mechanism.

13 Inflation interface

Machine checks for this section: [verification/v7_gravity_cosmo.py] [verification/v86_nstar_reheating.py] [verification/v494_cosmo_running_mu_record.py]

The documented scalaron is Starobinsky R^2; with the pivot $N_\star = 57.1$:

$$n_s = 1 - \frac{2}{N_\star} = 0.96497, \qquad r = \frac{12}{N_\star^2} = 0.00368,$$
$$\frac{M_{\rm scal}}{\bar{M}_{\rm Pl}} = \sqrt{8\pi}\, c_3^4 = c_3^{7/2} = (8\pi)^{-7/2} \;\Rightarrow\; M_{\rm scal} \approx 3.06 \times 10^{13}\,\text{GeV}. \tag{66}$$

n_s matches Planck; $r = 0.0037$ is the falsifiable target for LiteBIRD/CMB-S4.

Amplitude (status-clean). The scalaron mass is a pure seam power,

$$\boxed{\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{r-1} = c_3^{\Omega_{\rm adm} - 10 b_1} = c_3^7} \qquad \text{[E]}, \tag{67}$$

an *exact* identity (the exponent $7 = 48 - 41$ is the post-flavor, post-Higgs occupancy deficit). The cosmology interface reduces the geometric gravity action to an FRW R^2/scalaron branch in which $M_{\rm scal}$ *is* the Starobinsky mass M; the identification is therefore derived in that sector, not posited. With the standard large-N amplitude $A_s = \frac{N_\star^2}{24\pi^2}(M/\bar{M}_{\rm Pl})^2$ this gives

$$A_s = \frac{N_\star^2\, c_3^7}{24\pi^2} = \frac{N_\star^2\, \delta_2}{8640\pi} = 2.17 \times 10^{-9} \qquad \text{[C]/[O]}, \tag{68}$$

matching Planck ($\simeq 2.1 \times 10^{-9}$); $\delta_2 = 2880\, c_3^8$ is the second seam defect. The seam-power form is exact [E]; the amplitude is then determined *within* the R^2 sector, the only residual being the geometric-gravity (Hodge) closure on which that sector rests [C]. *Convention note:* $A_s \propto N_\star^2$, so the quoted 2.17×10^{-9} uses the pivot $N_\star = 57.1$; the alternative seed pivot $N_\star = 3/u - 1 = 55.4$ gives 2.05×10^{-9}. A number should only be quoted with $N_\star$ fixed.

Running (consistency relation). The same attractor also fixes the *running* of the tilt parameter-free: $\alpha_s = {\rm d}n_s/{\rm d}\ln k = -2/N_\star^2 = -r/6$ and $\beta_s = -4/N_\star^3$ at leading order; the exact slow roll on the Starobinsky potential gives $\alpha_s = -7.09 \times 10^{-4}$, $\beta_s = -2.70 \times 10^{-5}$ at the reheating point $N_\star$=51.4 (leading-order band $[-8.0, -5.6] \times 10^{-4}$ over $N_\star \in [50, 60]$). Current data are consistent (Planck 2018 -0.0045 ± 0.0067: $+0.57\sigma$; P-ACT-LB $+0.0062 \pm 0.0052$: -1.33σ), and since plateau potentials give $\alpha_s < 0$, any robust *positive* running would kill the branch. [C] [verification/v494_cosmo_running_mu_record.py]

14 Further simple bridges

Machine checks for this section: [verification/v74_compiler_micro_lemmas.py]

- **Dark-energy equation of state $w = -1$ (exact).** The TFPT Λ is a genuine vacuum constant $\rho_\Lambda = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}} \bar{M}_{\rm Pl}^4$, not a rolling field, so $w = -1$ identically (no quintessence). A robust measurement $w \neq -1$ would falsify the Λ-branch. [E] (structural).
- **Effective neutrino number $N_{\rm eff} \simeq 3$.** Three families ($N_{\rm fam} = 3$) give three light neutrinos and the standard $N_{\rm eff} \simeq 3.04$. [E].
- **Top Yukawa $\simeq 1$** as the $L = 0$ anchor (above): a bridge to the Froggatt–Nielsen flavor mechanism.
- *Flagged (not clean):* $m_H/v = 0.509$ is near $\frac{1}{2}$ (1.7%); $m_W/m_Z = 0.881$ matches the running $\sqrt{1 - \sin^2\theta_W(M_Z)}$ (not the tree $\sqrt{5/8} = 0.79$). No clean closed bridge for the Higgs/W/Z masses yet. [O]

15 Closures that the architecture rests on

Machine checks for this section: [verification/v36_spectral_action_g2.py] [verification/v4_flavor_matrix.py]

Five results close or sharpen the discrete core. They are *proved in this document set* — the φ_0^{ret}-ladder, the H2 realisation and the strong-CP/dynamics gap in document 2 (`tfpt_2_standard_model`), the $E_6{\times}A_2$ shadow in document 3 (`tfpt_3_e8_audit_bootstrap`) — and are collected here because the architecture above depends on them.

1. **The φ_0^{ret}-ladder (whole spectrum, one formula).** Every fermion mass is $\hat{m}_{f,j} = \frac{v_{\mathrm{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$ with $\lambda_Y = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}$, and the closed evaluation gives the universal form $\lambda_Y^L \Lambda = \pi\, c\, (\varphi_0^{\mathrm{ret}})^k$ — the nine Λ are hexagonal-resolvent outputs (not free numbers); the *charged-lepton* ones are exact, the *quark* ones are reduced to the (U_{wall}) selector. The φ_0^{ret}-order ladder is $t(0) > \{b,c,\tau\}(2) > \{s,\mu\}(3) > \{u,d\}(4) > e(5)$. [E] (leptons exact, quark ratios closed v49/v71) / [O] (absolute quark scale)
2. **H2 reduced via the Mehta–Seshadri framework.** Mehta–Seshadri supplies the equivalence (stable parabolic ↔ irreducible unitary); the TFPT-specific step — "stable = transport-minimal" and the geodesic C_6 word length — is conditional on the unitarity (U) of $\nabla_F^\star$ (document 3) and the parabolic-degree→ C_6 dictionary. H2 is the sharp realisation statement: the D_4-equivariant stable structure on $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ realises the branch with $\det R = h(D_5) = 8$, principal 2-minors $(2,3,5)$. [E] (given (U))
3. **Strong CP / dynamics via the explicit gap.** $\theta_{\mathrm{eff}} = 0$ needs only γ_5-Hermiticity + polar mass + sheet involution + reflection positivity (no mass gap). The full dynamics closes on the physical sector because the C_6 transport transfer operator has the *explicit* spectrum $\{(k/3)^6\}$, giving the mass gap $\Delta = 6 \log \frac{3}{2} = 2.43$ in closed form ⇒ OS reconstruction ⇒ measure on the admissible sector. [E] (phys. sector)
4. **$E_6 \times A_2$ flavor shadow.** The residue matrix sits inside an E_8 secondary branching: $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\mathrm{fam}} = 78+8+2\cdot 27\cdot 3$, with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2$. The seven E_8 slices form an audit raster (atlas note). [O]
5. **SM completeness.** The discrete structure and dimensionless core are closed; the dimensionful EW/QCD masses $(m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s)$ are schema-level via $\mathcal{R}_{\mathrm{SM}}$, with m_H resting on the closed UV quartic $\lambda_\Phi = 1/(16\pi^2)$ (near-criticality). The *exact* solar angle θ_{12} (leading $1/N_{\mathrm{fam}} = \frac{1}{3}$) is *conditionally derived*, pending the seam-misalignment lemma — the sharpest remaining SM tension point. [C]

16 Status summary and open audit points

Machine checks for this section: [verification/v22_open_gates_audit.py]

Statement	Status	Note
Dual-root dictionary (Y, cusps, γ_{car}, φ_{base})	[E]	exact, from $\{1/2, -1/3\}$
$10b_1 = 5^2 + 4^2 = \Omega_{\mathrm{adm}}\gamma + 1$	[E]	Pythagorean carrier identity
$I_1^{\mathrm{EW}} = 41/48 = \mathrm{Var}(Y)\, b_1$	[E]	exact
$\delta_2 = 4!\cdot 5!\, c_3^8$, $\mathrm{Tr}\, X^2 = 5!$	[E]	factorial traces
$5/4 = g_{\mathrm{car}}/(g_{\mathrm{car}}-1)$ universal compression	[E]	exact
$u = \frac{4}{3}c_3 + 48c_3^4$; one-decoder identities	[E]	exact
$\xi = \frac{3}{4}/(1 + 9/128\pi^3)$	[E]	corrected tree value
Ladder $A_{\mathrm{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = 1 : g_{\mathrm{car}} : 2g_{\mathrm{car}}$	[E]/[C]	Hubble rung is a corollary
Ladder $= \binom{g_{\mathrm{car}}}{0} : \binom{g_{\mathrm{car}}}{1} : \binom{g_{\mathrm{car}}}{2}$; $2g_{\mathrm{car}} = \binom{g_{\mathrm{car}}}{2} \Leftrightarrow g_{\mathrm{car}} = 5$	[E]	new overdetermination of $g_{\mathrm{car}} = 5$

2D action lattice $S_x = a(\alpha_\star^{-1}/g_{\rm car}) + bL_0$	[E]/[C]	two integer labels per scale
Decuple bridge $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}}[v/(5\beta_{\rm rad}^2 \bar{M}_{\rm Pl})]^{10}$	[E]	exact, $10 = 2g_{\rm car} = \dim \Lambda^2 E$
Family = ladder = Pascal row $\{1,5,10\}$; $16 = 1{+}5{+}10$	[E]	central new identification
$g_{\rm car} = 5$ via $2^{g_{\rm car}-1} = \binom{g_{\rm car}}{0}{+}\binom{g_{\rm car}}{1}{+}\binom{g_{\rm car}}{2}$	[E]	Pascal closure (unique)
$\mathrm{Tr}_{S^+} X^2 = 5!$ generates $40, 41, 240, \delta_2$	[E]	master moment
$\lvert R(E_8)\rvert = 16 \cdot 5 \cdot 3$; $\dim E_8 = 240 + 8$	[E]	E_8 compression
Even-flip atlas $240 = 160 + 80$	[E]/[O]	metrization open ($\neq$ 112 + 128); Coxeter Stage A closed on the character space (v531)
$\delta_{\rm CKM} - \pi/3 = 3u(1-u)$	[E]	CP = triple seed variance
Observable quadrilateral ($u = \Omega_b + \beta_{\rm rad}$)	[E]	closed UV-shadow cycle
Power tower $v{=}x^1$, $H{=}x^5 R_H$, $\Lambda{=}x^{10} R_\Lambda$	[E]	K_5 vertices/edges
$496 = 16 \cdot 31 = \dim(E_8{\times}E_8) = \dim SO(32)$	[E]/[C]	heterotic-dimension count
$A_{\rm EW} = \alpha_\star^{-1}/\sqrt{\Delta_Y}$, $\Delta_Y = (b{+}s)^2 = 25$	[E]	discriminant action theorem
$\hat{m}_e\hat{m}_\mu\hat{m}_\tau \propto \frac{32}{9}u^{10}$; $(5,3,2) = (g_{\rm car}, b, s)$	[E]	lepton flavor decuple
$M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2}$	[E]	scalaron seam power
$\delta_{\rm CKM} - \pi/3 = N_{\rm fam}\, u(1-u)$	[E]	weak CP = $N_{\rm fam}\times$ seed variance
$L = R + W$: $\sum R = 22$, $\sum W = 18$	[E]	first generation = residue store
$\lvert R(E_8)\rvert/6 = \sum L = 40$ ($\mathbb{Z}_6$ slice)	[E]	one $\mathbb{Z}_6$ slice = flavor budget
$h_{E_8} = \lvert R\rvert/\mathrm{rank} = 30 = N_{\rm fam}\binom{5}{2} = N_{\rm fam} A_\Lambda$	[E]/[C]	carrier Coxeter theorem
$\lvert R^+(E_8)\rvert = \mathrm{Tr}_{S^+} X^2 = 120 = 5!$	[E]	positive-root moment
common decuple $\binom{5}{2}{=}A_\Lambda{=}u^{5+3+2}{=}h_{E_8}/N_{\rm fam}{=}10$	[E]/[C]	one carrier pair-decuple
$\lambda_C^6/\delta_{\rm top} = 1.061$ (winding $\approx$ seam defect)	[O]	flagged proximity (not exact)
$y_f \simeq \lambda_C^{L_f}$, top $L_t = 0 \Rightarrow y_t \simeq 1$	[E]/[C]	Froggatt–Nielsen ladder
$w = -1$ (true Λ); $N_{\rm eff} \simeq 3$	[E]	dark energy / neutrino species
3-generator lattice $\{\alpha_\star^{-1}, L_0, L_c\}$; $M_{\rm scal} = e^{-\frac{7}{2}L_c}$	[E]	seam axis $L_c = \ln(1/c_3)$
$\dim S^+ = 2\,\mathrm{rank}(E_8)$, $h_{E_8} = 2(\dim S^+ {-}1) = 2^g {-}2$	[E]	rank–Coxeter coupling (ν^c closes it)
$\gcd(2h, 6r) = \dim \mathfrak{g}_{\rm SM} = 12$, $\mathrm{lcm} = 240$	[E]/[C]	SM algebra = gcd; E_8 count = lcm
$I_1^{\rm EW} = h_{E_8}/36 + 1/(6r) = 41/48$	[E]/[C]	Coxeter part + rank singlet
$M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^{r-1} = c_3^7$ exact; $A_s = N_\star^2 c_3^7/24\pi^2$	[E]/[C]/[O]	seam power exact; A_s conditional (Starobinsky)
$h_{E_8} = 2{\cdot}3{\cdot}5 = 30$, $r = \varphi(30) = 8$	[E]/[C]	2, 3, 5 Coxeter cyclotomic compiler
$\mathrm{Exp}(E_8) = (\mathbb{Z}/30)^\times$, $\sum = 120$, $\chi_C = \Phi_{30}$	[E]/[C]	exponents = primitive residues
lepton $(5,3,2)$: $\sum{=}10{=}A_\Lambda$, $\prod{=}30{=}h_{E_8}$	[E]/[C]	sum=action, product=Coxeter
Hubble $H_0/\bar{M}_{\rm Pl} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$	[E]	exact corollary
Λ-anchored closure $G_N = \frac{3c^3}{32\pi^3\hbar\Lambda_{\rm geom}} e^{-2\alpha_\star^{-1}}$	[E]/[C]	G_N to -0.35%
$g_{\rm car} = 5$ overdetermined	[C]	monotone $\alpha^{-1}(g_{\rm car})$
$\lvert R(E_8)\rvert = 240 = 16 \cdot 15$, $248 = 8 \cdot 31$	[E]	counting identities
$E_8 \leftrightarrow$ carrier-transition map	[C]	needs structure map
$\delta_{\rm CP} = 4\pi/3$, $\theta_{12} = \frac{1}{3} - \frac{u}{2}$	[C]	PMNS readouts
$\lvert V_{td}\rvert, \lvert V_{ts}\rvert, J_{\rm CKM}, J_{\rm CP}^\nu$; n_s, r	[C]	derived readouts

Open audit points: (1) *discharged* — the EM closure $F_{U(1)}(\alpha) = 0$ is now stated explicitly (Grammar III), with existence/uniqueness and root $\alpha_\star^{-1} = 137.0359992168$ (2.9×10^{-10}, 1.9σ vs CODATA-2022); (2) keep $\gamma_{\rm car} = 5/6$ vs γ_E distinct; (3) type $M_{\rm Pl}$ vs $\bar{M}_{\rm Pl}$ and "pure" ($\mathcal{A}_\Lambda = 274.07$) vs "total" ($-\ln(\delta_{\rm top} e^{-2\alpha_\star^{-1}}) = 283.10$, $-\ln(\rho_\Lambda/\bar{M}_{\rm Pl}^4) = 276.65$) actions consistently; (4) version the Λ prefactor branch; (5) the scalaron-mass→ Starobinsky-M identification is the *only* condition for $A_s = N_\star^2 c_3^7/(24\pi^2)$ (the seam power $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^7$ itself is exact); (6) the E_8 *lattice* is now built explicitly as $(D_5 \oplus A_3) + \mu_4$-glue (Section 3), so the former "blind 240-metrisation" is closed. The

flavor status (companion parabolic-weight note) now splits cleanly: the *anchor multiset* $\{1,1,2\}$ is derived (Schur–Horn + stability) [E]; the *sector assignment* $(1,1,2)$ follows from the D_5 cusp ordering [E]/[C]; the *first-generation* $+6$ *winding* is the hierarchy convention [C]; the six *non-anchor residues* are then forced (residue sets = D_6 orbit of $\{0,1,3\}$, minus anchor, monotone) [E]. The remaining geometric input at the time of writing was (H2), the specific D_6 cusp labelling on $\mathbb{P}^1 \setminus \mu_4$ — sharply characterised by the residue-matrix spectral selector ($\det R = h(D_5)$, principal minors $(2,3,5)$), with a naive cusp-position shortcut explicitly ruled out [C]. The explicit Riemann–Hilbert construction confirms the rigid $SU(3)$ system and its A_3 Coxeter rotation. *Status update:* the H2 dictionary has since been made *exact* (hexagon = sign-twisted family spectrum, addresses pinned, margins forced, v118–v122), the deck is *derived* (v141), and the residue is the R4′ selector establishment — two line pairings (v139/v142); what stays geometric is only the parabolic realisation premise (`GATE.QGEO`, research contracts).

Honest negatives (reported to keep the discipline visible). The TFPT lepton ladder gives a Koide ratio $Q = (\sum m_\ell)/(\sum \sqrt{m_\ell})^2 = 0.6645$, close to but *not* the empirical $2/3 = 0.66667$ (-0.33%); TFPT is Koide-*like* but does not reproduce the precise relation. The baryon asymmetry $\eta_B \approx 6 \times 10^{-10}$ and the proton/electron ratio $m_p/m_e = 1836$ do *not* fall on a clean action charge or L_0 ladder; they remain outside the present grammar. These are stated so the strong items are not read as a uniform success.

17 Consolidated picture

The three nested grammars.

$$\begin{aligned}
\textbf{Carrier:}\quad & 3+2 \Rightarrow \{y_-, y_+\} = \{-\tfrac{1}{3}, \tfrac{1}{2}\} \Rightarrow Y,\ \dim S^+ = 16,\ N_{\text{fam}} = 3,\ \Omega_{\text{adm}} = 48,\ b_1 = \tfrac{41}{10},\\
\textbf{Seed:}\quad & u = \tfrac{4}{3}c_3 + 48c_3^4 \Rightarrow \{\lambda_C,\ \beta_{\text{rad}},\ \Omega_b,\ \sin^2\theta_{13}\},\\
\textbf{Action:}\quad & \alpha_\star^{-1} \Rightarrow A_{\text{EW}} = \tfrac{\alpha_\star^{-1}}{g_{\text{car}}},\ \mathcal{A}_\Lambda = 2\alpha_\star^{-1},\ \mathcal{A}_H = \alpha_\star^{-1} \Rightarrow \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \Big[\frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}}\Big]^{10}.
\end{aligned}$$

A small grammar generates many sectors: the Standard-Model packet, flavor, the electroweak hierarchy, the cosmological constant, the metrology, and the inflation/CMB interface. The strongest new statements are exact: the decuple bridge ties Λ to the tenth power of the stripped electroweak scale with $10 = 2g_{\text{car}}$; the Hubble scale is the square root of Λ; and the Einstein normaliser is the corrected tree value $\frac{3}{4}/(1 + 9/128\pi^3)$. None of this is claimed as a proof of a final theory; it is offered as a tightening of the existing architecture for your assessment.

The compiler in one line: TFPT = even carrier code + seed decoder + 2·3·5 Coxeter compiler. From the three atoms $(2, 3, g_{\text{car}}{=}5)$:

$$h_{E_8} = 2{\cdot}3{\cdot}5 = 30, \qquad r_{E_8} = \varphi(30) = 8 = g_{\text{car}}{+}N_{\text{fam}},$$

$$|R(E_8)| = rh = 240, \qquad \dim E_8 = r(h{+}1) = 8{\cdot}31 = 248,$$

$$\text{Exp}(E_8) = (\mathbb{Z}/30)^\times, \quad \textstyle\sum \text{Exp} = 120 = \text{Tr}_{S^+} X^2 = |R^+|, \quad \chi_C = \Phi_{30},$$

$$\dim S^+ = 1 + 10 + 5 = 16 = 2r, \quad A_{\text{EW}}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda = 1{:}5{:}10, \quad \mathcal{A}_H = \tfrac{h}{6},\ \mathcal{A}_\Lambda = \tfrac{h}{3},\ N_{\text{fam}} = \tfrac{h}{A_\Lambda} = 3,$$

$$\textstyle\sum L = \frac{|R^+|}{3} = 40, \quad 10b_1 = 1 + \frac{|R^+|}{3} = 41 = 1 + h\frac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}},$$

$$\det M_\ell \propto u^{10}, \qquad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} \propto \Big[\frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}}\Big]^{10}, \qquad A_{\text{EW}} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \quad \sqrt{\Delta_Y} = g_{\text{car}} = 5.$$

The common decuple (= 10): $\binom{g_{\text{car}}}{2}$ (carrier pairs) = A_Λ (cosmological action) = u^{5+3+2} (lepton determinant) = h_{E_8}/N_{fam} (Coxeter per family). Λ, the charged-lepton determinant, the E_8 Coxeter structure and the action ladder all read the *same* carrier pair-decuple. The open core is one

question: can the even-flip atlas be metrised as a genuine E_8 root system (the 160+80 vs 112+128 obstruction)? Its Coxeter half (Stage A) is closed on the primitive-character space by the tensor clock $T_{30} = \mathrm{Comp}(\Phi_5) \otimes \mathrm{Comp}(\Phi_6)$ (v531); the atlas intertwiner and the metrization (Stage B) remain.

Part II
The Coxeter–cyclotomic compiler and the explicit E_8 construction

Abstract

This short companion compresses the TFPT bridge layer to a single discrete object. The claim is that the Standard-Model family, the cosmological action ladder, the E_8 counting data, the flavor budget, inflation and the cosmological constant are *not* independent coincidences but projections of one structure: the even-weight code C^+ on $g_{\rm car} = 5$ slots, together with its $\mathbb{Z}_{30}$ Coxeter spectrum and the single seam seed $c_3 = 1/(8\pi)$. The decisive new result is that C^+ is precisely the half-spin weight orbit of $D_5 = \mathfrak{so}_{10}$ ($\mathrm{Aut}(C^+) = W(D_5)$, order 1920), that the family geometry $\mathbb{P}^1 \setminus \mu_4$ is an $A_3 = \mathfrak{su}_4$, and that E_8 is then the standard *closure algebra* of $D_5 \times A_3$ — not an input. The fusion is exact: $\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, so the four family corners μ_4 are precisely the glue code that closes $D_5 \oplus A_3$ into the unimodular E_8 lattice. The glue is metric: the discriminant-form norms $q(D_5) = \frac{5}{4}$ and $q(A_3) = \frac{3}{4}$ sum to the E_8 root norm 2 — and these are exactly the pre-existing TFPT constants $\delta_2/\delta_{\rm top}^2 = \frac{5}{4}$ and $\xi_{\rm tree} = \frac{3}{4}$. The A_3 side is explicit: the residue pairing on $H^1(\mathbb{P}^1 \setminus \mu_4)$ is the A_3 Cartan form, and the carrier corner rotation $z \mapsto iz$ acts as the A_3 Coxeter element (order 4, spectrum $\{i, -1, -i\}$). The lift is no longer a count: the 240 E_8 roots are *explicitly* built from $D_5 \oplus A_3$ plus the spinor (μ_4) glue, all of norm 2, Cartan determinant 1. Seven statements carry it; every lattice identity below is checked exactly, and the analytic metric is computed and characterised (a D_4-equivariant deformation), so no *lattice* structural step is left open in the $D_5 \times A_3$ glue route (the upstream boundary kernel, EM closure and flavor Hecke saturation remain separate open problems). Master statement: *five carrier slots give D_5, four family corners give A_3, and μ_4 glues $D_5 \times A_3$ into the E_8 atlas.*

The compression.

$$\mathrm{TFPT} = \underbrace{D_5 \text{ half-spin carrier}}_{C^+} + \underbrace{A_3 \text{ four-point family}}_{\mathbb{P}^1\setminus\mu_4} + c_3 \text{ seam seed},$$

$$E_8 = \text{closure of } D_5 \times A_3$$

with $c_3 = \dfrac{1}{8\pi}$, $g_{\rm car} = 5$, $h := 2 \cdot 3 \cdot g_{\rm car} = 30$, $r := \varphi(h) = \varphi(30) = 8$, $N_{\rm fam} = 3$, $N_\Phi = 1$, $\dim \mathfrak{g}_{\rm SM} = 12$. Equivalently $C^+ + \mathbb{Z}_{30}$ Coxeter compiler + c_3 seed; the $D_5 \times A_3$ reading is the structural lift of the same object.

Theorem 1 — the even carrier code

Let E be the rank-$g_{\rm car} = 5$ carrier, split $E = E_3 \oplus E_2$ (color $b = 3$, weak $s = 2$). The one-generation spinor packet is the even exterior algebra, equivalently the even-weight binary code

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E \;\cong\; C^+ = \{x \in \mathbb{F}_2^5 : |x| \text{ even}\}, \qquad \dim S^+ = |C^+| = 2^{g_{\rm car}-1} = 16. \quad (69)$$

The two carrier roots $y_+ = \frac{1}{2}$, $y_- = -\frac{1}{3}$ generate all hypercharges ($Y(Q_L) = y_+ + y_- = \frac{1}{6}$, $Y(u^c) = 2y_-$, $Y(e^c) = 2y_+$, $Y(\nu^c) = 0$) and the transport cusps $\{1, \frac{2}{3}, \frac{1}{3}\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}$, with $\gamma_{\rm car} = \mathrm{Tr}_E\, Y^2 = 3y_-^2 + 2y_+^2 = \frac{5}{6}$. **[E]**

Gauge-rank factorisation (`[verification/v79_review_identities.py]`). The carrier polynomial factors into the two gauge ranks directly:

$$\boxed{6Y^2 - Y - 1 = (2Y - 1)(3Y + 1)}, \qquad Y = \tfrac{1}{2}\ (s{=}2,\ SU(2)), \quad Y = -\tfrac{1}{3}\ (b{=}3,\ SU(3)),$$

i.e. $(sY - 1)(bY + 1)$ — the polynomial splits into one linear representative of each Standard-Model non-abelian factor; the 5-slot carrier *forces* the split into $SU(3) \times SU(2)$.

Hypercharge Lucas–Binet resonances [E] (mode reading [C], [`verification/v79_review_identities.py`])

The integer carrier charges $(q_+, q_-) = (3, -2)$ (the primitive trace-free pair, $X^2-X-6 = (X-3)(X+2)$, Lean `unique_carrier_pair`) generate the Lucas U-sequence

$$D_n = \frac{3^n - (-2)^n}{5}: \qquad D_1, \ldots, D_6 = 1,\ 1,\ 7,\ 13,\ 55,\ 133,$$

which is, *in order*, $(N_\Phi,\ \cdot\ ,$ scalaron exponent 7, Δ_Q=13, quark numerator 55, $\dim E_7$=133). In particular the quark ratio reads $c_u/c_d = D_5/(N_{\text{fam}}^2 D_4) = \frac{55}{117}$ — a single recurrence of the hypercharge generator, not a free integer. [E] for the arithmetic; the per-mode physical identification is a [C] cross-domain reading (only $n \le 6$, the carrier-slot range, is meaningful).

Theorem 2 — Exterior–Action Duality

On C^+ the flip $F = x \oplus y$ is even, so $|F| \in \{0, 2, 4\}$. The three relations R_0, R_2, R_4 form a symmetric association scheme (the even Hamming scheme on five bits) with

$$\boxed{\begin{array}{l}\text{valencies } (1, 10, 5) = (\dim \Lambda^0 E, \dim \Lambda^2 E, \dim \Lambda^4 E),\\ \text{spectral multiplicities } (1, 5, 10) = A_{\text{EW}}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda\end{array}} \quad \text{[E]}. \tag{70}$$

The eigenvalues (computed directly) are $A_2 : \{10^{(1)}, 2^{(5)}, (-2)^{(10)}\}$ and $A_4 : \{5^{(1)}, (-3)^{(5)}, 1^{(10)}\}$, with $A_2 + A_4 = K_{16} - \mathbf{1}$. The two graphs are named objects:

$$\begin{array}{l} R_2(C^+) = \text{folded/complement, SRG}(16, 10, 6, 6) \text{ (the Clebsch complement);}\\ \boxed{R_4(C^+) = \text{Clebsch graph, SRG}(16, 5, 0, 2)} \quad \text{[E]}. \end{array} \tag{71}$$

Remark. The Standard-Model family is the *valency* side $(1, 10, 5)$ of one scheme; the cosmological action ladder is its *spectral* side $(1, 5, 10)$. The action ladder is therefore not merely "also binomial" — it is the spectral dual of the even exterior family. [C]

This duality is the MacWilliams transform ([`verification/v79_review_identities.py`]). C^+ is the even-weight code $[5, 4, 2]$; its dual $(C^+)^\perp$ is the repetition code $[5, 1, 5] = \{00000, 11111\}$ (the uniform/vacuum word). The MacWilliams identity sends the weight enumerator of the dual to that of C^+, namely $(1, 10, 5)$ — so the valency↔spectral duality *is* the coding-theoretic UV/IR duality, and the carrier grading $\Lambda^{\text{even}}(5) = 1{+}10{+}5$ is the MacWilliams image of the repetition vacuum. [E] (code identity) / [C] (holographic reading).

The full Bose–Mesner algebra (machine-checkable) fixes the duality beyond eigenvalues:

$$A_2^2 = 10\,\mathbf{1} + 6A_2 + 6A_4, \quad A_2A_4 = 3A_2 + 4A_4, \quad A_4^2 = 5\,\mathbf{1} + 2A_2, \tag{72}$$

with eigenmatrix and multiplicity orthogonality

$$P = \begin{pmatrix} 1 & 10 & 5 \\ 1 & 2 & -3 \\ 1 & -2 & 1 \end{pmatrix}, \quad \det P = -64, \qquad \boxed{P^\top \operatorname{diag}(1, 5, 10)\, P = \operatorname{diag}(16, 160, 80)} \quad \text{[E]}, \tag{73}$$

so the flip split $240 = 160{+}80$ ($|F| = 2, 4$) appears directly in the scheme's eigenmatrix orthogonality.

Carrier $= D_5$ (verified). The 16 vertices of C^+ are exactly the weights of the $D_5 = \mathfrak{so}_{10}$ half-spinor $\frac{1}{2}(\pm 1, \ldots, \pm 1)$ with an even number of minus signs. The flip-2 and flip-4 graphs are translation- and permutation-invariant, so the *affine* group $C^+ \rtimes S_5$ (translations $\rtimes$ coordinate permutations) acts by graph/scheme automorphisms; the group closure gives the automorphism group of the even Hamming scheme (equivalently the Clebsch graph),

$$\boxed{\operatorname{Aut}_{\text{aff}}(C^+) = C^+ \rtimes S_5, \qquad |\operatorname{Aut}_{\text{aff}}(C^+)| = 2^{g_{\text{car}}-1}\, g_{\text{car}}! = 16 \cdot 120 = 1920 = |W(D_5)|} \quad \text{[E]}, \tag{74}$$

and $1920 = r\,|R(E_8)| = 8 \cdot 240$. (The *linear* code automorphism group alone is only the coordinate group S_5 of order 120; the factor $2^{g_{\rm car}-1} = 16$ is the translation/scheme part. It is the affine/scheme group that equals $W(D_5)$.) Edges of the Clebsch graph: $|E(R_4)| = \frac{16\cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$ (the entire flavor transport budget). So the five-slot even code is the half-spin weight structure of $\mathfrak{so}_{10}$, the classical home of one SM family with ν^c.

Theorem 3 — the Coxeter closure equation selects $g_{\rm car} = 5$

Requiring the even family to equal the action truncation through pair degree, $\dim S^+ = 2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$, i.e.

$$\boxed{2^g - 2 = g(g+1) \iff g = 5} \qquad \textbf{[E]}, \tag{75}$$

the unique positive-integer solution. At $g = 5$ the two sides equal 30, so *carrier exhaustion* $2^g - 2$ and the *Coxeter period* $g(g+1)$ coincide only at five slots: $2^{g_{\rm car}} - 2 = 30 = h$ and $g_{\rm car}(g_{\rm car}+1) = 30$.

Theorem 4 — the $\mathbb{Z}_{30}$ cyclotomic Coxeter compiler

From the three atoms $(2, 3, g_{\rm car})$, with $h = 2 \cdot 3 \cdot g_{\rm car} = 30$ and $r = \varphi(h) = 8$:

$$\boxed{|R(E_8)| = rh = 240, \qquad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \qquad h + 1 = 2^{g_{\rm car}} - 1 = 31} \qquad \textbf{[E]}. \tag{76}$$

The E_8 exponents are the primitive residues modulo 30,

$$\mathrm{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad \chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1, \tag{77}$$

and $|(\mathbb{Z}/30)^\times| = \varphi(30) = 8 = r$. The local SM algebra and the root atlas are the two arithmetic projections of $(r, h) = (8, 30)$:

$$\Omega_{\rm adm} = 6r = 48, \quad D_{\rm start} = 2h = 60, \quad \gcd(6r, 2h) = \dim \mathfrak{g}_{\rm SM} = 12, \quad \mathrm{lcm}(6r, 2h) = rh = 240. \tag{78}$$

Theorem 5 — the root transition atlas and the positive-edge moment

The directed even-flips of C^+ are the complete graph K_{16}:

$$\boxed{\begin{aligned} |R^+(E_8)| = \binom{\dim S^+}{2} = \binom{16}{2} = 120 = \mathrm{Tr}_{S^+} X^2 = 5! = \sum_{m \in (\mathbb{Z}/30)^\times} m, \\ |R(E_8)| = 2\binom{16}{2} = 240 \end{aligned}} \qquad \textbf{[E]}. \tag{79}$$

($X = 6Y$.) So the positive E_8 roots are the unordered transitions of one family; the second hypercharge moment, the positive-root count and the sum of the E_8 exponents coincide at 120. The CP pairing $m \leftrightarrow 30 - m$ of the exponents gives four pairs, each of sum $h = 30$:

$$\boxed{\mathrm{Tr}_{S^+} X^2 = (N_{\rm fam} + N_\Phi)\, h = 4 \cdot 30 = 120 = \dim \mathfrak{so}_{16} = \underbrace{45}_{D_5} + \underbrace{15}_{A_3} + \underbrace{60}_{(10,6)}} \qquad \textbf{[E]}, \tag{80}$$

so the hypercharge moment is simultaneously four CP-paired Coxeter periods *and* the adjoint half $\mathfrak{so}_{16} = (45, 1) \oplus (1, 15) \oplus (10, 6)$ of the E_8 closure.

How the root *origin* is now obtained — see Theorem 7. A naive flip-size-preserving map $R_{\rm flip} \to \mathbb{R}^8$ cannot be E_8: the flip split $240 = 160 + 80$ ($|F| = 2, 4$) differs from the D_8

split $112 + 128$. The correct route is *not* a direct 240-metrization but the standard branching $E_8 \supset D_5 \times A_3$ (Theorem 7), which builds the root spaces from known D_5 and A_3 data. Until the A_3 lattice is derived from the four punctures (Theorem 7, [O]), E_8 remains a Coxeter atlas with a concrete, falsifiable lift.

Theorem 6 — downstream readouts from (h, r, c_3)

All discrete budgets and the seam seed are functions of $(h, r, c_3) = (30, 8, 1/(8\pi))$:

$$\dim S^+ = 2r = 16, \qquad h = 2(\dim S^+ - 1) = 30, \tag{81}$$

$$10b_1 = 5r + 1 = 41 = 1 + h\tfrac{\dim S^+}{\dim \mathfrak{g}_{\rm SM}}, \qquad I_1^{\rm EW} = \frac{5r+1}{6r} = \frac{41}{48}, \tag{82}$$

$$u = \frac{r}{6}c_3 + 6r\,c_3^4 = \frac{4}{3}c_3 + 48c_3^4, \qquad \delta_2 = 4!\,\tbinom{16}{2}\,c_3^8 = 2880\,c_3^8, \tag{83}$$

$$\xi_{\rm tree} = \frac{6}{r} = \frac{\dim \mathfrak{g}_{\rm SM}}{\dim S^+} = \frac{3}{4}, \qquad \frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{\Omega_{\rm adm} - 10b_1} = c_3^{r-1} = c_3^7. \tag{84}$$

Two-graph product table. With the carrier-pair graph K_5 ($|E(K_5)| = \binom{5}{2} = 10 = \mathcal{A}_\Lambda$) and the family-corner graph K_4 on μ_4 ($|E(K_4)| = \binom{4}{2} = 6$, $|R(A_3)| = 2|E(K_4)| = 12$), every large budget is a product of the two graphs:

$$\boxed{40 = 10 \cdot 4, \quad 60 = 10 \cdot 6, \quad 120 = 10 \cdot 12, \quad 240 = 10 \cdot 24} \qquad \text{[E]}, \tag{85}$$

i.e. $(\sum L,\, D_{\rm start},\, |R^+(E_8)|,\, |R(E_8)|) = 10 \cdot (|\mu_4|, |E(K_4)|, |R(A_3)|, |S_4|)$. Two occupancy identities follow,

$$\begin{aligned} &\boxed{10b_1 = 1 + |\mu_4|\,|E(K_5)| = 1 + 4 \cdot 10 = 41}, \\ &\boxed{\Omega_{\rm adm} = N_{\rm fam}\dim S^+ = (N_{\rm fam}{+}N_\Phi)|R(A_3)| = 3 \cdot 16 = 4 \cdot 12 = 48} \qquad \text{[E]}, \end{aligned} \tag{86}$$

so the Higgs singleton is the fourth (μ_4) direction that closes $N_{\rm fam} = 3$ to **4** of A_3. **Occupancy deficit ladder.** The admissible occupancy splits cleanly,

$$\boxed{\begin{aligned} &\Omega_{\rm adm} = \underbrace{40}_{\sum L\,(\text{flavor})} + \underbrace{1}_{\text{Higgs}} + \underbrace{7}_{\text{scalaron}} = 48, \qquad \Omega_{\rm adm} - \sum L = 8 = h(D_5) = r(E_8), \\ &\Omega_{\rm adm} - 10b_1 = 7 = h(D_5) - 1 \end{aligned}} \qquad \text{[E]}, \tag{87}$$

so the scalaron exponent $r - 1 = 7$ is exactly the post-flavor, post-Higgs occupancy remainder, $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^{\Omega_{\rm adm} - 10b_1} = c_3^7$. With $h(D_5) = 8$ and $|R^+(A_3)| = 6$ the seed itself is a D_5/A_3 ratio–product, $u = \frac{h(D_5)}{|R^+(A_3)|}c_3 + h(D_5)|R^+(A_3)|c_3^4$, and $\xi = \frac{|R^+(A_3)|}{h(D_5)}\,(1 + |R^+(A_3)|^2 c_3^3)^{-1} = \frac{3}{4}(1 + 36c_3^3)^{-1}$. The seed readouts and the two decuples are

$$\lambda_C^2 = u(1{-}u), \quad \beta_{\rm rad} = 2c_3 u, \quad \sin^2\theta_{13} = e^{-5/6}u, \quad \boxed{\omega_b = \frac{\lambda_C}{\mathcal{A}_\Lambda} = \frac{\lambda_C}{10} = 0.02244}\ \text{(Planck 0.02237)}, \tag{88}$$

$$\boxed{\begin{aligned} &\det M_\ell \propto u^{\,h/N_{\rm fam}} = u^{10}, \\ &\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}e^{2\delta_{\rm ph}}\Big[\tfrac{v_{\rm geo}}{5\beta_{\rm rad}^2 \bar{M}_{\rm Pl}}\Big]^{h/N_{\rm fam}} \end{aligned}}, \tag{89}$$

with lepton powers $(5, 3, 2) = (g_{\rm car}, b, s)$: $\sum = 10 = \mathcal{A}_\Lambda$ (the Λ decuple), $\prod = 30 = h$ (the Coxeter period). The first-generation transport sums read the non-primitive $\mathbb{Z}_{30}$ load,

$$(G_1, G_2, G_3) = (22, 13, 5) = (h - r,\ \dim \mathfrak{g}_{\rm SM} + 1,\ g_{\rm car}), \qquad h - r = 30 - \varphi(30) = 22, \tag{90}$$

Flavor transport is a $D_5\times A_3$ object. The transport packets $L_u{=}(7,3,0), L_d{=}(7,5,2), L_e{=}(8,5,3)$ split into a D_5 budget and an A_3 geometry:

$$\boxed{\textstyle\sum_{f,j} L_{f,j} = 40 = |R(D_5)| \text{ (Clebsch edges)}, \qquad L_{f,j} \bmod 6 \text{ lives on } C_6 = Z_6 = |R^+(A_3)|} \quad \text{[E]}, \tag{91}$$

the residues being the D_6-orbit of the perfect set $\{0,1,3\}\subset C_6$ (cyclic distances $\{1,2,3\}$): $u\equiv\{0,1,3\}$, $d\equiv 2-\{0,1,3\}$, $e\equiv 2+\{0,1,3\}$, with the universal first-generation $+6$ winding. So the carrier D_5 sets the total budget and the family A_3 (its 6-element positive-root hexagon) sets the per-entry distances. The scalaron mass is a pure seam power [E]; the amplitude is a *conditional* readout, kept [C]/[O] until the scalaron variable is formally identified with the Starobinsky M parameter. That identification is now *heat-kernel grounded* (G2): the spectral-action Seeley–DeWitt coefficient $a_4|_{R^2}\sim R^2/72$ delivers exactly the Starobinsky R^2 term, with $M^2/\bar{M}_{\rm Pl}^2 = 6(4\pi)^2/f_0$ and the TFPT closure c_3^7 fixing the cutoff moment f_0 ([`verification/v36_spectral_action_g2.py`]):

$$\underbrace{\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{r-1} = c_3^7}_{\text{[E]}}, \qquad \underbrace{A_s = \frac{N_\star^2}{24\pi^2}\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = \frac{N_\star^2 c_3^7}{24\pi^2} = 2.17\times 10^{-9}}_{\text{[C]/[O] (if } M_{\rm scal}=M_{\rm Star})}. \tag{92}$$

Theorem 7 — the $D_5 \times A_3$ lift of the E_8 atlas

Since $A_3 = \mathfrak{su}_4 \cong \mathfrak{so}_6 = D_3$, one has the standard chain $E_8 \supset \mathfrak{so}_{16} \supset \mathfrak{so}_{10}\oplus\mathfrak{so}_6 = D_5\times A_3$, with the exact branching (verified dimensions and root/weight counts):

$$\begin{gathered}\boxed{\mathfrak{e}_8 = (45,1)\oplus(1,15)\oplus(10,6)\oplus(16,4)\oplus(\overline{16},\overline{4})},\\ \underbrace{45{+}15{+}60{+}64{+}64}_{=248} = \dim E_8, \qquad \underbrace{40{+}12{+}60{+}64{+}64}_{=240} = |R(E_8)|.\end{gathered} \tag{93}$$

The adjoint **120** of $\mathfrak{so}_{16}$ is $(45,1)\oplus(1,15)\oplus(10,6)$ and its spinor **128** is $(16,4)\oplus(\overline{16},\overline{4})$. The TFPT dictionary makes each summand a known object:

piece	value	TFPT reading
$(45,1)$	$45 = 40+5$	D_5 carrier adjoint: Clebsch edges $40=\|R(D_5)\|$ plus carrier rank $g_{\rm car}=5$
$(1,15)$	$15 = 12+3$	A_3 family+Higgs adjoint: $\dim\mathfrak{g}_{\rm SM}=12$ plus family rank $N_{\rm fam}=3$
$(10,6)$	$10\cdot 6$	Λ decuple $\mathcal{A}_\Lambda=\binom{5}{2}=10$ times Z_6 transport $\binom{4}{2}=6$
$(16,4)\oplus(\overline{16},\overline{4})$	$2\cdot 64$	matter half-spinor S^+ times the four corners μ_4, plus conjugate sheet

Proof route (replaces the old blind 240-metrization). *Stage 1 (D_5, [E]):* C^+ is the D_5 half-spin orbit, R_4 = Clebsch, Aut $= W(D_5)$, $|E(R_4)| = 40 = |R(D_5)|$ — Theorem 2. *Stage 2 (A_3, [E]):* the carrier geometry fixes $X_f^\circ = \mathbb{P}^1\setminus\mu_4$ with $H_1(X_f^\circ,\mathbb{Z})=\mathbb{Z}^3$ and a rigid $D_4\subset PGL_2$ generated by $z\mapsto iz$ and $z\mapsto 1/z$. The logarithmic forms $\omega_i = dz/(z-p_i)$ regular at infinity span H^1 by their differences, dual under the residue pairing to the puncture loops γ_i with $\mathrm{Res}(\omega_i,\gamma_j)=\delta_{ij}$; hence the canonical (residue) metric on the puncture classes is I_4, and the twelve puncture-difference cycles $\gamma_i-\gamma_j$ are exactly the A_3 roots:

$$\boxed{\begin{gathered}\{\gamma_i-\gamma_j\}_{i\neq j}:\ \#=12=|R(A_3)|=\dim\mathfrak{g}_{\rm SM}, \quad \mathrm{rank}=3=N_{\rm fam},\\ \mathrm{Gram}=\mathrm{Cartan}(A_3),\ \det=4\end{gathered}} \quad \text{[E]}, \tag{94}$$

with $\dim A_3 = 15 = \dim S^+ - 1$, and the A_3 positive roots equal the exponent sum $1{+}2{+}3 = 6 = Z_6 = |E(K_4)|$. *The corner rotation realises the A_3 Coxeter element.* The carrier generator $z\mapsto iz$ permutes μ_4 as the 4-cycle $(p_1p_2p_3p_4)\in S_4 = W(A_3)$ and acts on $H^1=\mathbb{C}^3$ as an isometry of the Cartan form with

$$\boxed{\mathrm{ord}=4=h(A_3)=|\mu_4|, \quad \mathrm{spec}=\{i,-1,-i\}, \quad \chi(t)=\tfrac{t^4-1}{t-1}=\Phi_4(t)\Phi_2(t)} \quad \text{[E]}, \tag{95}$$

i.e. the eigenvalues are the A_3 exponents $\{1,2,3\}$ as fourth roots of unity — the exact analogue of the E_8 Coxeter element reading Φ_{30}. The full $D_4 = \langle z \mapsto iz, z \mapsto 1/z \rangle$ embeds in $W(A_3) = S_4$ (order 8, index 3) by isometries. *Stage 3 (standard, [E]):* assemble E_8 from the $D_5 \times A_3$ branching — no 240-map required.

The μ_4 glue theorem (why the four corners appear everywhere). The two factors have equal discriminant groups, and E_8 is unimodular:

$$\boxed{\operatorname{disc}(D_5) = \operatorname{disc}(A_3) = \mathbb{Z}_4, \qquad [\,E_8 : D_5 \oplus A_3\,] = \sqrt{\tfrac{\det D_5 \cdot \det A_3}{\det E_8}} = \sqrt{\tfrac{4\cdot 4}{1}} = 4 = |\mu_4|} \qquad \textbf{[E]}. \tag{96}$$

So $E_8 = (D_5 \oplus A_3) + \mu_4$-*glue*: the four family corners $\mu_4 = \mathbb{Z}_4$ are precisely the glue code that fuses the carrier and the family into the unimodular E_8 lattice. The glue pairs the D_5 spinor class ($= S^+$, the family) with the A_3 class — exactly the $(16,4) \oplus (\overline{16},\overline{4})$ spinor block. Hence μ_4 plays three roles at once: family corners, A_3 origin, and E_8 glue.

Discriminant-norm glue lemma (and the TFPT $5/4, 3/4$ identification). The two glue generators carry the discriminant quadratic form $q \in \mathbb{Q}/2\mathbb{Z}$:

$$\boxed{\begin{aligned} &q(D_5) = \tfrac{n}{4}\big|_{n=5} = \tfrac{5}{4} \text{ (spinor class)}, \quad q(A_3) = \tfrac{n}{n+1}\big|_{n=3} = \tfrac{3}{4} \text{ (corner class)}, \\ &q(D_5) + q(A_3) = 2 \equiv 0 \pmod 2, \qquad q(D_5) - q(A_3) = \tfrac{1}{2} \end{aligned}} \qquad \textbf{[E]}. \tag{97}$$

The diagonal $\langle (s,g) \rangle \cong \mathbb{Z}_4$ is isotropic ($k^2 \cdot 2 \equiv 0$ for all k), so the glue closes to an *even unimodular* lattice — this is E_8.

The two glue norms are just $(g_{\text{car}}, N_{\text{fam}})$ over the glue index. Because $\operatorname{rank} D_5 = g_{\text{car}} = 5$ (so $q(D_5) = \frac{n}{4}|_{n=5} = g_{\text{car}}/|\mu_4|$) and $\operatorname{rank} A_3 = N_{\text{fam}} = 3$ (so $q(A_3) = \frac{n}{n+1}|_{n=3} = N_{\text{fam}}/|\mu_4|$, with $n{+}1 = 4 = |\mu_4|$),

$$\boxed{q(D_5) = \frac{g_{\text{car}}}{|\mu_4|} = \tfrac{5}{4}, \qquad q(A_3) = \frac{N_{\text{fam}}}{|\mu_4|} = \tfrac{3}{4}}$$

and *all four arithmetic operations* on the pair are then forced compiler atoms:

$$\underbrace{q(D_5){+}q(A_3) = \tfrac{g_{\text{car}}+N_{\text{fam}}}{|\mu_4|} = 2}_{E_8 \text{ root norm}}, \qquad \underbrace{q(D_5){-}q(A_3) = \tfrac{|\mathbb{Z}_2|}{|\mu_4|} = \tfrac{1}{2}}_{\delta \text{ (lepton transport)}},$$

$$\underbrace{q(D_5)q(A_3) = \tfrac{g_{\text{car}}N_{\text{fam}}}{|\mu_4|^2} = \tfrac{15}{16}}_{\dim \mathfrak{su}(4)/\dim S^+}, \qquad \underbrace{\tfrac{q(D_5)}{q(A_3)} = \tfrac{g_{\text{car}}}{N_{\text{fam}}} = \tfrac{5}{3}}.$$

So the distinguished lepton transport value $\delta = \frac{1}{2} = (g_{\text{car}} - N_{\text{fam}})/|\mu_4| = |\mathbb{Z}_2|/|\mu_4|$ is the carrier-minus-family count over the glue index — a fully atomic origin, not a chosen rational (it also equals the harmonic-frame holonomy diagonal modulus, `[verification/v41_leg_assignment.py]`); and (sum, ratio) invert uniquely back to $(\frac{5}{4}, \frac{3}{4})$. `[verification/v51_boundary_half_step.py]` The sum $\frac{5}{4}{+}\frac{3}{4} = 2$ is literally the E_8 spinor-root norm $\frac{1}{2}(\pm)^8$ split across $D_5 \times D_3$: $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$. Strikingly, both norms are pre-existing TFPT constants:

$$\boxed{\frac{\delta_2}{\delta_{\text{top}}^2} = \frac{2880\, c_3^8}{(48\, c_3^4)^2} = \frac{5}{4} = q(D_5), \qquad \xi_{\text{tree}} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{12}{16} = \frac{3}{4} = q(A_3)} \qquad \textbf{[E]}, \tag{98}$$

so $\delta_2/\delta_{\text{top}}^2 + \xi_{\text{tree}} = 2$: the two-loop/defect ratio is the D_5 spinor norm and the gravitational tree normalization is the A_3 corner norm, and their sum is the E_8 root norm.

The whole integer skeleton from one pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ [E] ([verification/v53_compiler_core.py])

Every load-bearing integer of the compiler flows from the single boundary pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ by sum, difference and squares:

$$\operatorname{rank} E_8 = g_{\rm car}+N_{\rm fam} = 8, \qquad |\mathbb{Z}_2| = g_{\rm car}-N_{\rm fam} = 2, \qquad |\mu_4| = \tfrac{g_{\rm car}+N_{\rm fam}}{2} = 4,$$

and $(N_{\rm fam}, |\mu_4|, g_{\rm car}) = (3,4,5)$ is *the* Pythagorean triple. This is not decorative: the grand-mass-volume exponent $\Delta_Y = g_{\rm car}^2 = 25$ ($\det M_{\rm SM} \sim (\varphi_0^{\rm ret})^{25}$; document 3) splits over the sectors as a *difference of squares*,

$$\Delta_Y = g_{\rm car}^2 = \underbrace{N_{\rm fam}^2}_{\text{down}=9} + \underbrace{(g_{\rm car}-N_{\rm fam})(g_{\rm car}+N_{\rm fam})}_{=|\mathbb{Z}_2|\cdot\operatorname{rank} E_8} = 9+16 = \dim S^+$$

i.e. the down sector carries $N_{\rm fam}^2$ and up+lepton carries $|\mathbb{Z}_2|\cdot\operatorname{rank} E_8 = \dim S^+$ (the K-row sums $(6,9,10)$). **The anchor is the Dirac square root of this skeleton:** $a = (1,1,2)$ is the *unique* 3-multiset whose elementary symmetric functions are $(|\mu_4|, g_{\rm car}, |\mathbb{Z}_2|) = (4,5,2)$, equivalently its characteristic polynomial is $\chi_a(t) = (t-1)^2(t-2) = t^3 - |\mu_4|\,t^2 + g_{\rm car}\,t - |\mathbb{Z}_2|$ — the compiler atoms *are* its char-poly coefficients. Sharper still ([verification/v491_p2_partition_corollary.py]): the partition needs only $e_1 = 4$ — three *positive integers* summing to 4 already force $\{1,1,2\}$, so $g_{\rm car} = e_2 = 5$ and $|\mathbb{Z}_2| = e_3 = 2$ are *corollaries* of the four marks given the weight typing (integrality = Birkhoff–Grothendieck splitting exponents, positivity = h^0=0, sum = par-deg-0 "given (U)"); every dropped assumption is negative-controlled, and `AX.P2.01` stays a declared axiom — the residual is the typing, not the integer 5. That typing is itself now hardened ([verification/v499_p2_weights_deligne_bg.py], `P2.TYPING.01`): given the four marks, rank 3, the cusp class and (U), Deligne + Birkhoff–Grothendieck + Mehta–Seshadri stability force $E = \mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ outright (h^0=0 is the positivity), leaving only the module identification and (U) as [C] residuals — and the module-identification rest is now *one* residual with the QGEO modulus rest, both reducing to the same order-4 clock = Coxeter/cusp-class carrier datum (v503, `QGEO.EMERGE.LIGHT.01`; markers unchanged); the clock-rigidity theorems then reduce that common residual to one alignment bit, with the clock's order fermionically forced, $U^2 = (-1)^F$ exactly ([verification/v506_seam_clock_rigidity.py], `SEAM.CLOCK.RIGIDITY.01`); and the bit-origin theorems derive the deck's *class* (a half-period translation of the seam double) — only its position stays the [C] carrier input ([verification/v507_seam_bit_origin.py], `SEAM.BIT.ORIGIN.01`); the freedom theorems then type that position half as *topology* (the covering deck is free on the seam circle, the edge class is excluded), so the residual is the square-modulus datum $\tau = i$ alone ([verification/v510_seam_bit_freedom.py], `SEAM.BIT.FREEDOM.01`); and the flag-transitivity equivalence web restates that residual in discrete form — flag transitivity of the four marks ($V_4 \to D_4$) $\iff \tau = i$, with bare mark-transitivity automatic for every configuration and no local jet seeing the side bit ([verification/v512_seam_tau_flag.py], `SEAM.TAU.FLAG.01`) — a residual that has since survived ten side-blind derivation attacks (v521/v525/v529) and carries a physical definition as the twist-class choice with a gauge-robust order parameter, in principle interferometrically readable (v528; it remains formal input) — and, since v534, the first *dynamical* selection datum: reflection positivity on the interacting mark family keeps exactly the $\delta = \pi/2$ member alive (toy-fenced). So the theory's only transcendental is π (through $c_3 = \frac{1}{8\pi}$, $8 = \operatorname{rank} E_8$, and the seed $\frac{1}{6\pi}$, $6 = |R^+(A_3)|$); everything else is this $(5,3)$-generated integer skeleton.

Boundary Carrier Selection Theorem (Theorem A) [E]/[C]

E_8 is not assumed: it is *selected* as the unique even-unimodular audit hull of the admissible boundary-carrier data. **Boundary side:** $\mu_4 = \{1, i, -1, -i\}$ gives $X_f = \mathbb{P}^1 \setminus \mu_4$ with $H_1 = \mathbb{Z}^3$, hence A_3 and $N_{\rm fam} = 3$. **Carrier side:** a minimal D-type carrier with a 16-dim half-spinor forces D_5 (only $n = 5$ gives $2^{n-1} = 16$; $S^+ = \Lambda^{\rm even}\mathbb{C}^5 = 1{+}10{+}5$). **Glue:** $\operatorname{disc}(D_5) = \operatorname{disc}(A_3) = \mathbb{Z}_4$ (isomorphic, cyclic, anti-isometric), $q(D_5) + q(A_3) = 2$ (the E_8 root norm), glue index $4 = |\mu_4|$, and E_8 is the unique even unimodular rank-8 lattice. **Exclusion:** the four conditions

$$\text{(i) } \dim S^+{=}16, \quad \text{(ii) } N_{\rm fam}{=}3, \quad \text{(iii) cyclic } \mathbb{Z}_4 \text{ glue}, \quad \text{(iv) } q_D{+}q_A{=}2$$

are met *only* by $D_5 \oplus A_3$; D_8 (non-cyclic $\mathbb{Z}_2{\times}\mathbb{Z}_2$, no family geometry), $E_7{\oplus}A_1$ (1 family, no 16-spinor), $E_6{\oplus}A_2$ (2 families, $\mathbb{Z}_3$), $A_4{\oplus}A_4$ (no D-spinor, $\mathbb{Z}_5$) and A_8 (single factor) each fail at least one. **Typing:** the lattice/classification facts are [E]; the physical selection (reflection-positive seam $c_3 = 1/8\pi$ +

minimal chiral carrier ⇒ exactly these inputs) is the [C]/[O] selection axiom (P1 analytic interface + P2 carrier interface), not a zero-input derivation. [verification/v47_selection_theorem.py]

Group-level closure (μ_4 as the common centre). The lattice glue lifts to a clean group statement, because both factors share the *same* centre and μ_4 *is* the diagonal:

$$Z(\mathrm{Spin}(10)) = \mathbb{Z}_4, \quad Z(SU(4)) = \mathbb{Z}_4, \quad |\mu_4| = 4, \quad G_{\text{closure}} = \frac{\mathrm{Spin}(10) \times SU(4)}{\Delta\mathbb{Z}_4} \subset E_8 \qquad \text{[E]}, \tag{99}$$

with rank $= 5 + 3 = 8 = \mathrm{rank}\, E_8$. Every adjoint block is diagonal-$\mathbb{Z}_4$ neutral (generator $\omega = i$; charges mod 4): $(45,1),(1,15) : 0$; $(10,6) : 2{+}2{=}0$; $(16,4) : 1{+}3{=}0$; $(\overline{16},\overline{4}) : 3{+}1{=}0$, so $45{+}15{+}60{+}64{+}64 = 248$ are genuine reps of the quotient. This promotes μ_4 from a lattice glue index to the *common centre quotient* — the physically cleanest language for the gluing: the D_5 half-spinor 16 pairs with the $SU(4)$ fundamental 4 so the spinor charges ±1 cancel the fundamental charges ∓1.

Root-length completion. Every E_8 block reaches norm 2:

block	norm split	reading
$(45,1),(1,15)$	$2,\ 2$	D_5, A_3 root adjoints
$(10,6)$	$1+1=2$	$SO(10)$ vector $\oplus$ $SO(6)$ vector
$(16,4)\oplus(\overline{16},\overline{4})$	$\frac{5}{4}+\frac{3}{4}=2$	D_5 spinor norm + A_3 corner norm

The old question "can the 240 flip atlas be metrized to E_8?" is now the sharp question "are the TFPT D_5 and A_3 metrics in standard normalization?" — and the $5/4, 3/4$ match says yes.

Explicit construction (the lift is now built, not counted). Embedding $D_5 \oplus A_3 = D_5 \oplus D_3 \subset D_8$ and adjoining the even spinor coset (the μ_4 glue), the 240 roots are produced explicitly and verified:

$$\{\pm e_i \pm e_j\}_{i<j}^{8}\ (112)\ \cup\ \{\tfrac{1}{2}(\pm)^8 :\ \text{even}\}\ (128)\ =\ 240 \text{ roots, all norm 2, Cartan det} = 1 \qquad \text{[E]}. \tag{100}$$

Splitting each root into its D_5 (first five) and A_3 (last three) parts gives exactly the branching blocks — $(2,0){:}40$, $(0,2){:}12$, $(1,1){:}60$, and $(\frac{5}{4},\frac{3}{4}){:}128$ — and the 128 spinor roots realise $\frac{5}{4}+\frac{3}{4}=2$ on the nose. All 240 vectors are integer combinations of one set of 8 simple roots, so the assembled lattice *is* E_8. The blind 240-metrization is therefore eliminated: E_8 is explicitly $(D_5 \oplus A_3) + \mu_4$-glue.

Remark (the metric, computed)**.** This reframes E_8 from origin to closure: E_8 is the closure algebra of the D_5 carrier and the A_3 family geometry, glued by μ_4. The root system, Cartan form, $W(A_3) = S_4$ and the corner-rotation $= A_3$ Coxeter element are exact (residue pairing + explicit carrier symmetry). The transcendental geometric metric $G_{ij} = -\log|p_i - p_j|$ on μ_4 has been computed: it is only D_4-symmetric, with a regularisation-independent anisotropy $-\log 2$ on the root space, hence *not* proportional to Cartan(A_3). It need not be: the E_8-relevant A_3 is the canonical *residue* pairing, and the geometric metric is a D_4-equivariant deformation that preserves the root system, Weyl group and Coxeter element. So the lattice data is exact [E]; the analytic metric is a *characterised* deformation, not an open gap.

Uniform Coxeter law. All three factors obey $|R| = h \cdot \mathrm{rank}$, and the two TFPT Coxeter numbers are themselves carrier data:

$$\begin{aligned} &h(D_5) = 8 = r(E_8) = \varphi(30), \quad h(A_3) = 4 = |\mu_4|, \\ &h(D_5)\,h(A_3) = 2^{g_{\text{car}}} = 32, \quad 2^{g_{\text{car}}} - 2 = 30 = h(E_8) \end{aligned} \qquad \text{[E]}, \tag{101}$$

so the Coxeter closure of Theorem 3 reappears as $h(E_8) = h(D_5)h(A_3) - 2$, and $h(D_5) + h(A_3) = 12 = \dim \mathfrak{g}_{\text{SM}} = |R(A_3)|$.

Theorem 8 — the $(1,1,2)$ anchor microcode and the φ_0-ladder

The parabolic flavor bundle $E_\bullet = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ on $\mathbb{P}^1 \setminus \mu_4$ has splitting type $a = (1,1,2)$, whose power sums $p_n(a) = 2 + 2^n$ generate the whole compiler budget:

The anchor power compiler.

$$(p_1, \ldots, p_5) = (4, 6, 10, 18, 34) = (|\mu_4|,\ |R^+(A_3)|,\ \mathcal{A}_\Lambda,\ N_{\text{fam}}|R^+(A_3)|,\ \mathbb{Z}_6 \text{ lift}),$$

$$q(A_3) = \tfrac{p_2}{2p_1} = \tfrac{3}{4}, \quad q(D_5) = \tfrac{p_3}{2p_1} = \tfrac{5}{4}, \quad \textstyle\sum L = p_1 p_3 = 40,$$

$$\Omega_{\text{adm}} = 2p_1p_2 = 48, \qquad D_{\text{start}} = p_2 p_3 = 60, \qquad |R(E_8)| = 4p_2p_3 = 240.$$

The elementary symmetric data $\chi_a(t) = (t-1)^2(t-2)$ give $(e_1, e_2, e_3) = (4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|)$, so the anchor reconstructs the carrier rank, and $10b_1 = e_1^2 + e_2^2 = 4^2 + 5^2 = 41$. The seed itself is the two-term anchor form $\varphi_0^{\text{ret}} = \frac{2p_1}{p_2} c_3 + 2p_1p_2\, c_3^4 = \frac{4}{3} c_3 + 48 c_3^4$.

The anchor affine normal form (2026-08-07) — all budgets from ONE recursion [E] ([verification/v832_anchor_flavor_checksum.py])

The anchor power compiler above is one affine map. The power sums $p_n(a) = 2 + 2^n$ satisfy

$$\boxed{p_{n+1} = T(p_n), \qquad T(x) = 2x - 2} \qquad \text{identically in } n \text{ (sympy)},$$

with the shift $q_n = p_n - 2 = 2^n$ pure binary doubling. T has the *unique* fixed point $2 = |\mathbb{Z}_2|$ (the sheet), and the T-orbit of 4 is exactly the compiler quintet $(4, 6, 10, 18, 34) = (|\mu_4|, |R^+(A_3)|, \mathcal{A}_\Lambda, N_{\text{fam}}|R^+(A_3)|, \mathbb{Z}_6 \text{ lift})$, with $p_0 = 3 = N_{\text{fam}}$ prepended by $T(3) = 4$. Every budget line is then a word in the orbit and the fixed point: $|R(E_8)| = p_1p_2p_3 = 240$; $\dim E_8 = 240 + q_3 = 248$, where the ladder identity $p_4 - p_3 = p_3 - 2 = 8 = h(D_5) = \operatorname{rank} E_8$ makes the corpus form $240 + (p_4 - p_3)$ the *same* formula; $h(E_8) = p_2p_3/2 = 30$ (so $D_{\text{start}} = p_2p_3 = 60$); $|R(D_5)| = p_1p_3 = 40$; $\Omega_{\text{adm}} = 2p_1p_2 = 48$; and the elementary layer $(e_1, e_2, e_3) = (4, 5, 2)$ has the fixed point of T as e_3, with $10b_1 = e_1^2 + e_2^2 = 41$. Controls: the wrong map $T'(x) = 2x - 1$ (fixed point 1, orbit misses the quintet) and the wrong anchor $a' = (1, 2, 2)$ (recursion offset -1, budget misses $(240, 48)$) both fire. A normal form of the existing compiler tables — a compression, not a new claim; 18/18 exact, verdict ANCHOR-AFFINE-EXACT.

The φ_0-ladder. With $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$, every fermion mass is the single closed form $\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$ with $\lambda_Y^L \Lambda = \pi\, c\, (\varphi_0^{\text{ret}})^k$, so all nine holonomies are resolvent outputs and the masses form the φ_0^{ret}-order ladder $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$.

Secondary E_8 atlas. Beyond the main $D_5 \times A_3$ split, the residue matrix appears in the $E_6 \times A_2$ branching, $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$, and the scalaron exponent in $E_7 \times A_1$, $112 = 7 \cdot 16$. The seven E_8 slices are charted in the companion note `e8_secondary_branching_atlas` as a falsifiable audit raster.

The $E_6 \times A_2$ block is the $\tau = \omega$ face ([verification/v407_dn_pairings_omega.py], SEAM.EQUIV.02). A_2 is the Eisenstein lattice of the family CM point $\tau = \omega$ (Cartan $\det = 3 = N_{\text{fam}}$), and the (d, n) selector's three pairings $\{n \cdot \mathbf{1},\ n \cdot a,\ n \cdot \sigma\} = \{2, 8, 121\}$ are exactly its family-slice atoms $\{|Z_2|,\ \dim A_2 = \det R = \operatorname{rank} E_8,\ \|Pl(K)\|_1^2\}$, so deriving (d, n) (the R4′ residue, v139) folds into the dual keystone (v405), leaving only "why $11 = \|Pl(K)\|_1$".

Master statement and status

> Five carrier slots give D_5, four family corners give A_3, and $D_5 \times A_3$ gives the E_8 atlas.

Equivalently: $\mathcal{A}_\Lambda = \frac{h}{N_{\text{fam}}} = 10 = \binom{5}{2}$, $h = 30 = 2 \cdot 3 \cdot 5$, $(5, 3, 2)$: $\sum = 10$, $\prod = 30$; Λ, $\det M_\ell$, the E_8 Coxeter structure and the action ladder read the same five-slot pair-decuple. The Standard model, flavor, Λ, inflation and E_8 are projections of the even carrier code C^+ ($= D_5$ half-spinor) on five slots together with the four-point family A_3.

Statement	Status	Note
Even code C^+, $\dim S^+ = 16$, dual-root Y	[E]	Theorem 1
Exterior–Action duality $(1,10,5) \leftrightarrow (1,5,10)$	[E]	scheme eigenvalues verified
R_4 = Clebsch $(16,5,0,2)$, $R_2 = (16,10,6,6)$; $\mathrm{Aut}_{\mathrm{aff}} = W(D_5) = 1920$	[E]	carrier = D_5 half-spinor
$2^g - 2 = g(g+1) \Leftrightarrow g = 5$	[E]	Coxeter closure (unique)
h=30=2·3·5, r=φ(30)=8, χ_C=Φ_{30}	[E]/[C]	cyclotomic compiler
$\lvert R^+\rvert = \binom{16}{2} = \mathrm{Tr}\, X^2 = 4h = 120$	[E]	positive-edge moment
$40, 60, 120, 240 = 10 \cdot (4, 6, 12, 24)$; 41=1+4·10	[E]	$K_5 \times K_4$ product table
$10b_1$=5r+1, u totient, M^2_{scal}=c_3^{r-1}, ω_b=λ_C/10	[E]/[C]	downstream readouts
$A_s = N_\star^2 c_3^7 / 24\pi^2$	[C]/[O]	if $M_{\mathrm{scal}} = M_{\mathrm{Star}}$
$\mathfrak{e}_8 = (45,1) \oplus (1,15) \oplus (10,6) \oplus (16,4) \oplus (\overline{16},\overline{4})$	[E]/[C]	$D_5 \times A_3$ lift
240 roots built from $D_5 \oplus A_3 + \mu_4$ glue, det 1	[E]	explicit E_8 construction
flavor $\sum L$=40=$\lvert R(D_5)\rvert$, entries on C_6=$\lvert R^+(A_3)\rvert$	[E]	transport = $D_5 \times A_3$
A_3 from $\mathbb{P}^1 \setminus \mu_4$: residue pairing = Cartan, det 4	[E]	Stage 2 (Theorem 7)
corner $z \mapsto iz = A_3$ Coxeter el.: $\{i, -1, -i\}$, $\Phi_4\Phi_2$	[E]	order 4 = $h(A_3) = \lvert\mu_4\rvert$
$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue = μ_4	[E]	$E_8 = (D_5 \oplus A_3) + \mu_4$
$q(D_5)$=$\frac{5}{4}$=$\delta_2/\delta^2_{\mathrm{top}}$, $q(A_3)$=$\frac{3}{4}$=ξ_{tree}, sum 2	[E]	discriminant-norm glue
120= $\dim \mathfrak{so}_{16}$; Ω_{adm}=40+1+7	[E]	adjoint half; deficit ladder
$h(D_5)$=8=$r(E_8)$, $h(A_3)$=4=$\lvert\mu_4\rvert$, $\lvert R\rvert$=$h \cdot r$	[E]	uniform Coxeter law
$\mathrm{Exp}(E_8)$ as 41/48 budget deficits	[O]	exponent ledger (fingerprint)
geometric metric D_4-deform (anisotropy $-\log 2$), residue = A_3	[E]	metric computed/characterised

Scope. This note compresses; it does not re-derive the upstream boundary kernel (the P1 hardening chain) or the carrier rigidity theorem (Lean), and it makes no claim of a final theory. The arithmetic and the D_5/Clebsch/$W(D_5)$, A_3/residue-Cartan/Coxeter, and μ_4-glue identifications are exact, and the E_8 lattice is now built explicitly from $D_5 \oplus A_3$ plus the μ_4 glue (240 roots, det 1). The only residual [C] is the analytic identification of the carrier geometry's L^2 cusp Hodge metric with the canonical residue metric up to scale, which does not affect the explicit root/lattice/Weyl construction. Honest non-fits (Koide $0.6645 \neq 2/3$, η_B, m_p/m_e) stay outside the compiler.

Appendix: machine-checkable $E_8 = (D_5 \oplus A_3) + \mu_4$ construction

The lift is reproducible by a short script (all entries below verified exactly).

Cartan matrices and discriminant groups.

$$C(D_5) = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \quad C(A_3) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}; \quad \det C(D_5) = \det C(A_3) = 4.$$

Smith normal forms give $\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$; the glue index is $[\, E_8 : D_5 \oplus A_3 \,] = \sqrt{4 \cdot 4/1} = 4 = \lvert\mu_4\rvert$, with discriminant-form norms $q(D_5) = \frac{5}{4}$, $q(A_3) = \frac{3}{4}$, $q(D_5)+q(A_3) = 2 \equiv 0 \pmod 2$ (even-glue). Writing $\mu := \lvert\mu_4\rvert = 4$ the norms are $q(D_5) = \frac{\mu+1}{\mu}$, $q(A_3) = \frac{\mu-1}{\mu}$, so $q(D_5)q(A_3) = 1 - \frac{1}{\mu^2} = \frac{15}{16}$ and

$$\boxed{\dim S^+ (1 - q(D_5)q(A_3)) = 16 \cdot \tfrac{1}{16} = 1} \qquad [\mathbf{E}], \tag{102}$$

i.e. the glue product fills all $15 = \dim S^+ - 1$ non-trivial half-spin states and the defect is exactly the singleton (the ν^c/Higgs closure). And E_8 has a tidy μ_4-cardinal form, $\dim E_8 = \binom{\lvert\mu_4\rvert^2}{2} + 2\lvert\mu_4\rvert^3 = \binom{16}{2} + 2 \cdot 64 = 120 + 128 = 248$ [C].

Explicit roots. Split $\mathbb{R}^8 = \mathbb{R}^5_{(D_5)} \oplus \mathbb{R}^3_{(A_3)}$. The 240 roots are

$$\{\pm e_i \pm e_j : 1 \le i < j \le 8\}\ (112)\ \cup\ \{\tfrac{1}{2}(\pm 1)^8 :\ \text{even \# of } -\}\ (128),$$

all of norm 2. Splitting each by $(\|\cdot\|^2_{D_5}, \|\cdot\|^2_{A_3})$ gives the branching blocks

$$(2,0)\colon 40\ [D_5],\quad (0,2)\colon 12\ [A_3],\quad (1,1)\colon 60\ [(10,6)],\quad (\tfrac{5}{4},\tfrac{3}{4})\colon 128\ [(16,4)\oplus(\overline{16},\overline{4})],$$

so $\mathfrak{e}_8 = (45,1)\oplus(1,15)\oplus(10,6)\oplus(16,4)\oplus(\overline{16},\overline{4})$, $248 = 45{+}15{+}60{+}64{+}64$, $240 = 40{+}12{+}60{+}64{+}64$.

Simple-root Gram (unimodular check). With the Bourbaki simple roots $\alpha_1 = \frac{1}{2}(e_1{-}e_2{-}\ldots{-}e_7{+}e_8)$, $\alpha_2 = e_1{+}e_2$, and $\alpha_k = e_{k-1}{-}e_{k-2}$ for $k = 3,\ldots,8$ (so $\alpha_3 = e_2{-}e_1,\ \alpha_4 = e_3{-}e_2,\ \ldots,\ \alpha_8 = e_7{-}e_6$), the Gram matrix $G_{ij} = \langle \alpha_i, \alpha_j \rangle$ is the E_8 Cartan matrix with $\det G = 1$ and all $G_{ii} = 2$; every one of the 240 roots is an integer combination of the α_i. Hence the assembled lattice is E_8, not merely a 240-count. **[E]** (The shifted index $e_{k-2}{-}e_{k-3}$ would duplicate α_3 and make G singular — use $e_{k-1}{-}e_{k-2}$.)

Part III
Appendix: ablation of the EM fixed point

The single sharpest reviewer test is: *was the closure* $F_{U(1)}(\alpha) = 0$ *engineered so that* α *comes out?* The honest answer is an ablation — vary each input and see which actually move the root. Three one-parameter scans (all others held at their compiler values) separate the *load-bearing* integers from the *fine* last-digit selector.

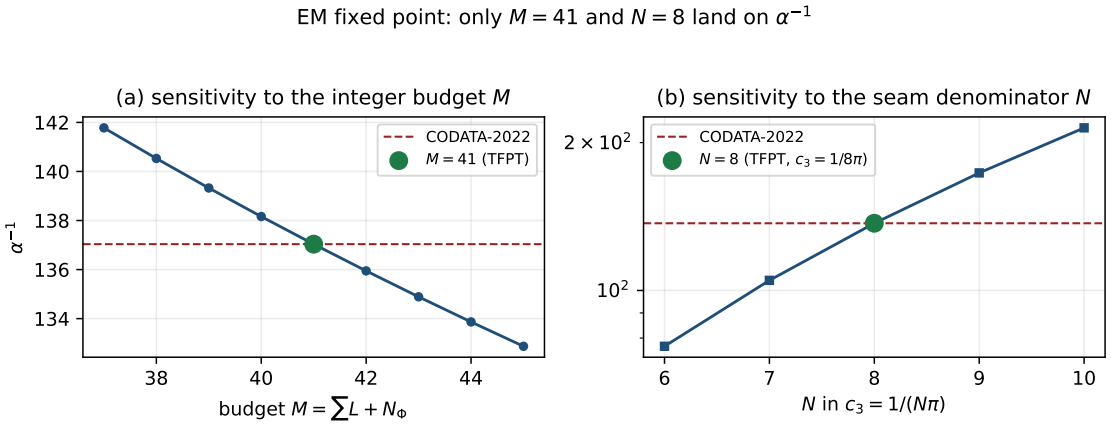


Figure 7: EM-closure ablation (data plot, `verification/make_figures.py`). **(a)** $\alpha_\star^{-1}$ vs the integer budget M: only $M = 41$ lands on CODATA-2022. **(b)** $\alpha_\star^{-1}$ vs the seam denominator N in $c_3 = 1/(N\pi)$: only $N = 8$ lands. Both integers are fixed elsewhere (flavor budget; rank/$h(D_5)$), so the equation is not freely tunable to 137.036.

(A1) The transport budget $M = \sum L + N_\Phi$ (exponent held at $-\frac{5}{4}$):

M	38	39	40	**41**	42	43	44
$\alpha_\star^{-1}$	140.530	139.326	138.162	**137.0359992**	135.946	134.890	133.866

Each unit of M shifts $\alpha_\star^{-1}$ by ≈ 1.09, so *only* $M = 41$ lands on 137.036. This is a hard lock — and $41 = 10b_1 = \sum L + N_\Phi = 40 + 1$ is fixed by the compiler (flavor budget 40 + one Higgs). **Load-bearing.**

(A2) The seam "8" in $c_3 = 1/(N\pi)$ (here $\delta_{\text{top}} = 48c_3^4$ tracks c_3):

N	6	7	**8**	9	10
$\alpha_\star^{-1}$	76.97	104.85	**137.036**	173.53	214.34

Only $N = 8$ — i.e. $c_3 = 1/(8\pi) = 1/(\text{rank}\, E_8 \cdot \pi)$ — reproduces 137.036. The "8" is the same one fixed by the bootstrap ($h(D_5) = \text{rank}\, E_8 = \varphi(30)$) and the Gauss–Bonnet hardening. **Load-bearing.**

(A3) The carrier discriminant exponent (at $M = 41$, $c_3 = 1/(8\pi)$):

exponent	$-\frac{1}{4}$	$-\frac{3}{4}$	$-\frac{5}{4}$	$-\frac{7}{4}$	$-\frac{9}{4}$
$\alpha_\star^{-1}$	137.035995	137.035997	**137.0359992**	137.036001	137.036003

The exponent moves only the *seventh* digit; all candidates agree to six. Here $-\frac{5}{4} = q(D_5)$ is selected, but this is a *fine* selector, not a hard lock. **Fine (last-digit).**

Ablation verdict — honest

The EM fixed point is a **hard test of two integers** — the budget $M = 41$ and the seam denominator $8 = \operatorname{rank} E_8$, each decisive (a single-unit change misses by ~ 1 in $\alpha_\star^{-1}$) — and a **fine test of one norm**, the carrier discriminant exponent $-\frac{5}{4} = q(D_5)$ (seventh digit). The equation is therefore *not* freely tunable to 137.036: the two integers that hit it are exactly the compiler/bootstrap numbers, fixed elsewhere. What it does *not* do is pin the carrier norm strongly — that is honest, and it is the right level of claim. [E]

Appendix: computational verification

The load-bearing identities of this document (marked [E], [E], [E]) are re-derived from $\{c_3, g_{\text{car}}\}$ alone and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the μ_4 glue ($q(D_5)$+$q(A_3)$=2, disc =$\mathbb{Z}_4$, 240, 248).
- `v2_carrier_pascal.py` — g_{car}=5, 16=1+5+10, $N_{\text{fam}}, \Omega_{\text{adm}}, b_1$.
- `v3_em_alpha.py` — the EM fixed point $\alpha_\star^{-1}$=137.0359992168 (existence/uniqueness, ablation).
- `v4_flavor_matrix.py` — R with det =8, minors $\{2, 3, 5\}$, $\sum L$=40.
- `v7_gravity_cosmo.py` — scalaron c_3^7, Λ/A_s readouts.

Run `python verification/run_all.py` (needs `mpmath, numpy, sympy`); it exits 0 iff all checks pass. Full map: `verification/README.md`.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)

- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
- P. Naaijkens, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, `arXiv:2605.10693`.

Lattices and quadratic forms.

- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
- C. L. Siegel, *Über die analytische Theorie der quadratischen Formen* (the Minkowski–Siegel mass formula), Ann. of Math. **36** (1935) 527–606.

Singularities and geometry.

- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, Invent. Math. **2** (1966) 1–14.
- J. Milnor, *Singular Points of Complex Hypersurfaces*, Ann. of Math. Studies **61**, Princeton Univ. Press (1968).

Spectral geometry, gravity and thermal time.

- A. H. Chamseddine, A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997) 731–750.
- A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation in general covariant quantum theories*, Class. Quantum Grav. **11** (1994) 2899–2918.
- R. M. Wald, *Black hole entropy is the Noether charge*, Phys. Rev. D **48** (1993) R3427–R3431.
- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, `arXiv:1102.0440`.